

# P-gp DDI Qualification

| Version                         | Yuri05-patch-1-OSP12.2                                                                                                                                                                                          |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Qualification Plan Release      | <a href="https://github.com/Open-Systems-Pharmacology/Qualification-DDI-P-gp/releases/tag/vYuri05-patch-1">https://github.com/Open-Systems-Pharmacology/Qualification-DDI-P-gp/releases/tag/vYuri05-patch-1</a> |
| OSP Version                     | 12.2                                                                                                                                                                                                            |
| Qualification Framework Version | 3.5                                                                                                                                                                                                             |

This qualification report is filed at:

<https://github.com/Open-Systems-Pharmacology/OSP-Qualification-Reports>

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# 1 Introduction

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## 1.1 Objective

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This **qualification report** evaluates for the PBPK platform **PK-Sim** (as part of the open systems pharmacology (OSP) suite) the ability to perform simulations with the intended purpose to predict Permeability-glycoprotein (**P-gp**)-mediated drug-drug interactions (**DDI**).

To demonstrate the level of confidence, the predictive performance of the platform for this indented purpose is assessed via a network of PBPK models of selected P-gp DDI perpetrators, and a respective sensitive P-gp victim drug (digoxin) and a comprehensive dataset from published clinical DDI studies. All PBPK models represent whole-body PBPK models, which allow dynamic DDI simulations in organs expressing P-gp.

The respective *qualification plan* to produce this *qualification report* is transparently provided open-source (<https://github.com/Open-Systems-Pharmacology/Qualification-DDI-P-gp>). The same applies for all presented PBPK models including *evaluation reports* on model building and evaluation of each model (<https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library>).

*Evaluation reports* including descriptions on model building and detailed evaluations of the included models are documented separately (see [Section 1.2](#)).

Please refer to the [Appendix](#) to learn more details:

- An overview over the Open Systems Pharmacology Suite is given in chapter [Section 5.1](#)
- [Section 5.2](#) shows the implementation of the underlying mathematical equations for drug-drug interactions in the OSP suite.
- A detailed general description of the performed qualification workflow (*qualification plan*, *qualification report*, etc.) can be found in chapter [Section 5.3](#).

## 1.2 Pgp DDI Network

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To qualify the OSP suite for the prediction of the P-gp DDI potential of new drugs, a set of verified PBPK models of perpetrators, and respective P-gp DDI victim drugs is specified to set up a P-gp-mediated DDI modeling network.

The following perpetrator compounds were selected:

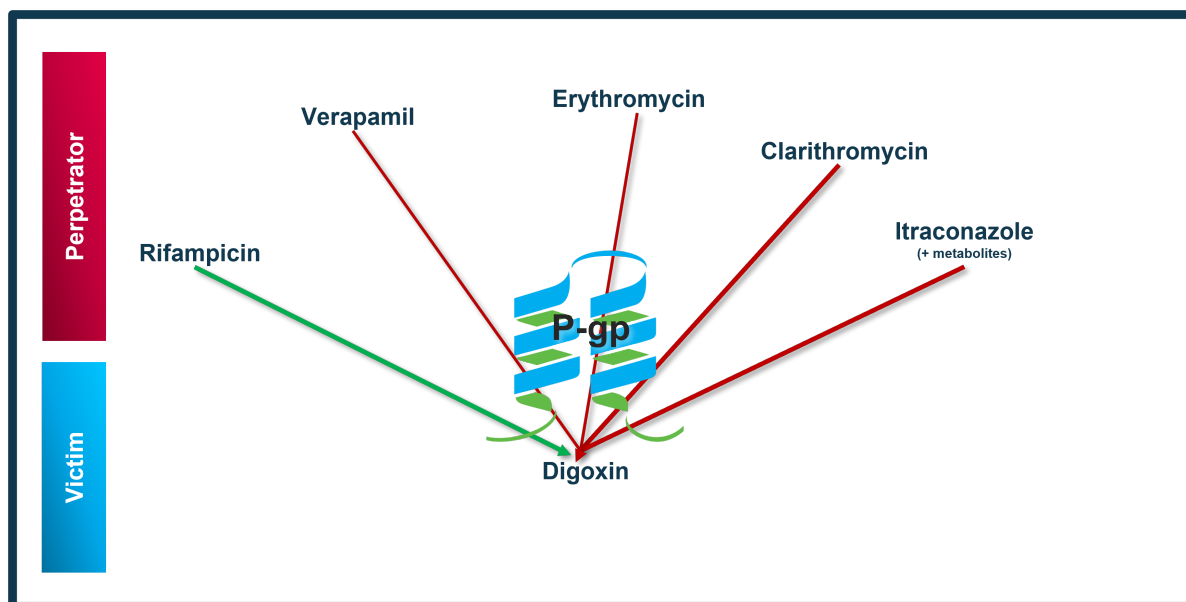
- **Rifampicin** Model snapshot and evaluation plan (*release v2.0*): <https://github.com/Open-Systems-Pharmacology/Rifampicin-Model/releases/tag/v2.0>
- **Verapamil** Model snapshot and evaluation plan (*release v2.1*): <https://github.com/Open-Systems-Pharmacology/Verapamil-Model/releases/tag/v2.1>
- **Erythromycin** Model snapshot and evaluation plan (*release v2.0*): <https://github.com/Open-Systems-Pharmacology/Erythromycin-Model/releases/tag/v2.0>
- **Clarithromycin** Model snapshot and evaluation plan (*release v2.0*): <https://github.com/Open-Systems-Pharmacology/Clarithromycin-Model/releases/tag/v2.0>
- **Itraconazole** including metabolites Model snapshot and evaluation plan (*release v2.0*): <https://github.com/Open-Systems-Pharmacology/Itraconazole-Model/releases/tag/v2.0>

The following sensitive P-gp substrates as victim drugs were selected:

- **Digoxin** Model snapshot and evaluation plan (*release v2.0*): <https://github.com/Open-Systems-Pharmacology/Digoxin-Model/releases/tag/v2.0>

**Figure 1** shows the prespecified and developed DDI modeling network of interacting perpetrator and victim drugs for the OSP suite qualification of predicting P-gp-mediated DDI.

**Figure 1: P-gp DDI modeling network**



The arrows indicate where at least one clinical DDI study between the two connected substances was available and included in the model network. Red indicates inhibition and green indicates induction as the primary type of interaction. Thin arrows indicate weak, mid-thick arrows moderate and thick arrows strong P-gp modulation by the perpetrator.

The published DDI studies between the respective perpetrators and victim drugs were simulated and compared to observed data. The following sections give an overview of the clinical studies being part of this qualification report. The respective data identifier (DataID) refers to the **ID** of the dataset in the OSP PK database, version 1.7 (<https://github.com/Open-Systems-Pharmacology/Database-for-observed-data/releases/tag/v1.7>).

### 1.2.1 Clarithromycin - Digoxin DDI

The release of the snapshot containing the respective simulations can be found here: <https://github.com/Open-Systems-Pharmacology/Clarithromycin-Digoxin-DDI/releases/tag/v1.0>

The Clarithromycin / digoxin interaction was evaluated using 6 clinical DDI studies including 21 different clinical settings (Gurley 2006, Gurley 2007, Gurley 2008b, Kurata 2002, Rengelshausen 2003, Tsutsumi 2002).



| <b>DataID</b> | <b>Enzyme</b> | <b>Perpetrator / victim</b> | <b>Study design</b>                                                                                                                                             | <b>Comment</b> | <b>Clinical study</b>        |
|---------------|---------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------|
| 16324         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>500</b> mg po twice daily (14 doses)<br>Digoxin: <b>0.4</b> mg po single dose, simultaneous with <b>12<sup>th</sup></b> clarithromycin dose  |                | <a href="#">Gurley 2006</a>  |
| 16328         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>500</b> mg po twice daily (14 doses)<br>Digoxin: <b>0.5</b> mg po single dose, simultaneous with <b>12<sup>th</sup></b> clarithromycin dose  |                | <a href="#">Gurley 2007</a>  |
| 229           | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>500</b> mg po twice daily (14 doses)<br>Digoxin: <b>0.25</b> mg po single dose, 2 hours after <b>12<sup>th</sup></b> clarithromycin dose     |                | <a href="#">Gurley 2008b</a> |
| 16305         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg po single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16307         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg iv single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16309         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg po single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16311         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg iv single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16313         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg po single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16315         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg iv single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16333         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg po single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16334         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg iv single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16335         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg po single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16336         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg iv single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |
| 16337         | P-gp          | Clarithromycin / digoxin    | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg po single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |                | <a href="#">Kurata 2002</a>  |

| DatalD | Enzyme | Perpetrator / victim     | Study design                                                                                                                                                    | Comment | Clinical study                     |
|--------|--------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------------------------------|
| 16338  | P-gp   | Clarithromycin / digoxin | Clarithromycin: <b>400</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg iv single dose, simultaneous with <b>10<sup>th</sup></b> clarithromycin dose |         | <a href="#">Kurata 2002</a>        |
| 3025   | P-gp   | Clarithromycin / digoxin | Clarithromycin: <b>250</b> mg po twice daily (6 doses)<br>Digoxin: <b>0.75</b> mg po single dose, 0.5 hours after <b>3<sup>rd</sup></b> clarithromycin dose     |         | <a href="#">Rengelshausen 2003</a> |
| 16409  | P-gp   | Clarithromycin / digoxin | Clarithromycin: <b>250</b> mg po twice daily (6 doses)<br>Digoxin: <b>0.01</b> mg/kg iv single dose, 2 hours after <b>3<sup>rd</sup></b> clarithromycin dose    |         | <a href="#">Rengelshausen 2003</a> |
| 16411  | P-gp   | Clarithromycin / digoxin | Clarithromycin: <b>250</b> mg po twice daily (6 doses)<br>Digoxin: <b>0.75</b> mg po single dose, 0.5 hours after <b>3<sup>rd</sup></b> clarithromycin dose     |         | <a href="#">Rengelshausen 2003</a> |
| 16413  | P-gp   | Clarithromycin / digoxin | Clarithromycin: <b>250</b> mg po twice daily (6 doses)<br>Digoxin: <b>0.75</b> mg po single dose, 0.5 hours after <b>3<sup>rd</sup></b> clarithromycin dose     |         | <a href="#">Rengelshausen 2003</a> |
| 3029   | P-gp   | Clarithromycin / digoxin | Clarithromycin: <b>200</b> mg po twice daily (10 doses)<br>Digoxin: <b>0.5</b> mg iv single dose, simultaneous with <b>3<sup>rd</sup></b> clarithromycin dose   |         | <a href="#">Tsutsumi 2002</a>      |
| 3032   | P-gp   | Clarithromycin / digoxin | Clarithromycin: <b>200</b> mg po twice daily (10 doses)<br>Digoxin: <b>0.5</b> mg iv single dose, simultaneous with <b>3<sup>rd</sup></b> clarithromycin dose   |         | <a href="#">Tsutsumi 2002</a>      |

## 1.2.2 Erythromycin - Digoxin DDI

The release of the snapshot containing the respective simulations can be found here: <https://github.com/Open-Systems-Pharmacology/Erythromycin-Digoxin-DDI/releases/tag/v1.1>

The erythromycin-digoxin interaction was evaluated using one clinical DDI study ([Tsutsumi 2002](#)).

| DatalD | Transporter | Perpetrator / victim   | Study design                                                                                                             | Clinical study                |
|--------|-------------|------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 3028   | P-gp        | Erythromycin / digoxin | Erythromycin: 200 mg po QID for 5 days<br>Digoxin: 0.5 mg iv, concomitantly with erythromycin on the 2 <sup>nd</sup> day | <a href="#">Tsutsumi 2002</a> |

## 1.2.3 Itraconazole - Digoxin DDI

The release of the snapshot containing the respective simulations can be found here: <https://github.com/Open-Systems-Pharmacology/Itraconazole-Digoxin-DDI/releases/tag/v1.0>

The itraconazole-digoxin interaction was evaluated using 2 clinical DDI studies including 3 different clinical settings ([Jalava 1997](#), [Partanen 1995](#)).

| DataID | Transporter | Perpetrator / victim   | Study design                                                                                                                                                                                                               | Comment | Clinical study                |
|--------|-------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------|
| 271    | P-gp        | Itraconazole / digoxin | Itraconazole: <b>200</b> mg po once daily (5 doses)<br>Digoxin: <b>0.5</b> mg po single dose, 1 h after the <b>3<sup>rd</sup></b> itraconazole dose                                                                        |         | <a href="#">Jalava 1997</a>   |
| 16329  | P-gp        | Itraconazole / digoxin | Itraconazole: <b>200</b> mg po once daily (10 doses)<br>Digoxin: <b>0.25</b> mg po once daily (20 doses), 2 hours after <b>12<sup>th</sup></b> itraconazole dose                                                           |         | <a href="#">Partanen 1995</a> |
| 16330  | P-gp        | Itraconazole / digoxin | Itraconazole: <b>200</b> mg po once daily (10 doses) starting with the <b>11<sup>th</sup></b> digoxin dose<br>Digoxin: <b>0.25</b> mg po once daily (20 doses), simultaneous with <b>10<sup>th</sup></b> itraconazole dose |         | <a href="#">Partanen 1995</a> |

## 1.2.4 Verapamil - Digoxin DDI

The release of the snapshot containing the respective simulations can be found here: <https://github.com/Open-Systems-Pharmacology/Verapamil-Digoxin-DDI/releases/tag/v1.1>

The verapamil-digoxin interaction was evaluated using a 11 clinical DDI studies including 12 different clinical settings ([Belz 1981](#), [Belz 1983](#), [Doering 1983](#), [Johnson 1987](#), [Klein 1982](#), [Pedersen 1981](#), [Pedersen 1982](#), [Pedersen 1983](#), [Rodin 1988](#), [Schwartz 1982](#), [Wiebe 2020](#)).

| DatalD | Transporter | Perpetrator / victim | Study design                                                                                                                          | Comment                                                                                                              | Clinical study                |
|--------|-------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 16635  | P-gp        | Verapamil / digoxin  | Verapamil: 80 mg po TID for 14 days<br>Digoxin: 0.125 mg po TID for 14 days                                                           |                                                                                                                      | <a href="#">Belz 1981</a>     |
| 16165  | P-gp        | Verapamil / digoxin  | Verapamil: 80 mg po TID for 14 days<br>Digoxin: 0.125 mg po TID for 14 days                                                           |                                                                                                                      | <a href="#">Belz 1983</a>     |
| 16167  | P-gp        | Verapamil / digoxin  | Verapamil: 120 mg po TID for 14 days<br>Digoxin: 0.125 mg po TID for 14 days                                                          |                                                                                                                      | <a href="#">Belz 1983</a>     |
| 16169  | P-gp        | Verapamil / digoxin  | Verapamil: 80 mg po TID for 14 days<br>Digoxin: 0.125 mg po TID for 14 days                                                           |                                                                                                                      | <a href="#">Doering 1983</a>  |
| 16159  | P-gp        | Verapamil / digoxin  | Verapamil: 80 mg po BID for 4 days followed by TID for 10 days<br>Digoxin: 1.0 mg iv (15 min) on day 13                               |                                                                                                                      | <a href="#">Johnson 1987</a>  |
| 16174  | P-gp        | Verapamil / digoxin  | Verapamil: 80 mg po TID for 14 days<br>Digoxin: 0.25 mg PO OD for 28 days                                                             |                                                                                                                      | <a href="#">Klein 1982</a>    |
| 16638  | P-gp        | Verapamil / digoxin  | Verapamil: 80 mg po TID for 10 days<br>Digoxin: 0.75 mg IV SD at day 10                                                               |                                                                                                                      | <a href="#">Pedersen 1981</a> |
| 16163  | P-gp        | Verapamil / digoxin  | Verapamil: 80 mg po TID followed by 120mg TID<br>Digoxin: initial dose of 1mg PO followed by 0.0625 mg po BID followed by 0.125mg BID |                                                                                                                      | <a href="#">Pedersen 1982</a> |
| 16162  | P-gp        | Verapamil / digoxin  | Verapamil: 120 mg po TID for 12 days<br>Digoxin: 1.0 mg iv (bolus) on day 8                                                           |                                                                                                                      | <a href="#">Pedersen 1983</a> |
| 3063   | P-gp        | Verapamil / digoxin  | Verapamil: 80mg BID for 4 days followed by 80mg TID for 10 days<br>Digoxin: 0.25 mg PO BID for 14 days                                |                                                                                                                      | <a href="#">Rodin 1988</a>    |
| 16175  | P-gp        | Verapamil / digoxin  | Verapamil: 80 mg po QID for 70 days<br>Digoxin: 0.46 mg PO OD for 85 days                                                             | Daily dose of digoxin varied between patients between 0.25 and 1 mg/day. 0.46 is the arith. mean of all dose levels. | <a href="#">Schwartz 1982</a> |
| 16172  | P-gp        | Verapamil / digoxin  | Verapamil: 120 mg po SD<br>Digoxin: 0.25 mg PO SD 1h after verapamil dosing                                                           |                                                                                                                      | <a href="#">Wiebe 2020</a>    |

## 1.2.5 Rifampicin - Digoxin DDI

The release of the snapshot containing the respective simulations can be found here: <https://github.com/Open-Systems-Pharmacology/Rifampicin-Digoxin-DDI/releases/tag/v1.0>

The rifampicin-digoxin interaction was evaluated using 9 clinical DDI studies including 11 different clinical settings (Drescher 2003, Gurley 2006, Gurley 2007, Gurley 2008b, Greiner 1999, Kirby 2012, Larsen 2007, Reitmann 2011, Wiebe 2020).

| DatalD | Transporter | Perpetrator / victim | Study design                                                                                                                                                              | Comment | Clinical study                |
|--------|-------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------------------------|
| 191    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (16 doses)<br>Digoxin: <b>1</b> mg po single dose, simultaneous with <b>11<sup>th</sup></b> rifampicin dose                       |         | <a href="#">Greiner 1999</a>  |
| 193    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (16 doses)<br>Digoxin: <b>1</b> mg iv single dose, simultaneous with <b>11<sup>th</sup></b> rifampicin dose                       |         | <a href="#">Greiner 1999</a>  |
| 227    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>300</b> mg po twice daily (16 doses)<br>Digoxin: <b>0.25</b> mg po single dose, 2 hours after <b>13<sup>th</sup></b> rifampicin dose                       |         | <a href="#">Gurley 2008b</a>  |
| 314    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (14 doses)<br>Digoxin: <b>0.5</b> mg po single dose, 12 hours after <b>14<sup>th</sup></b> rifampicin dose                        |         | <a href="#">Kirby 2012</a>    |
| 316    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (14 doses)<br>Digoxin: <b>0.5</b> mg iv single dose, 12 hours after <b>14<sup>th</sup></b> rifampicin dose                        |         | <a href="#">Kirby 2012</a>    |
| 337    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (6 doses)<br>Digoxin: <b>0.5</b> mg po single dose, simultaneous with <b>7<sup>th</sup></b> rifampicin dose                       |         | <a href="#">Larsen 2007</a>   |
| 396    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (28 doses)<br>Digoxin: <b>0.5</b> mg po single dose, 1 h after the <b>28<sup>th</sup></b> (last) rifampicin dose                  |         | <a href="#">Reitmann 2011</a> |
| 397    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (28 doses)<br>Digoxin: <b>0.5</b> mg po single dose, 169 h (1 week +1 h) after the <b>28<sup>th</sup></b> (last) rifampicin dose  |         | <a href="#">Reitmann 2011</a> |
| 398    | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (28 doses)<br>Digoxin: <b>0.5</b> mg po single dose, 337 h (2 weeks +1 h) after the <b>28<sup>th</sup></b> (last) rifampicin dose |         | <a href="#">Reitmann 2011</a> |
| 16301  | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po single dose<br>Digoxin: <b>0.25</b> mg po single dose, simultaneous with rifampicin dose                                                     |         | <a href="#">Wiebe 2020</a>    |
| 16319  | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (14 doses)<br>Digoxin: <b>1</b> mg iv single dose, 12 h after the <b>10<sup>th</sup></b> rifampicin dose                          |         | <a href="#">Drescher 2003</a> |
| 16322  | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (14 doses)<br>Digoxin: <b>0.4</b> mg po single dose, simultaneous with <b>7<sup>th</sup></b> rifampicin dose                      |         | <a href="#">Gurley 2006</a>   |
| 16326  | P-gp        | Rifampicin / digoxin | Rifampicin: <b>600</b> mg po once daily (14 doses)<br>Digoxin: <b>0.5</b> mg iv single dose, simultaneous with <b>7<sup>th</sup></b> rifampicin dose                      |         | <a href="#">Gurley 2007</a>   |

## 2 Qualification of Use Case Pgp-mediated DDI

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The following section shows the correlations between observed and model-predicted AUC and  $C_{\max}$  ratios, respectively.

Specifically, the PBPK model performance for the PK parameters **AUC ratio (AUCR)** and  **$C_{\max}$  ratio (CMAXR)** is assessed via:

- predicted (*Pred*) vs. observed (*Obs*) plots
- *Pred/Obs* vs. *Obs* plots
- geometric mean fold error (GMFE):

$$10^{\frac{\sum |\log(\frac{Pred}{Obs})|}{n}}$$

- number of AUCR and CMAXR falling within 2-fold error range and within the limits as suggested by [Guest et al. 2011](#)
- detailed table of results for each study

In the plots,

- the dotted lines denote 0.50–2.00 (2-fold) criterion,
- the solid lines denote the limits as suggested by [Guest et al. 2011](#),
- the bold solid line denotes the unity line,
- each color represents one combination of drugs,
- squares represent studies with intravenous administration of the victim drug and circles represent studies with oral administration of the victim drug.

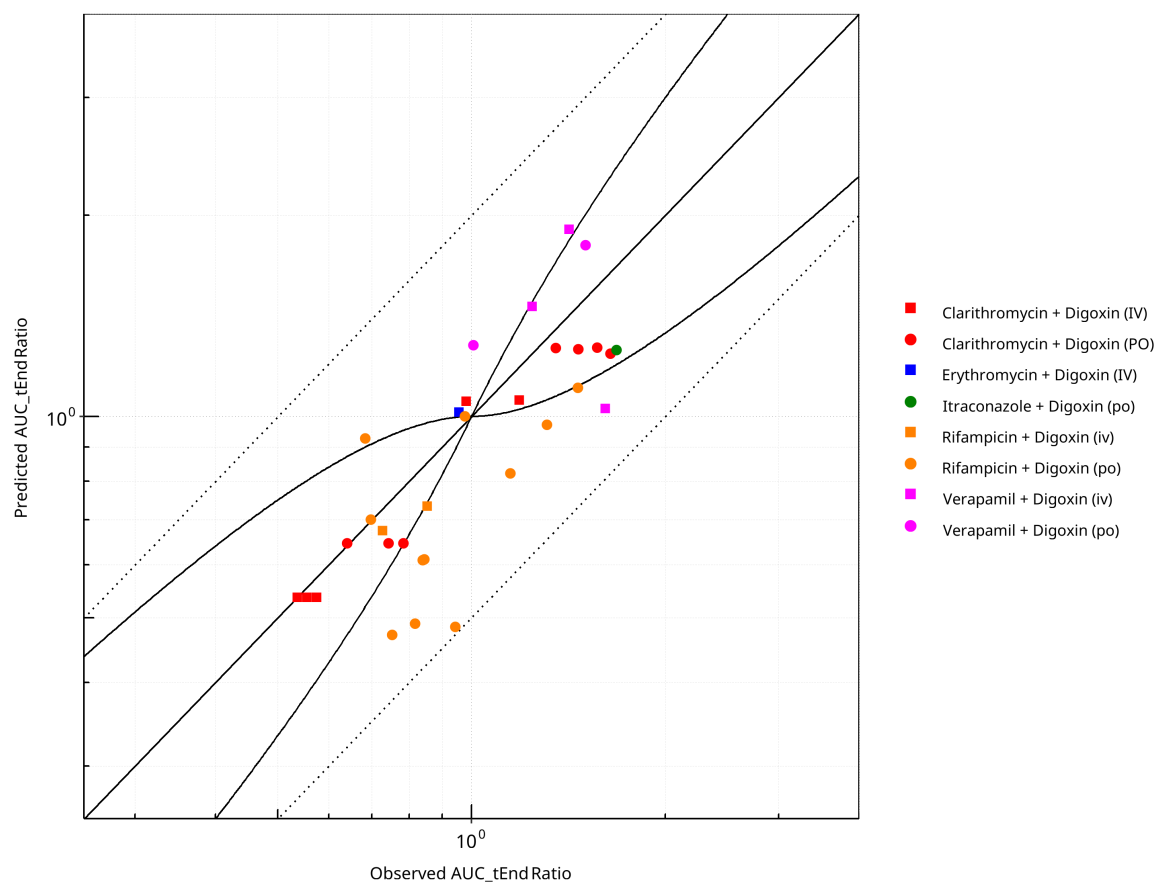


Figure 2-1: Pgp DDI. Predicted vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)

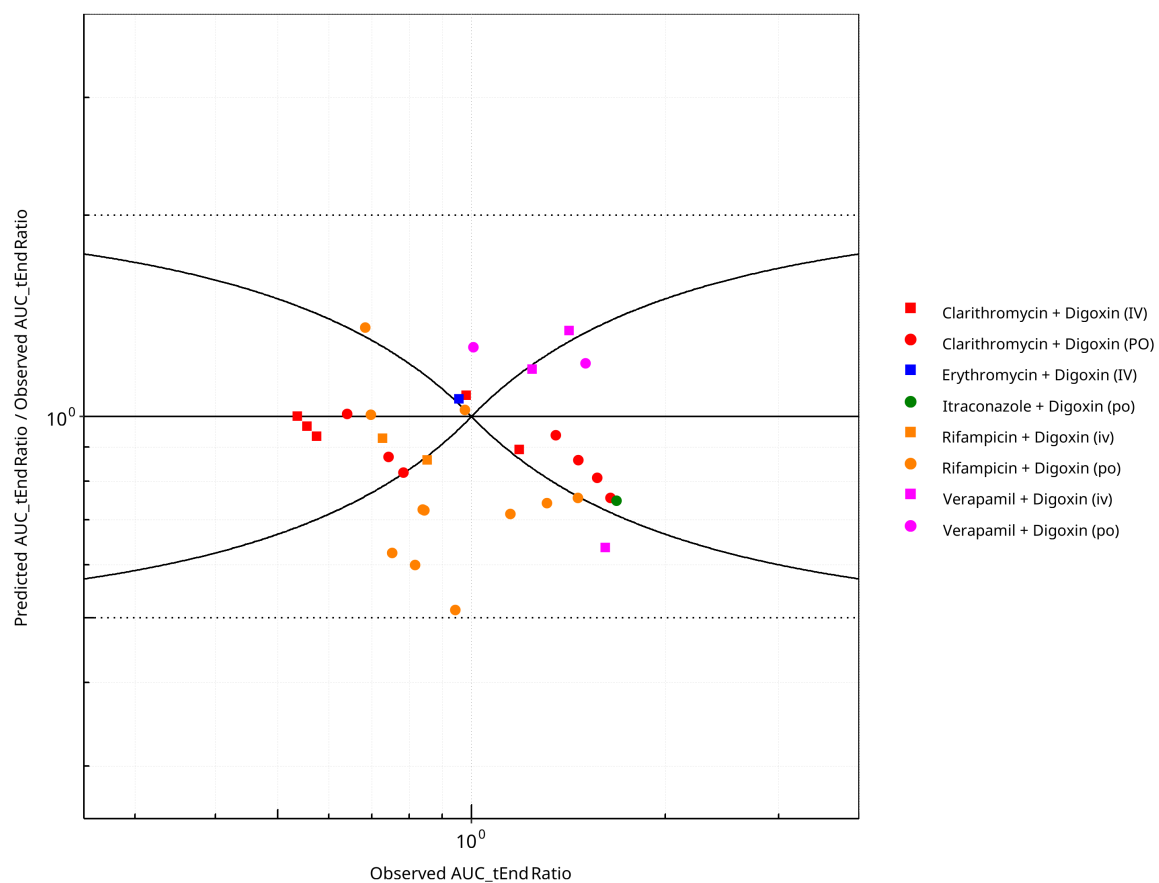


Figure 2-2: Pgp DDI. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)



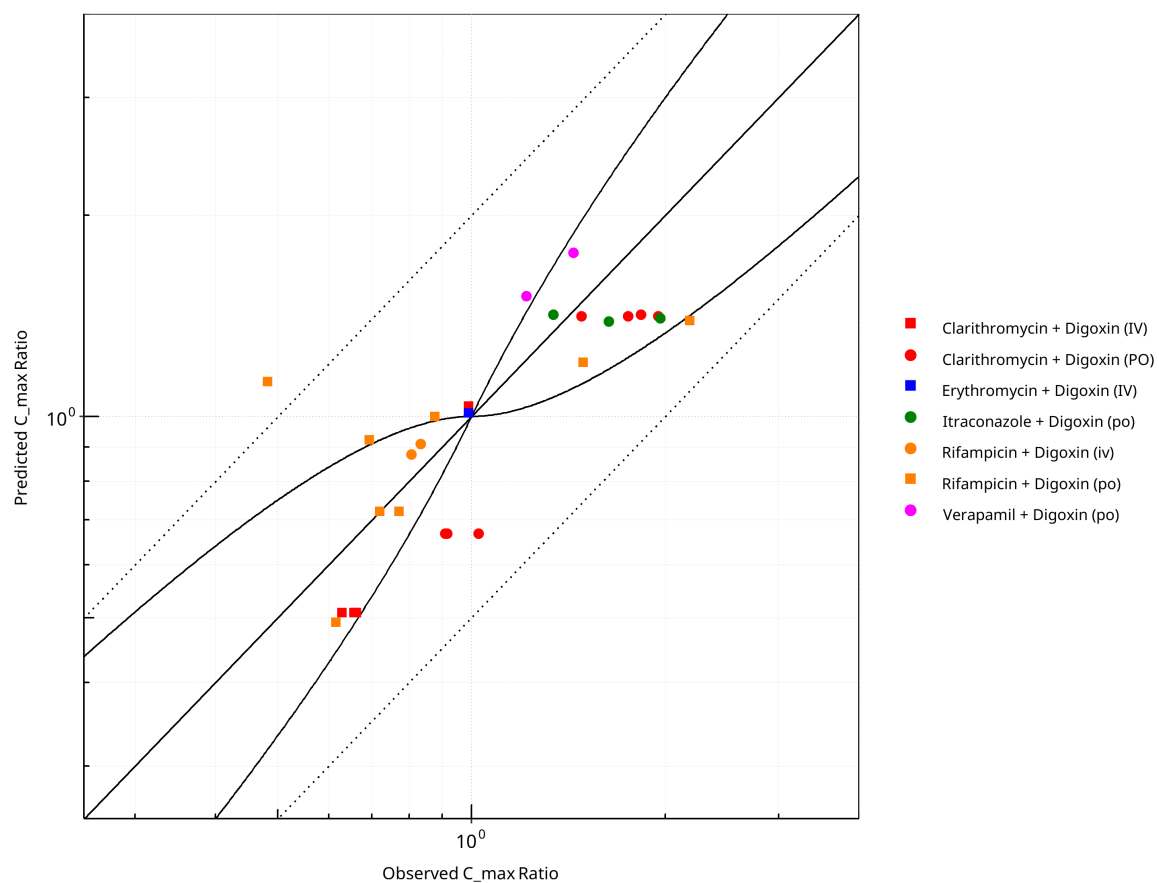


Figure 2-3: Pgp DDI. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

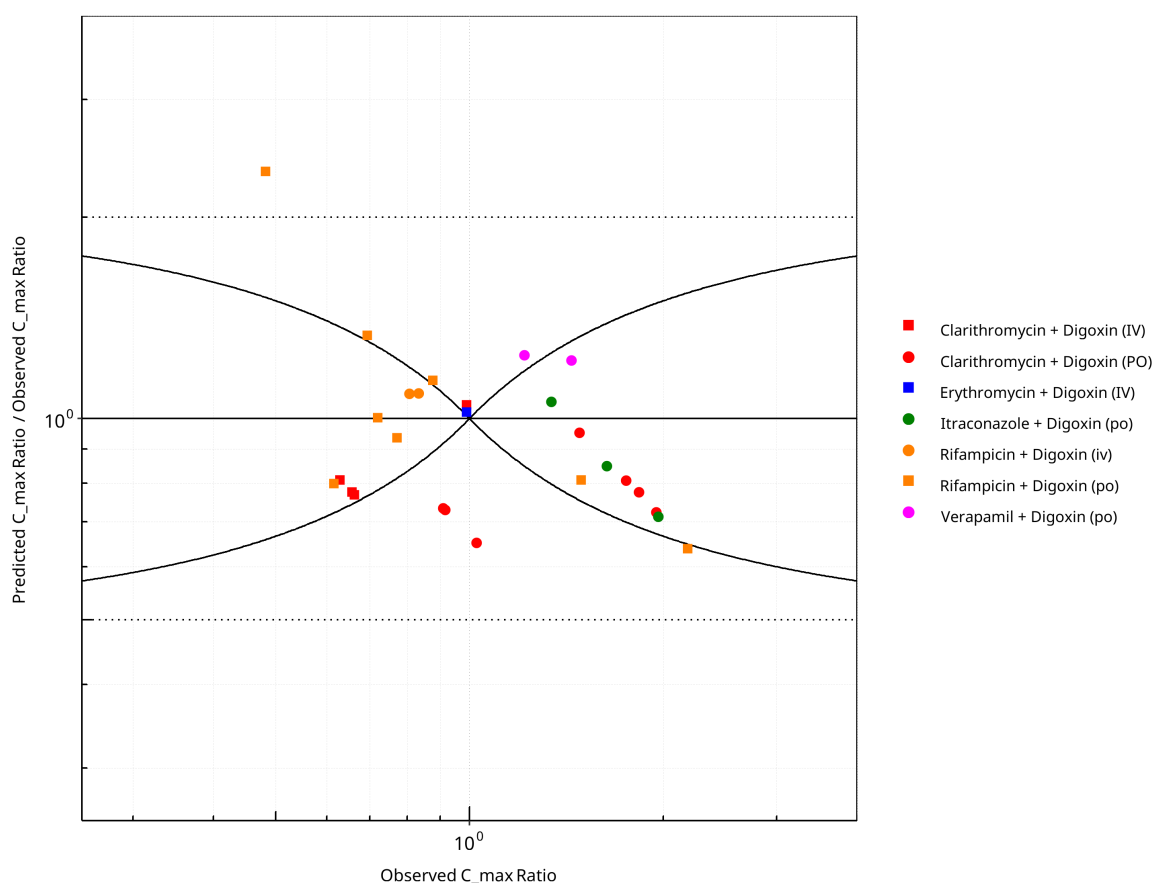


Figure 2-4: Pgp DDI. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

Table 2-1: GMFE for Pgp DDI Ratio

| PK parameter        | GMFE |
|---------------------|------|
| AUC <sub>tEnd</sub> | 1.24 |
| C <sub>max</sub>    | 1.25 |

Table 2-2: Summary table for Pgp DDI - AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| AUC <sub>tEnd</sub>               | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 32     | -         |
| Points within Guest <i>et al.</i> | 16     | 50        |
| Points within 2 fold              | 32     | 100       |

**Table 2-3: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 27     | -         |
| Points within Guest <i>et al.</i> | 17     | 62.96     |
| Points within 2 fold              | 26     | 96.30     |

**Table 2-4: Summary table for Pgp DDI**

| DataID | Perpetrator                                          | Victim         | Predicted<br>AUC_tEnd<br>Ratio | Observed<br>AUC_tEnd<br>Ratio | Pred/Obs<br>AUC_tEnd<br>Ratio | Predicted<br>C_max<br>Ratio | Observed<br>C_max<br>Ratio | Pred/Obs<br>C_max<br>Ratio | Reference    |
|--------|------------------------------------------------------|----------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|----------------------------|----------------------------|--------------|
| 191    | Rifampicin,<br>600 mg, PO,<br>MD OD (16<br>days)     | Digoxin,<br>PO | 0.70                           | 0.70                          | 1.01                          | 1.13                        | 0.48                       | 2.34                       | Greiner 1999 |
| 193    | Rifampicin,<br>600 mg, PO,<br>MD OD (16<br>days)     | Digoxin,<br>IV | 0.73                           | 0.85                          | 0.86                          | 0.91                        | 0.83                       | 1.09                       | Greiner 1999 |
| 227    | Rifampicin,<br>300 mg, PO,<br>MD BID (7<br>days)     | Digoxin,<br>PO | 0.47                           | 0.75                          | 0.63                          | 0.49                        | 0.62                       | 0.80                       | Gurley 2008b |
| 229    | Clarithromycin,<br>500 mg, PO,<br>MD BID (7<br>days) | Digoxin,<br>PO | 1.26                           | 1.47                          | 0.86                          | 1.42                        | 1.75                       | 0.81                       | Gurley 2008b |
| 271    | Itraconazole,<br>200 mg, PO,<br>MD OD (5<br>days)    | Digoxin,<br>PO | 1.26                           | 1.68                          | 0.75                          | 1.42                        | 1.34                       | 1.06                       | Jalava 1997  |
| 314    | Rifampicin,<br>600 mg, PO,<br>MD OD (14<br>days)     | Digoxin,<br>PO | 0.49                           | 0.94                          | 0.51                          | 0.52                        | -                          | -                          | Kirby 2012   |
| 316    | Rifampicin,<br>600 mg, PO,<br>MD OD (14<br>days)     | Digoxin,<br>PO | 0.82                           | 1.15                          | 0.72                          | 1.13                        | -                          | -                          | Kirby 2012   |
| 337    | Rifampicin,<br>600 mg, PO,<br>MD OD (6<br>days)      | Digoxin,<br>PO | 0.49                           | 0.82                          | 0.60                          | 0.52                        | -                          | -                          | Larsen 2007  |
| 396    | Rifampicin,<br>600 mg, PO,<br>MD OD (28<br>days)     | Digoxin,<br>PO | 1.11                           | 1.46                          | 0.76                          | 1.21                        | 1.49                       | 0.81                       | Reitman 2011 |
| 397    | Rifampicin,<br>600 mg, PO,<br>MD OD (28<br>days)     | Digoxin,<br>PO | 0.93                           | 0.68                          | 1.36                          | 0.92                        | 0.69                       | 1.33                       | Reitman 2011 |
| 398    | Rifampicin,<br>600 mg, PO,<br>MD OD (28<br>days)     | Digoxin,<br>PO | 1.00                           | 0.98                          | 1.03                          | 1.00                        | 0.88                       | 1.14                       | Reitman 2011 |

| DataID | Perpetrator                                                                 | Victim         | Predicted<br>AUC_tEnd<br>Ratio | Observed<br>AUC_tEnd<br>Ratio | Pred/Obs<br>AUC_tEnd<br>Ratio | Predicted<br>C_max<br>Ratio | Observed<br>C_max<br>Ratio | Pred/Obs<br>C_max<br>Ratio | Reference               |
|--------|-----------------------------------------------------------------------------|----------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|----------------------------|----------------------------|-------------------------|
| 3028   | Erythromycin,<br>200 mg, PO,<br>MD QID (5<br>days)                          | Digoxin,<br>IV | 1.02                           | 0.95                          | 1.06                          | 1.01                        | 0.99                       | 1.02                       | Tsutsumi 2002           |
| 3029   | Clarithromycin,<br>200 mg, PO,<br>MD BID (5<br>days)                        | Digoxin,<br>IV | 1.05                           | 0.98                          | 1.08                          | 1.04                        | 0.99                       | 1.05                       | Tsutsumi 2002           |
| 3063   | Verapamil, 80<br>mg, PO, MD:<br>80mg BID (4<br>days), 80mg<br>TID (10 days) | Digoxin,<br>PO | 1.81                           | 1.50                          | 1.20                          | 1.76                        | 1.44                       | 1.22                       | Rodin 1988              |
| 16159  | Verapamil, 80<br>mg, PO, MD<br>BID 4days,<br>TID 10days                     | Digoxin,<br>IV | 1.46                           | 1.24                          | 1.18                          | 1.05                        | -                          | -                          | Johnson et al.<br>1987  |
| 16162  | Verapamil, 120<br>mg, PO, TID, 7<br>days pre-<br>treatment                  | Digoxin,<br>IV | 1.91                           | 1.42                          | 1.34                          | 1.01                        | -                          | -                          | Pedersen et al.<br>1983 |
| 16172  | Verapamil, 120<br>mg, PO, SD                                                | Digoxin,<br>PO | 1.28                           | 1.01                          | 1.27                          | 1.52                        | 1.22                       | 1.25                       | Wiebe 2020              |
| 16301  | Rifampicin,<br>600 mg, PO,<br>SD                                            | Digoxin,<br>PO | 0.97                           | 1.31                          | 0.74                          | 1.39                        | 2.18                       | 0.64                       | Wiebe 2020              |
| 16305  | Clarithromycin,<br>400 mg, PO,<br>MD                                        | Digoxin,<br>PO | 0.65                           | 0.74                          | 0.87                          | 0.67                        | 1.02                       | 0.65                       | Kurata 2002             |
| 16307  | Clarithromycin,<br>400 mg, PO,<br>MD                                        | Digoxin,<br>IV | 0.54                           | 0.57                          | 0.93                          | 0.51                        | 0.66                       | 0.78                       | Kurata 2002             |
| 16309  | Clarithromycin,<br>400 mg, PO,<br>MD                                        | Digoxin,<br>PO | 0.65                           | 0.64                          | 1.01                          | 0.67                        | 0.91                       | 0.74                       | Kurata 2002             |
| 16311  | Clarithromycin,<br>400 mg, PO,<br>MD                                        | Digoxin,<br>IV | 0.54                           | 0.54                          | 1.00                          | 0.51                        | 0.66                       | 0.77                       | Kurata 2002             |
| 16313  | Clarithromycin,<br>400 mg, PO,<br>MD                                        | Digoxin,<br>PO | 0.65                           | 0.78                          | 0.83                          | 0.67                        | 0.92                       | 0.73                       | Kurata 2002             |
| 16315  | Clarithromycin,<br>400 mg, PO,<br>MD                                        | Digoxin,<br>IV | 0.54                           | 0.56                          | 0.97                          | 0.51                        | 0.63                       | 0.81                       | Kurata 2002             |

| DataID | Perpetrator                                          | Victim         | Predicted<br>AUC_tEnd<br>Ratio | Observed<br>AUC_tEnd<br>Ratio | Pred/Obs<br>AUC_tEnd<br>Ratio | Predicted<br>C_max<br>Ratio | Observed<br>C_max<br>Ratio | Pred/Obs<br>C_max<br>Ratio | Reference             |
|--------|------------------------------------------------------|----------------|--------------------------------|-------------------------------|-------------------------------|-----------------------------|----------------------------|----------------------------|-----------------------|
| 16319  | Rifampicin,<br>600 mg, PO,<br>MD                     | Digoxin,<br>IV | 0.68                           | 0.73                          | 0.93                          | 0.88                        | 0.81                       | 1.09                       | Drescher 2003         |
| 16322  | Rifampicin,<br>300 mg, PO,<br>MD                     | Digoxin,<br>PO | 0.61                           | 0.84                          | 0.73                          | 0.72                        | 0.77                       | 0.94                       | Gurley 2006           |
| 16324  | Clarithromycin,<br>500 mg, PO,<br>MD                 | Digoxin,<br>PO | 1.27                           | 1.35                          | 0.94                          | 1.42                        | 1.48                       | 0.95                       | Gurley 2006           |
| 16326  | Rifampicin,<br>300 mg, PO,<br>MD                     | Digoxin,<br>PO | 0.61                           | 0.84                          | 0.73                          | 0.72                        | 0.72                       | 1.00                       | Gurley 2007           |
| 16328  | Clarithromycin,<br>500 mg, PO,<br>MD                 | Digoxin,<br>PO | 1.27                           | 1.57                          | 0.81                          | 1.42                        | 1.95                       | 0.73                       | Gurley 2007           |
| 16329  | Itraconazole,<br>200 mg, PO,<br>MD                   | Digoxin,<br>PO | 1.35                           | -                             | -                             | 1.39                        | 1.63                       | 0.85                       | Partanen 1996         |
| 16330  | Itraconazole,<br>200 mg, PO,<br>MD                   | Digoxin,<br>PO | 1.37                           | -                             | -                             | 1.40                        | 1.97                       | 0.71                       | Partanen 1996         |
| 16409  | Clarithromycin,<br>250 mg, PO,<br>MD BID (3<br>days) | Digoxin,<br>IV | 1.06                           | 1.19                          | 0.89                          | 1.00                        | -                          | -                          | Rengelshausen<br>2003 |
| 16411  | Clarithromycin,<br>250 mg, PO,<br>MD BID (3<br>days) | Digoxin,<br>PO | 1.24                           | 1.64                          | 0.76                          | 1.42                        | 1.83                       | 0.78                       | Rengelshausen<br>2003 |
| 16638  | Verapamil, 80<br>mg, PO, TID                         | Digoxin,<br>IV | 1.03                           | 1.61                          | 0.64                          | 0.83                        | -                          | -                          | Pedersen 1981         |

## 2.1 Mechanism

### 2.1.1 Induction

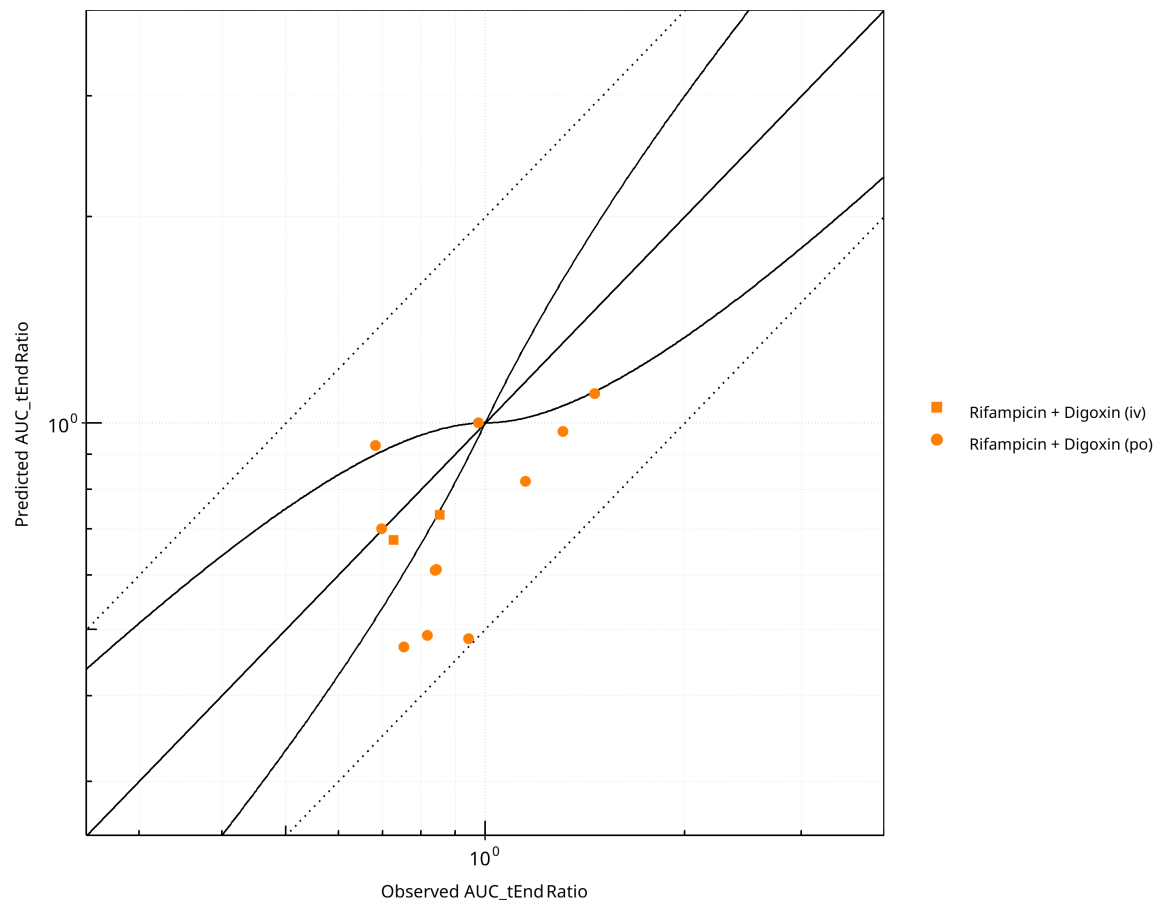
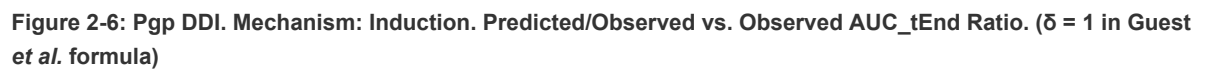


Figure 2-5: Pgp DDI. Mechanism: Induction. Predicted vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)







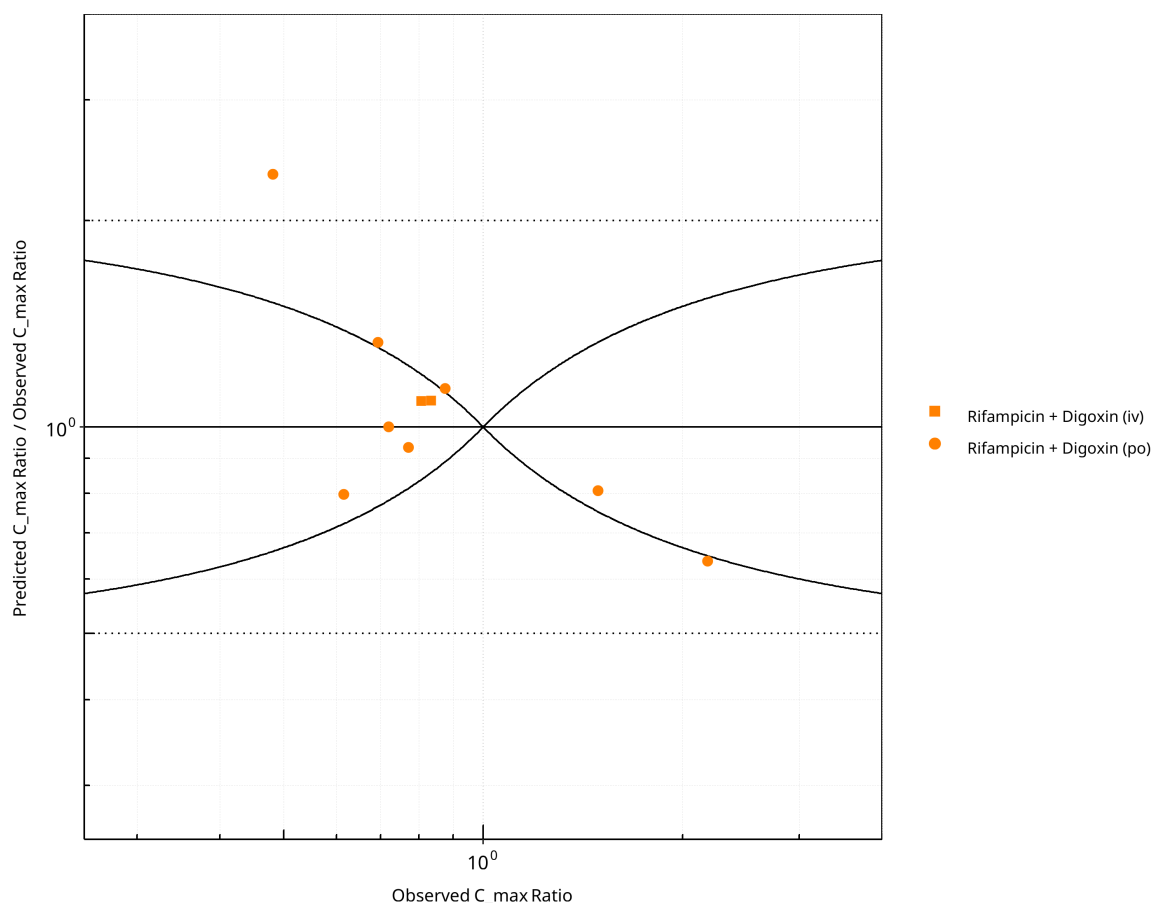


Figure 2-8: Pgp DDI. Mechanism: Induction. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

Table 2-5: GMFE for Pgp DDI Ratio

| PK parameter        | GMFE |
|---------------------|------|
| AUC <sub>tEnd</sub> | 1.33 |
| C <sub>max</sub>    | 1.27 |

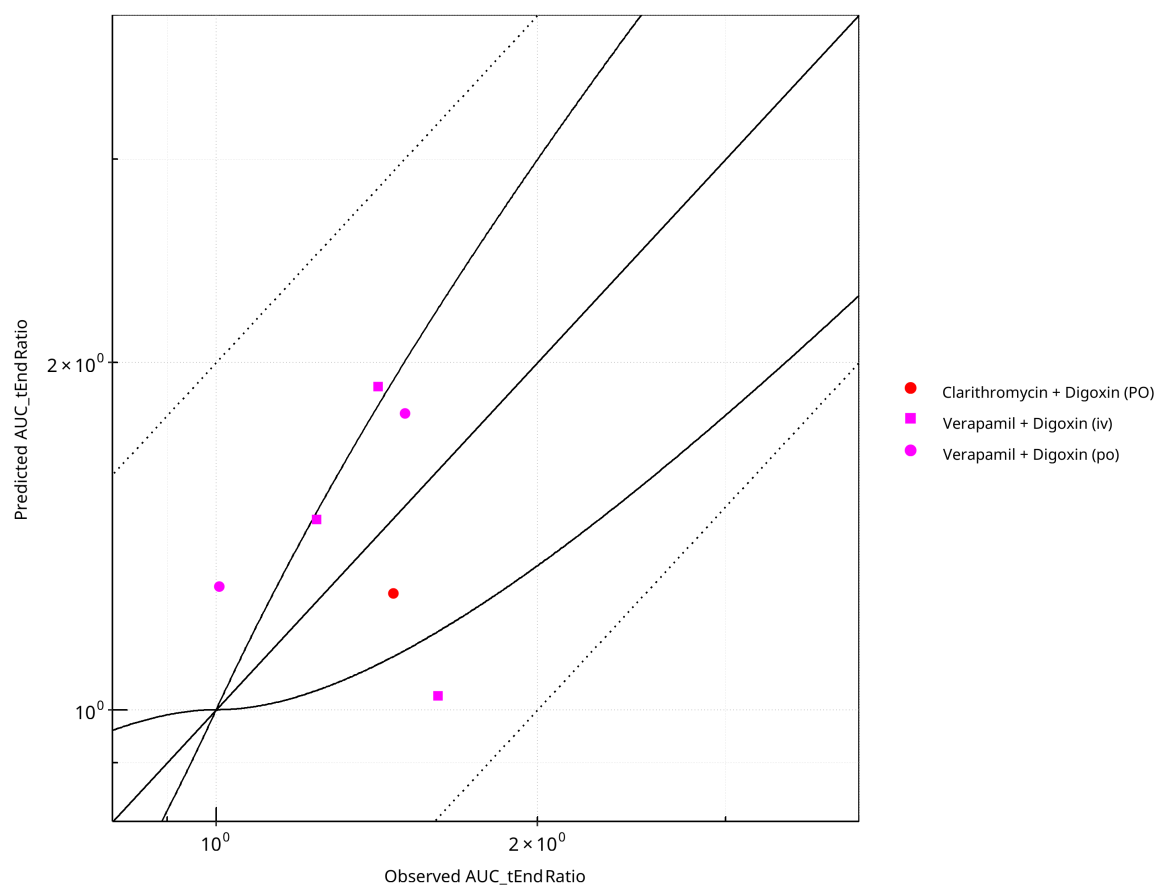
Table 2-6: Summary table for Pgp DDI - AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| AUC <sub>tEnd</sub>               | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 13     | -         |
| Points within Guest <i>et al.</i> | 2      | 15.38     |
| Points within 2 fold              | 13     | 100.00    |

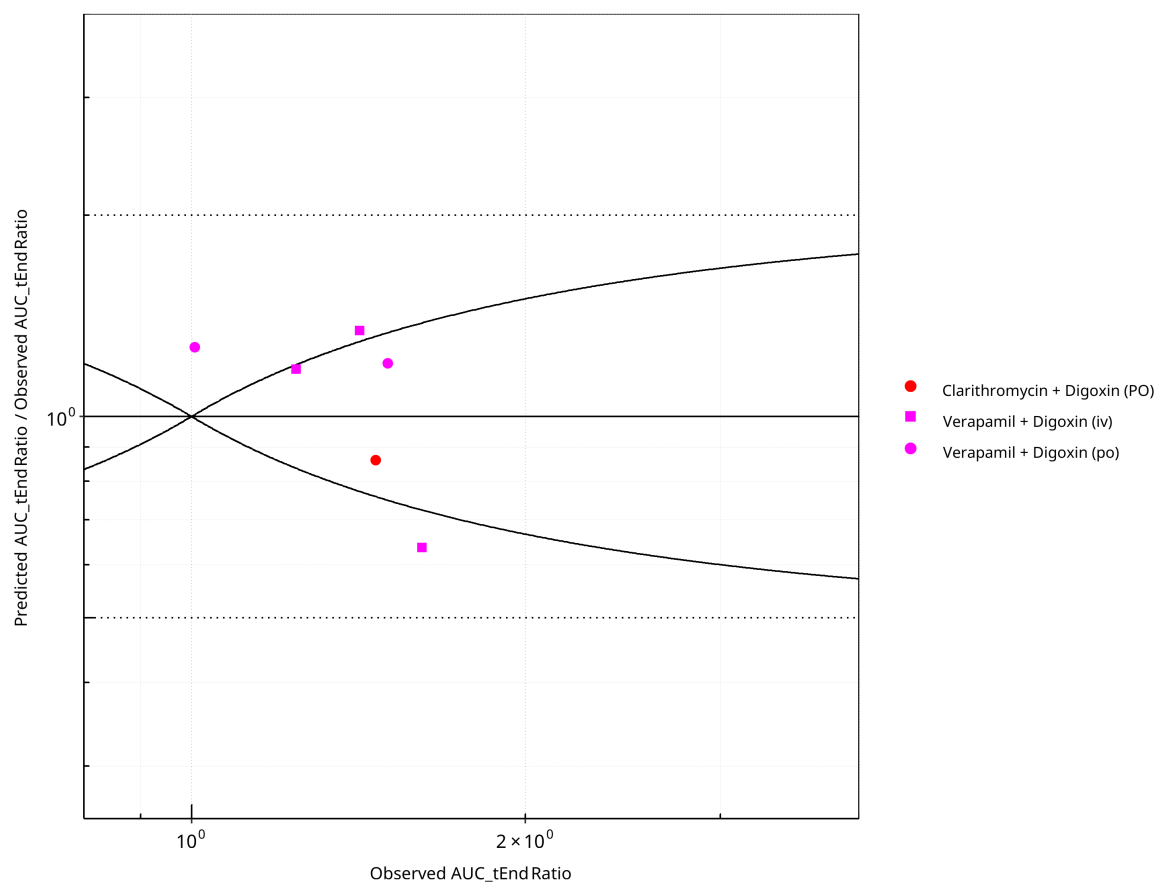
**Table 2-7: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 10     | -         |
| Points within Guest <i>et al.</i> | 6      | 60        |
| Points within 2 fold              | 9      | 90        |

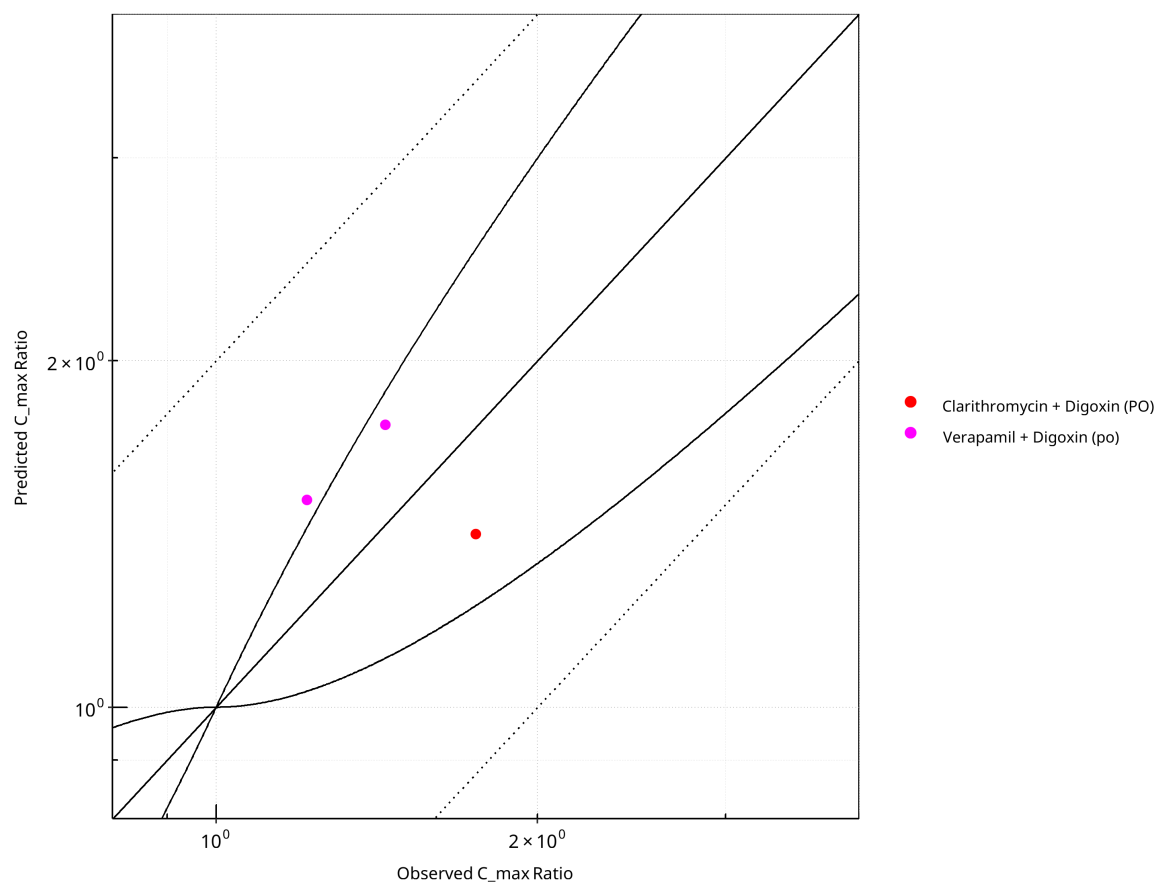
## 2.1.2 Mechanism-based Inactivation



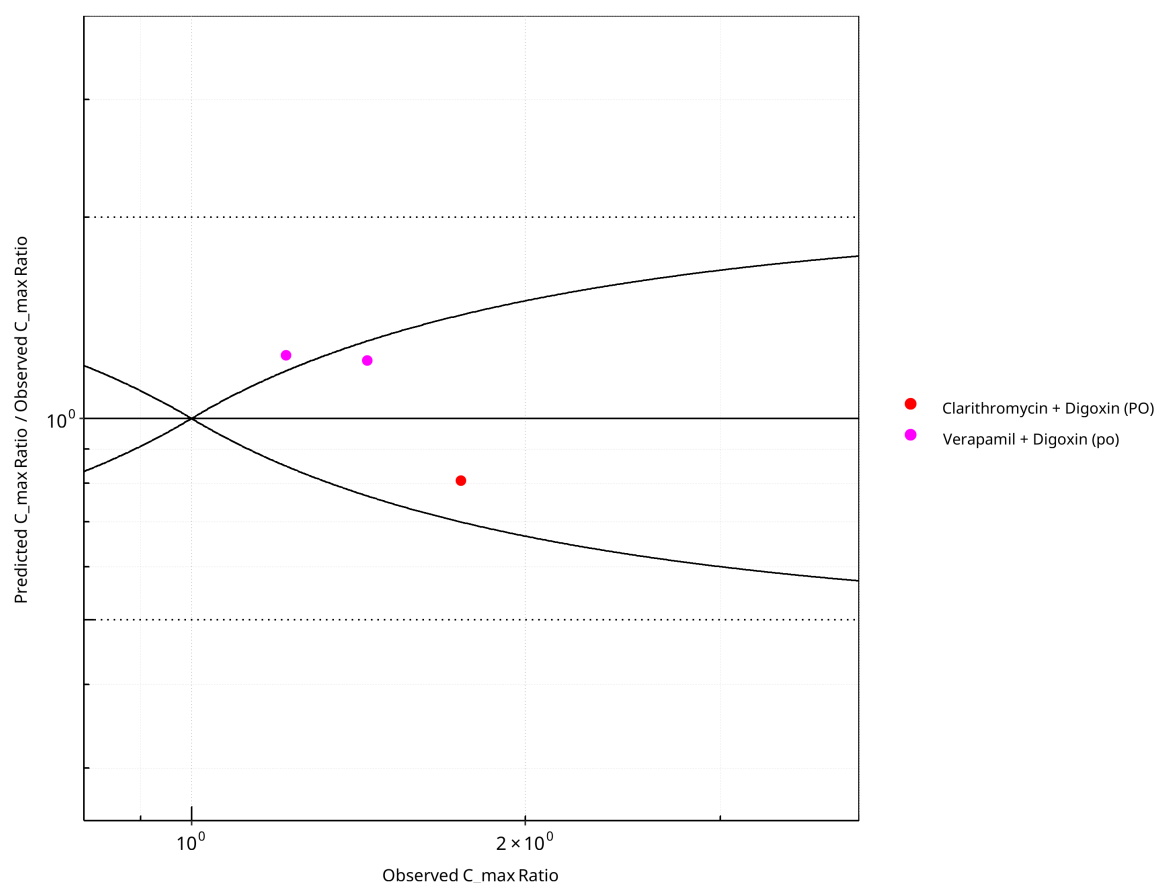
**Figure 2-9: Pgp DDI. Mechanism: Mechanism-based Inactivation. Predicted vs. Observed AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-10: Pgp DDI. Mechanism: Mechanism-based Inactivation. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-11: Pgp DDI. Mechanism: Mechanism-based Inactivation. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-12: Pgp DDI. Mechanism: Mechanism-based Inactivation. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

**Table 2-8: GMFE for Pgp DDI Ratio**

| PK parameter     | GMFE |
|------------------|------|
| AUC_tEnd         | 1.28 |
| C <sub>max</sub> | 1.24 |

**Table 2-9: Summary table for Pgp DDI - AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| AUC_tEnd                          | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 6      | -         |
| Points within Guest <i>et al.</i> | 3      | 50        |
| Points within 2 fold              | 6      | 100       |

Table 2-10: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 3      | -         |
| Points within Guest <i>et al.</i> | 2      | 66.67     |
| Points within 2 fold              | 3      | 100.00    |

### 2.1.3 Reversible Inhibition

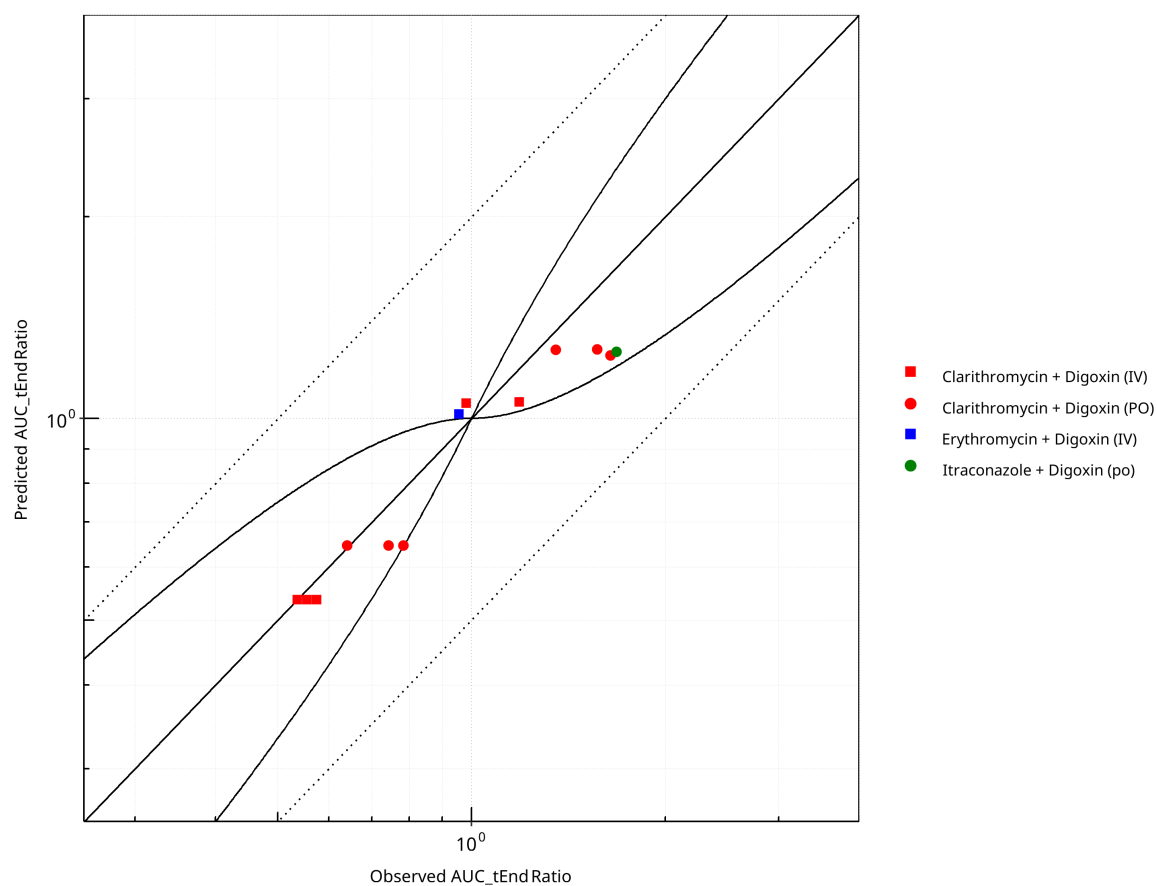
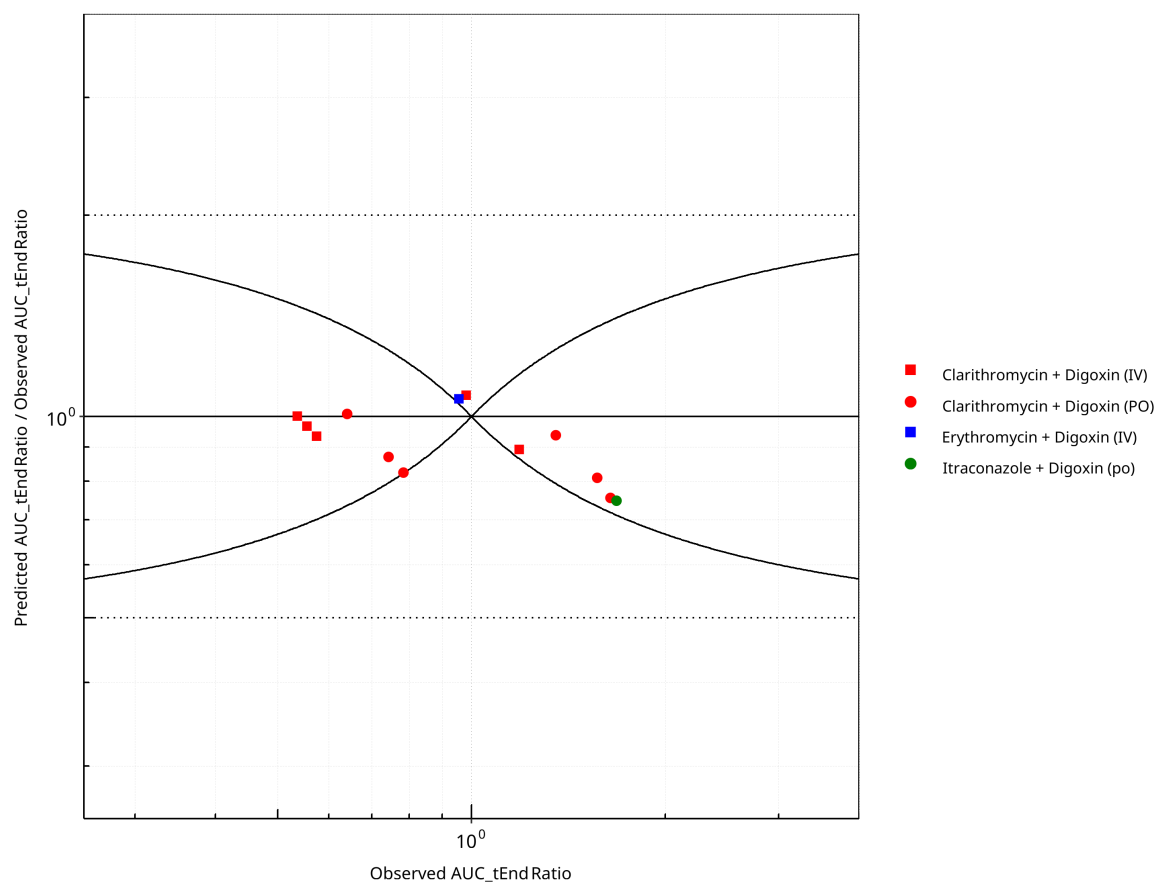
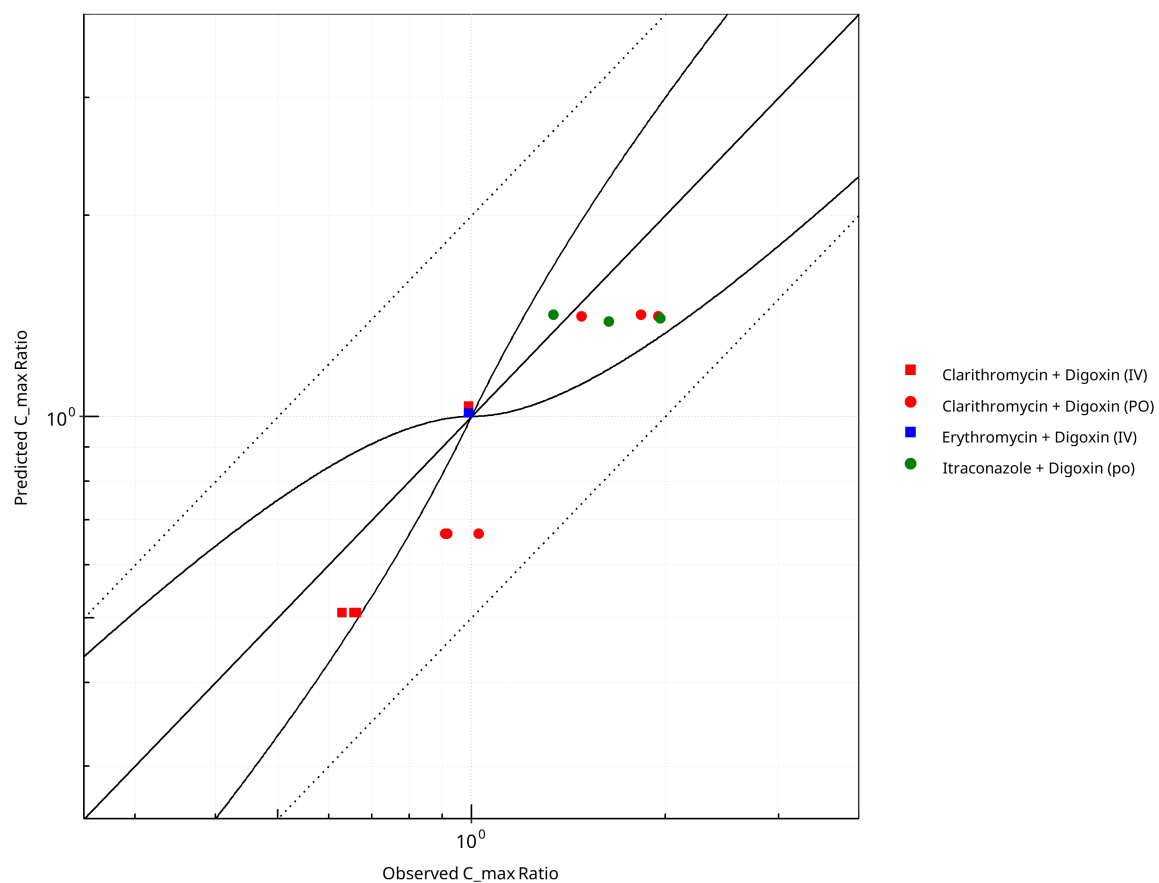


Figure 2-13: Pgp DDI. Mechanism: Reversible Inhibition. Predicted vs. Observed AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

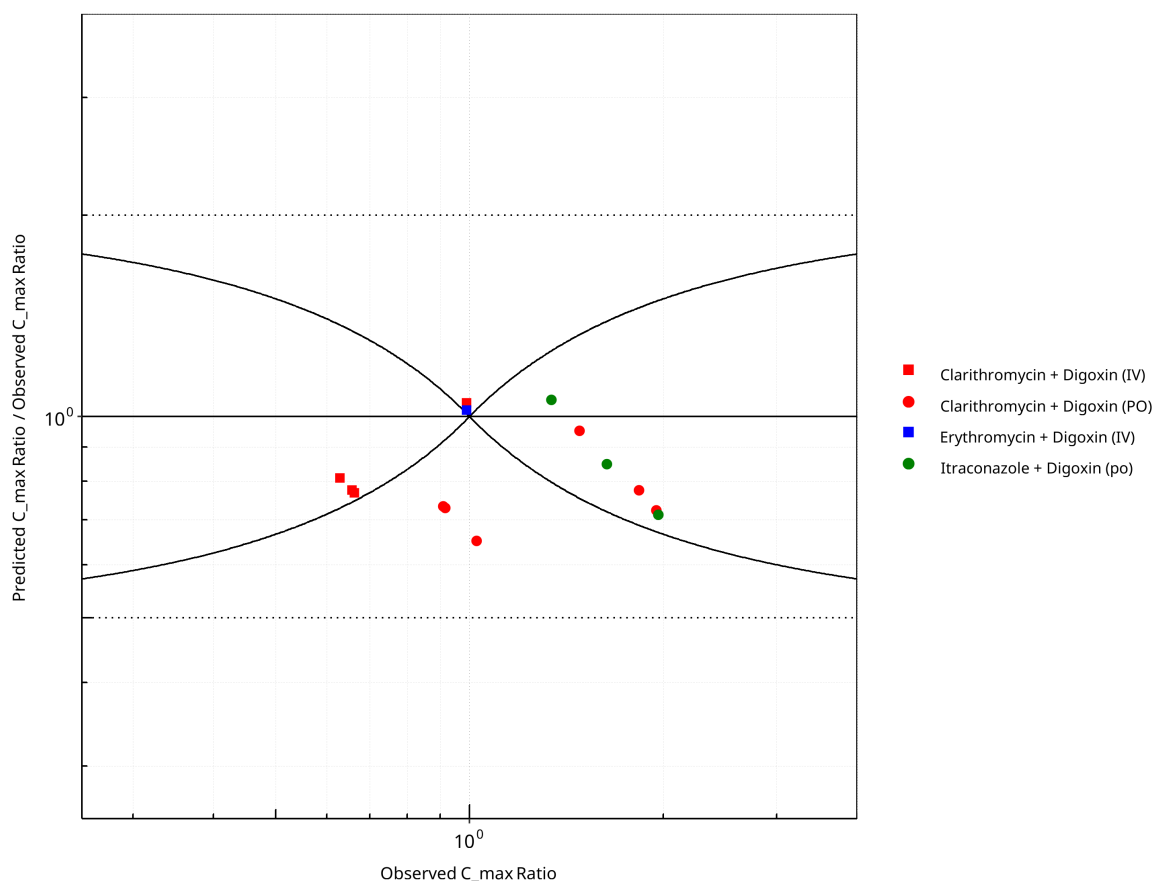


**Figure 2-14: Pgp DDI. Mechanism: Reversible Inhibition. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)**





**Figure 2-15: Pgp DDI. Mechanism: Reversible Inhibition. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-16: Pgp DDI. Mechanism: Reversible Inhibition. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

**Table 2-11: GMFE for Pgp DDI Ratio**

| PK parameter        | GMFE |
|---------------------|------|
| AUC <sub>tEnd</sub> | 1.12 |
| C <sub>max</sub>    | 1.24 |

**Table 2-12: Summary table for Pgp DDI - AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| AUC <sub>tEnd</sub>               | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 13     | -         |
| Points within Guest <i>et al.</i> | 11     | 84.62     |
| Points within 2 fold              | 13     | 100.00    |

Table 2-13: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 14     | -         |
| Points within Guest <i>et al.</i> | 9      | 64.29     |
| Points within 2 fold              | 14     | 100.00    |

## 2.2 Perpetrator

### 2.2.1 Clarithromycin

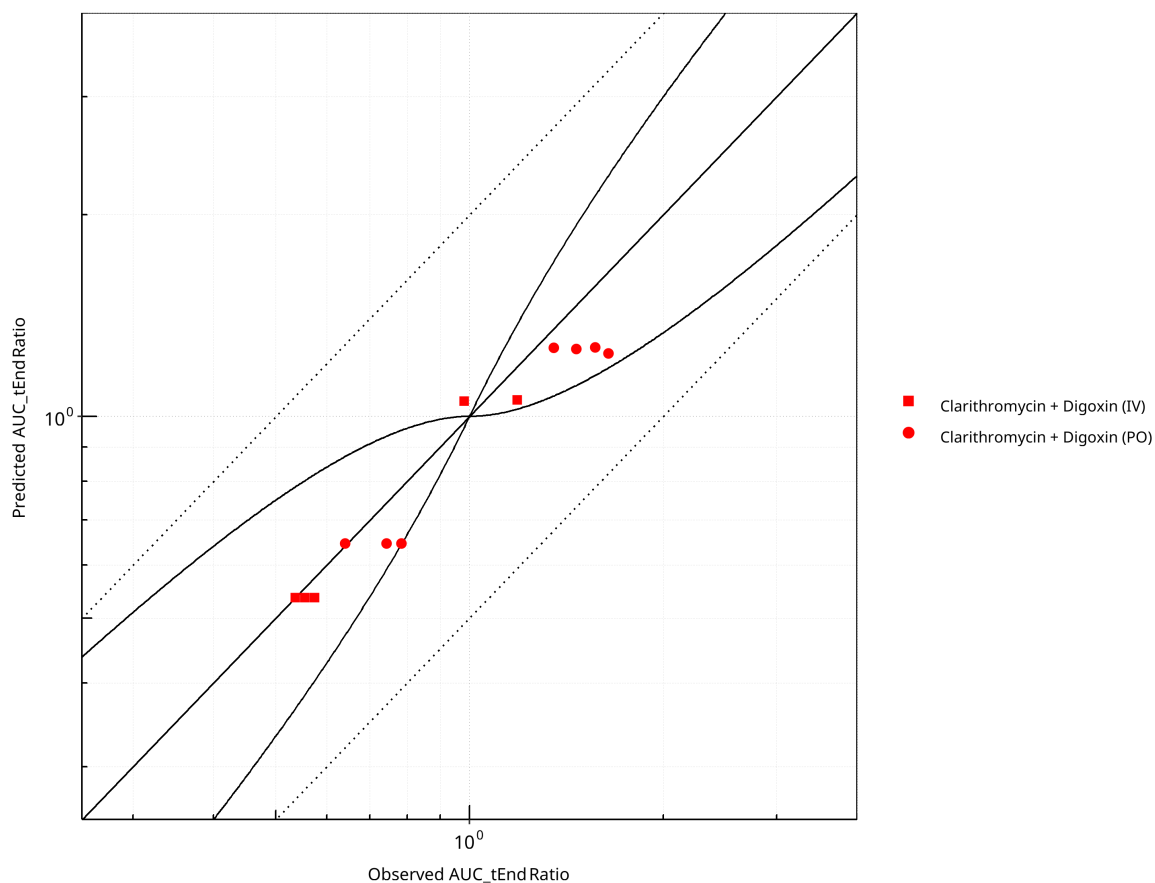
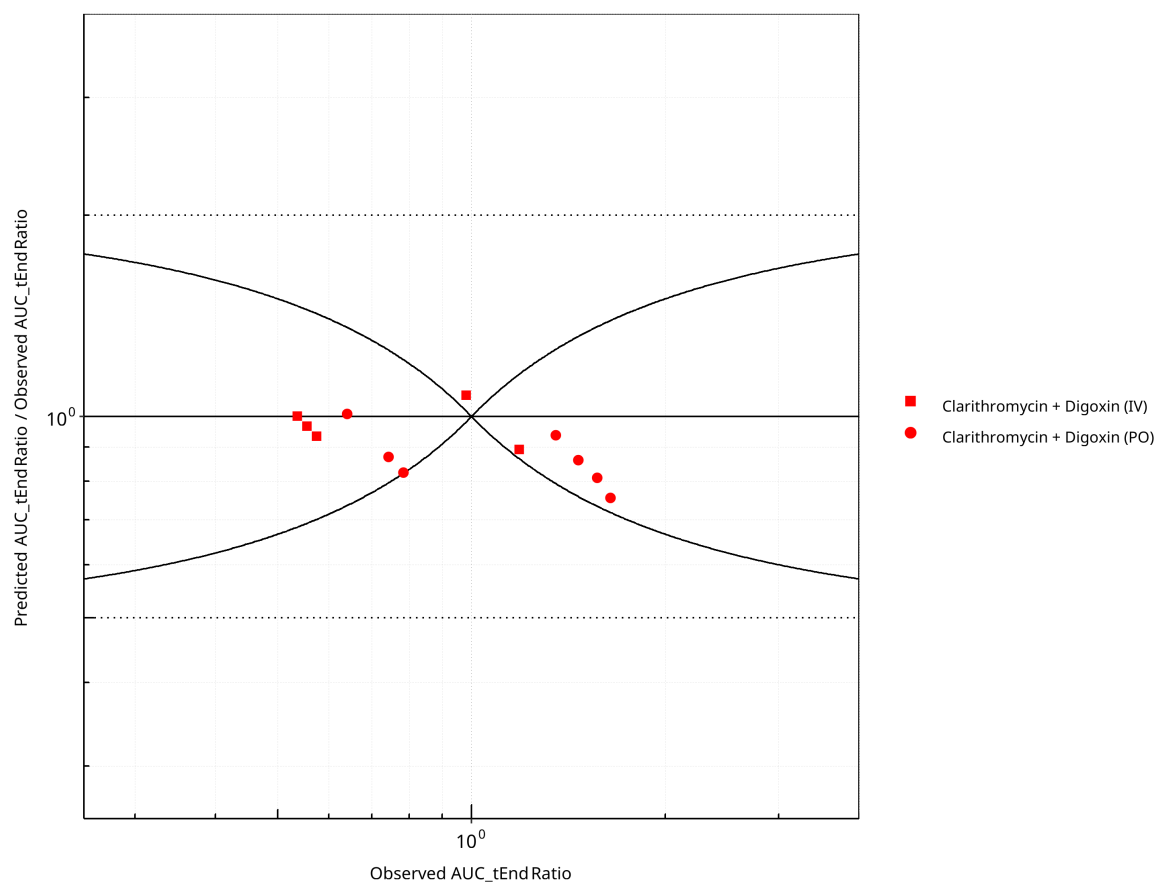


Figure 2-17: Pgp DDI. Perpetrator: Clarithromycin. Predicted vs. Observed AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)



**Figure 2-18: Pgp DDI. Perpetrator: Clarithromycin. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

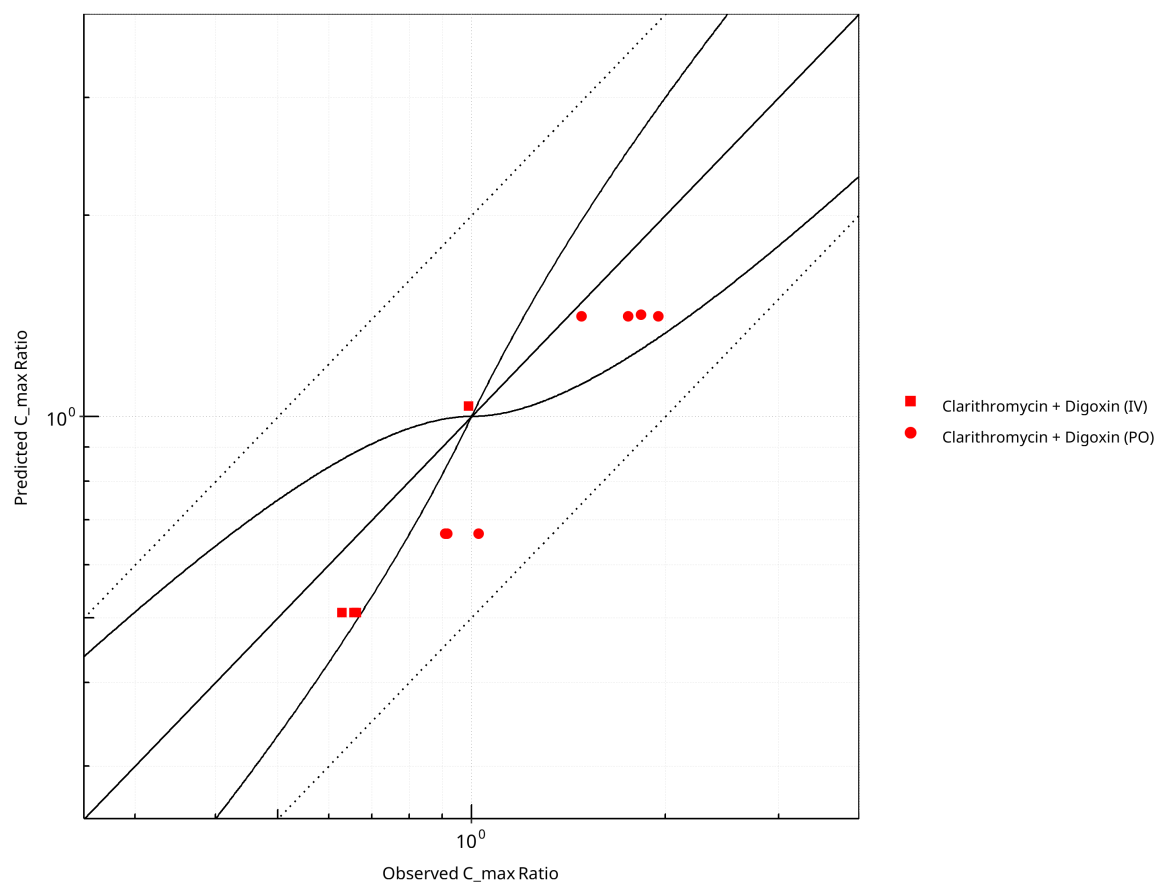
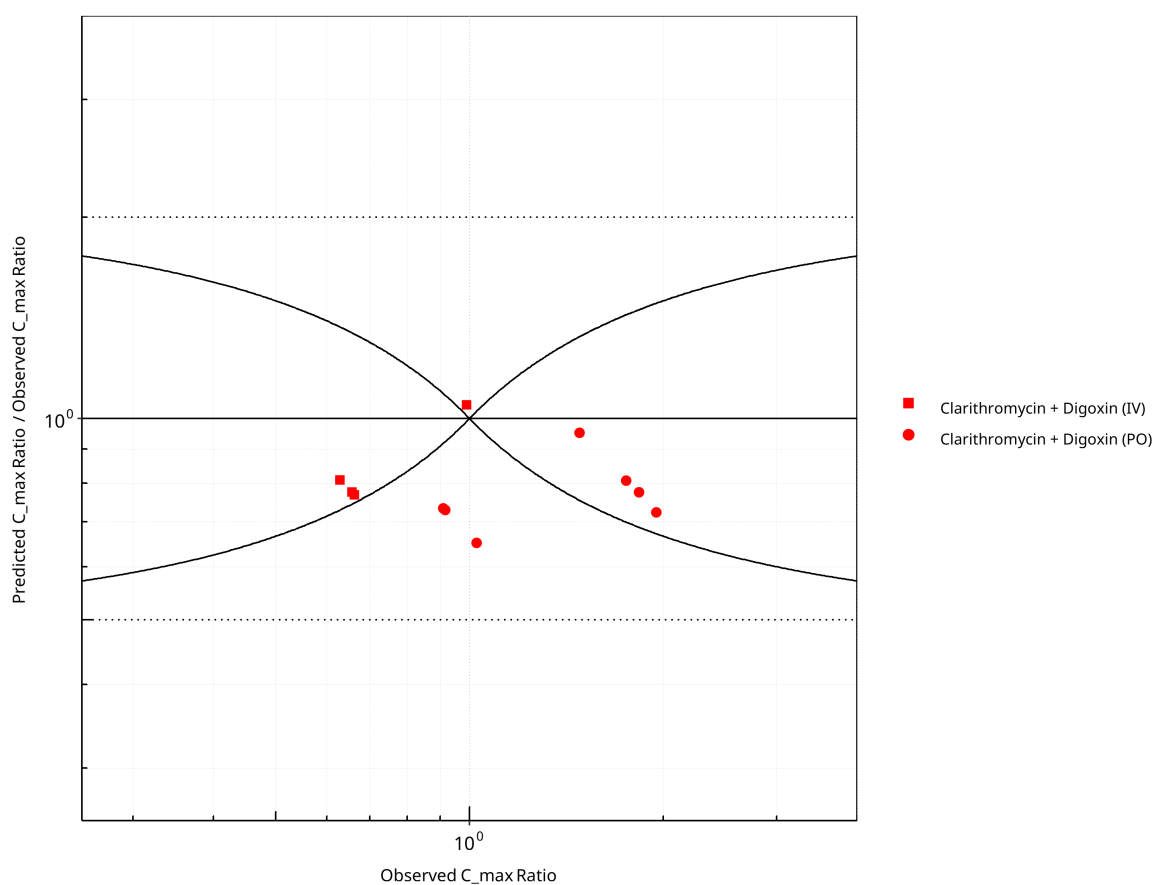


Figure 2-19: Pgp DDI. Perpetrator: Clarithromycin. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)



**Figure 2-20: Pgp DDI. Perpetrator: Clarithromycin. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

**Table 2-14: GMFE for Pgp DDI Ratio**

| PK parameter        | GMFE |
|---------------------|------|
| AUC <sub>tEnd</sub> | 1.12 |
| C <sub>max</sub>    | 1.27 |

**Table 2-15: Summary table for Pgp DDI - AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| AUC <sub>tEnd</sub>               | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 12     | -         |
| Points within Guest <i>et al.</i> | 11     | 91.67     |
| Points within 2 fold              | 12     | 100.00    |

Table 2-16: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 11     | -         |
| Points within Guest <i>et al.</i> | 7      | 63.64     |
| Points within 2 fold              | 11     | 100.00    |

## 2.2.2 Erythromycin

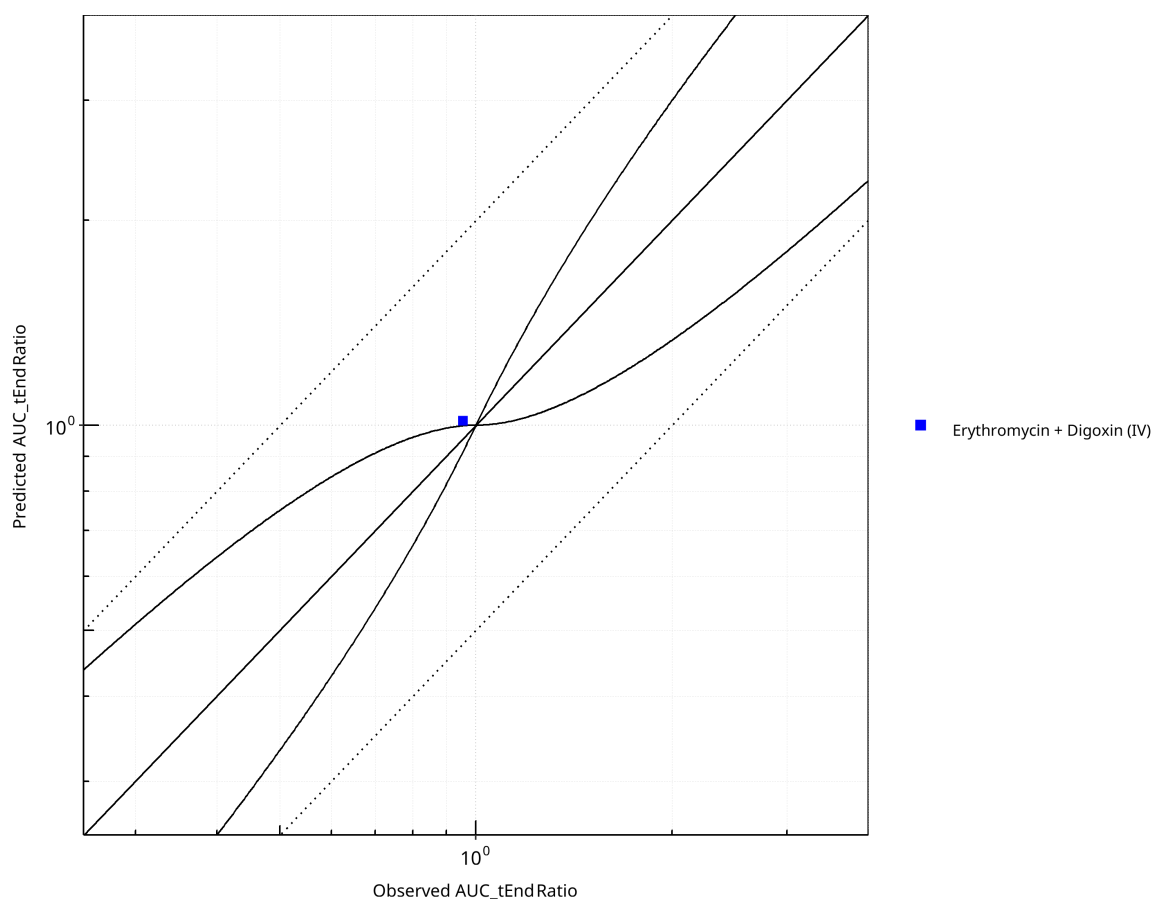
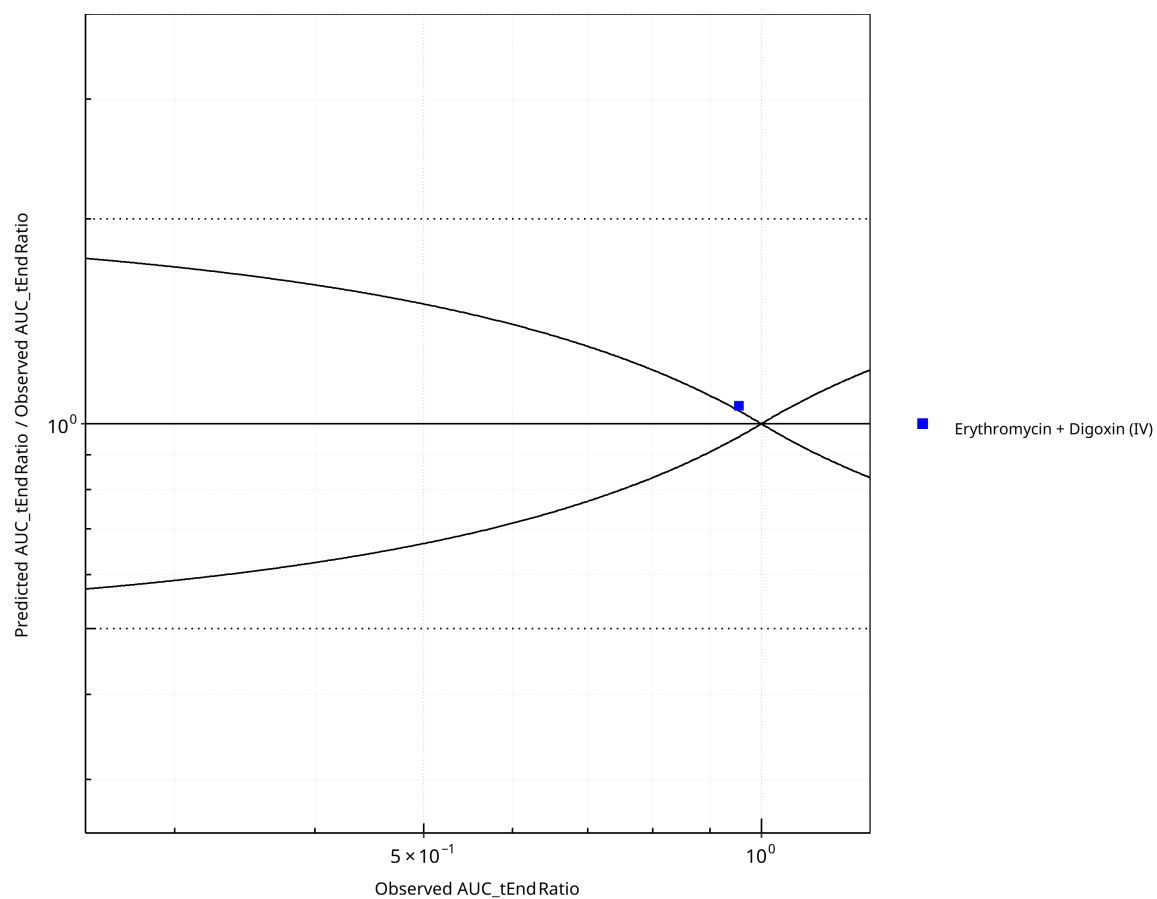
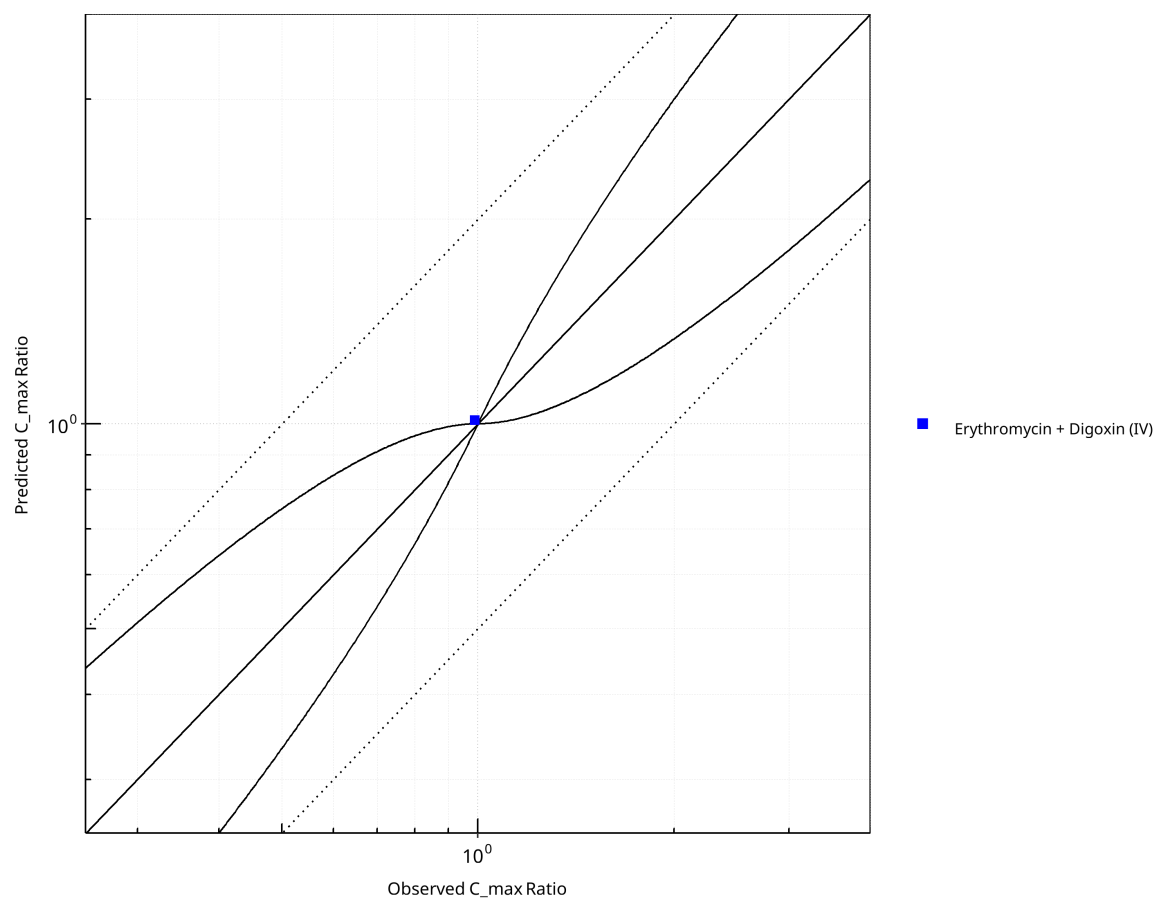


Figure 2-21: Pgp DDI. Perpetrator: Erythromycin. Predicted vs. Observed AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

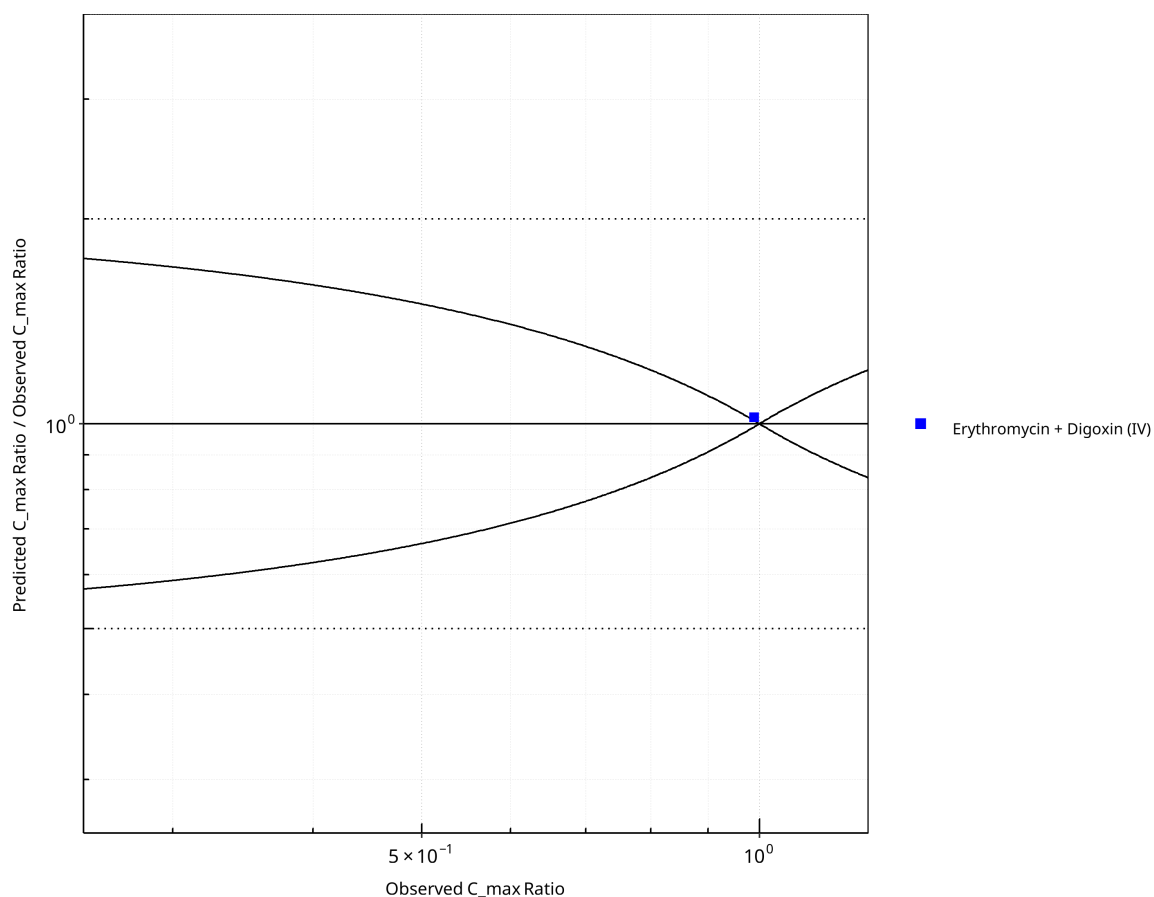


**Figure 2-22: Pgp DDI. Perpetrator: Erythromycin. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)**





**Figure 2-23: Pgp DDI. Perpetrator: Erythromycin. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-24: Pgp DDI. Perpetrator: Erythromycin. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

**Table 2-17: GMFE for Pgp DDI Ratio**

| PK parameter        | GMFE |
|---------------------|------|
| AUC <sub>tEnd</sub> | 1.06 |
| C <sub>max</sub>    | 1.02 |

**Table 2-18: Summary table for Pgp DDI - AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| AUC <sub>tEnd</sub>               | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 1      | -         |
| Points within Guest <i>et al.</i> | 0      | 0         |
| Points within 2 fold              | 1      | 100       |

Table 2-19: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 1      | -         |
| Points within Guest <i>et al.</i> | 0      | 0         |
| Points within 2 fold              | 1      | 100       |

### 2.2.3 Itraconazole

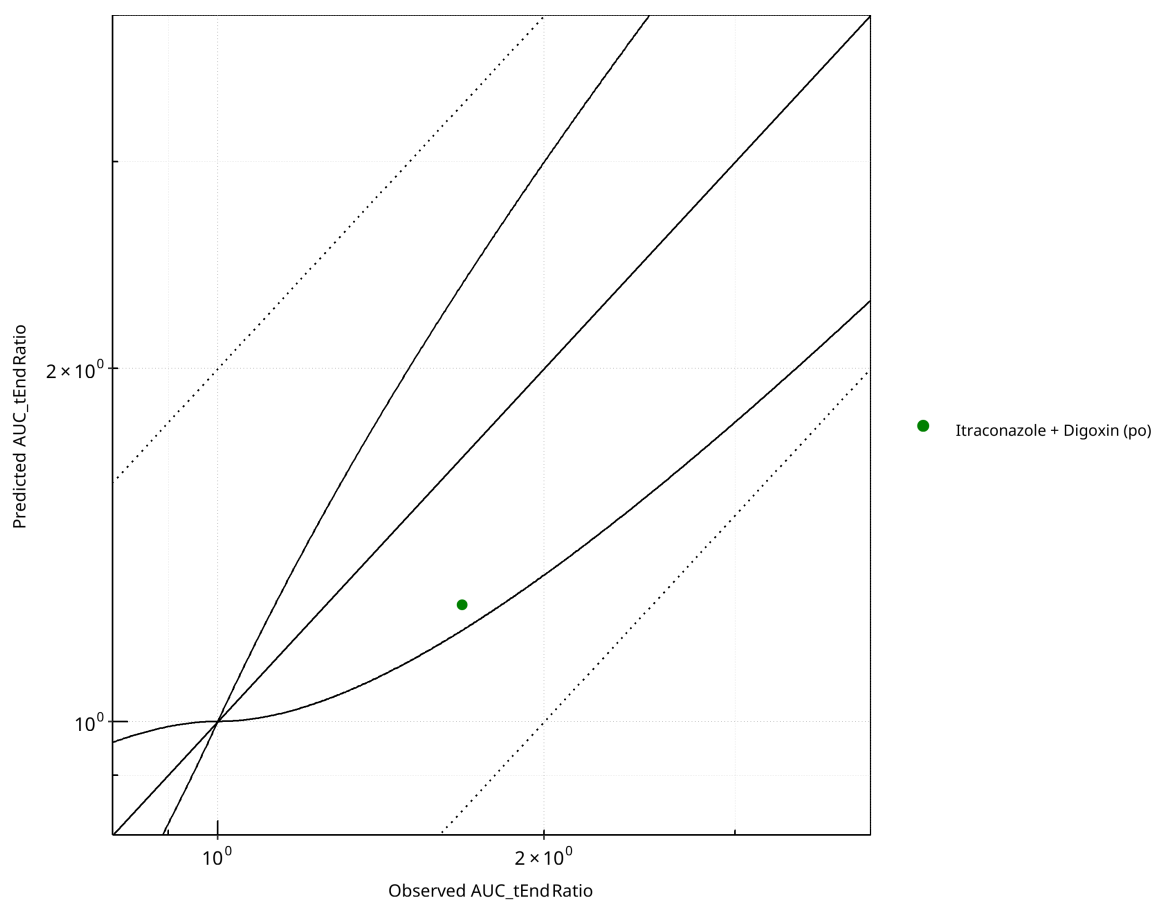
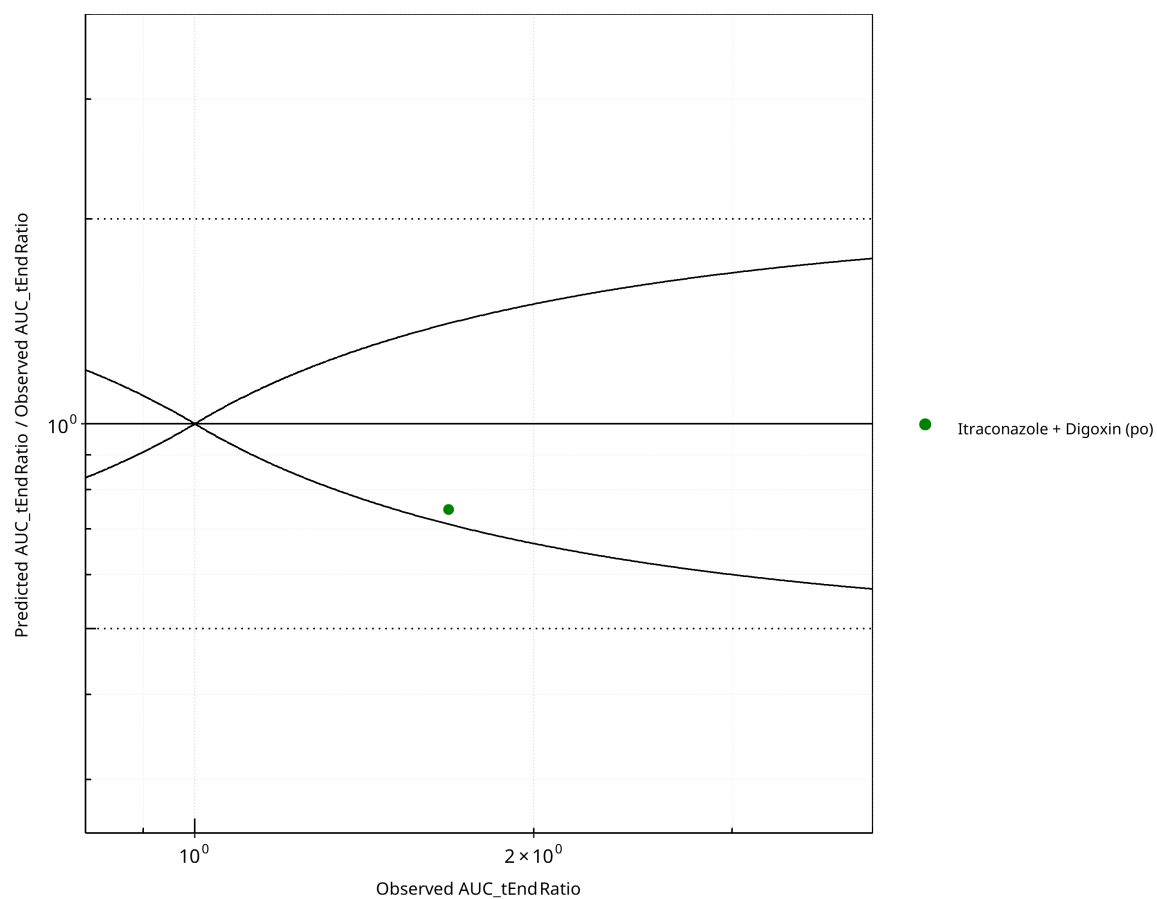
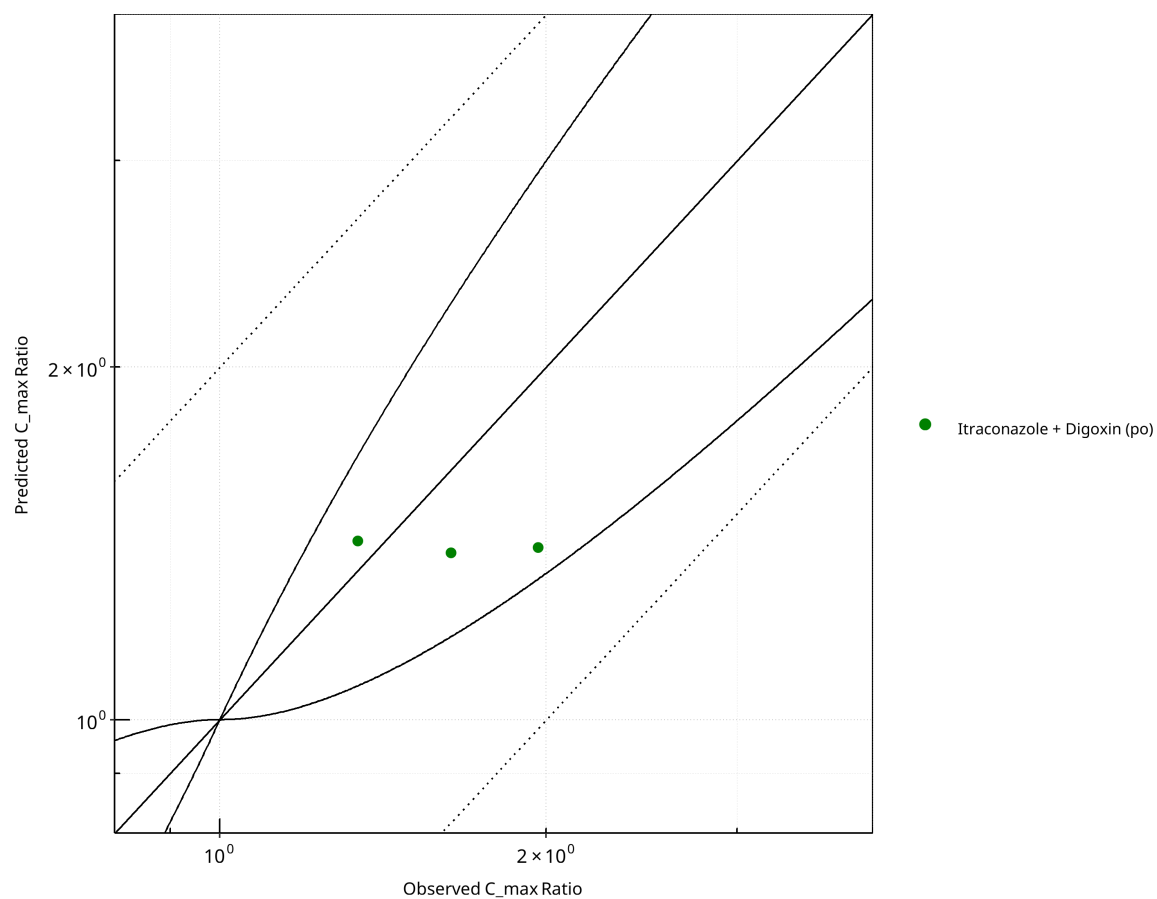


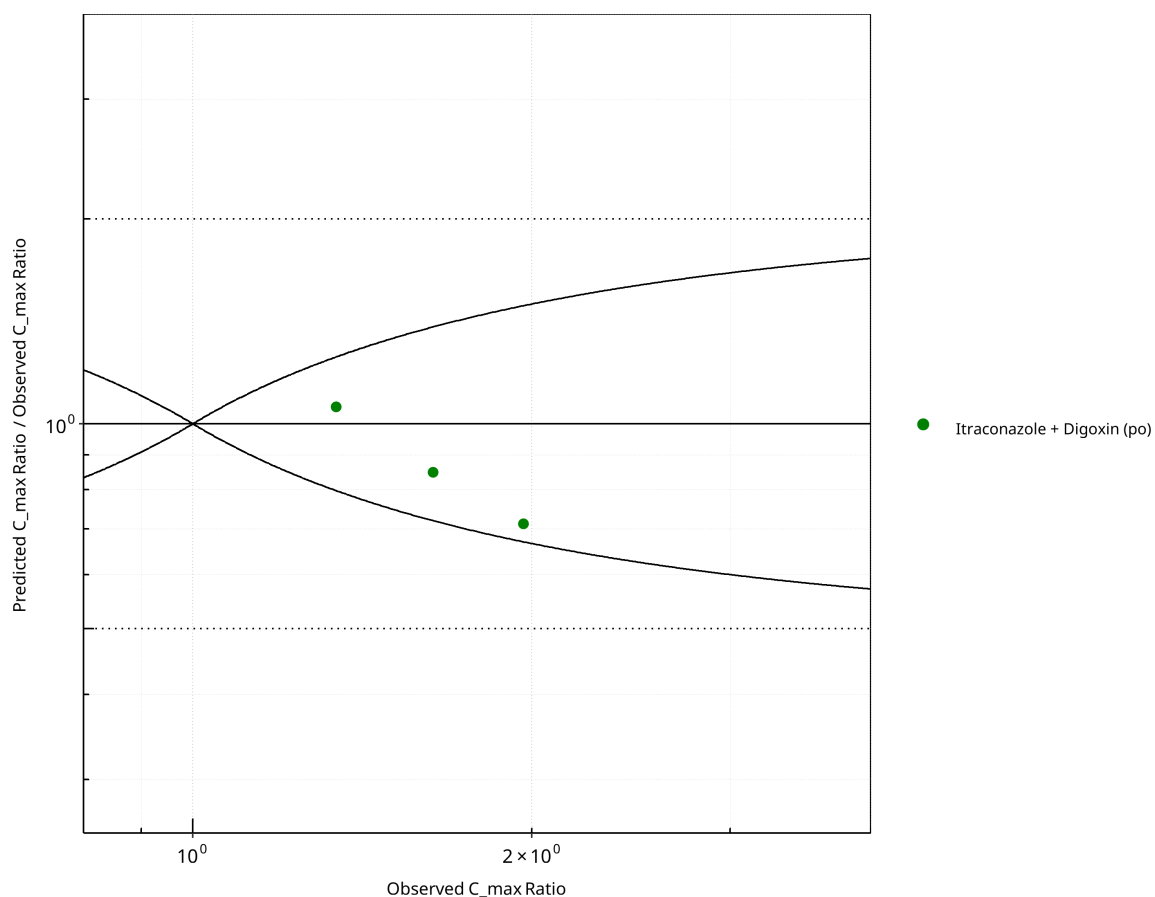
Figure 2-25: Pgp DDI. Perpetrator: Itraconazole. Predicted vs. Observed AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)



**Figure 2-26: Pgp DDI. Perpetrator: Itraconazole. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-27: Pgp DDI. Perpetrator: Itraconazole. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-28: Pgp DDI. Perpetrator: Itraconazole. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

**Table 2-20: GMFE for Pgp DDI Ratio**

| PK parameter     | GMFE |
|------------------|------|
| AUC_tEnd         | 1.33 |
| C <sub>max</sub> | 1.20 |

**Table 2-21: Summary table for Pgp DDI - AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| AUC_tEnd                          | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 1      | -         |
| Points within Guest <i>et al.</i> | 1      | 100       |
| Points within 2 fold              | 1      | 100       |

Table 2-22: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 3      | -         |
| Points within Guest <i>et al.</i> | 3      | 100       |
| Points within 2 fold              | 3      | 100       |

## 2.2.4 Rifampicin

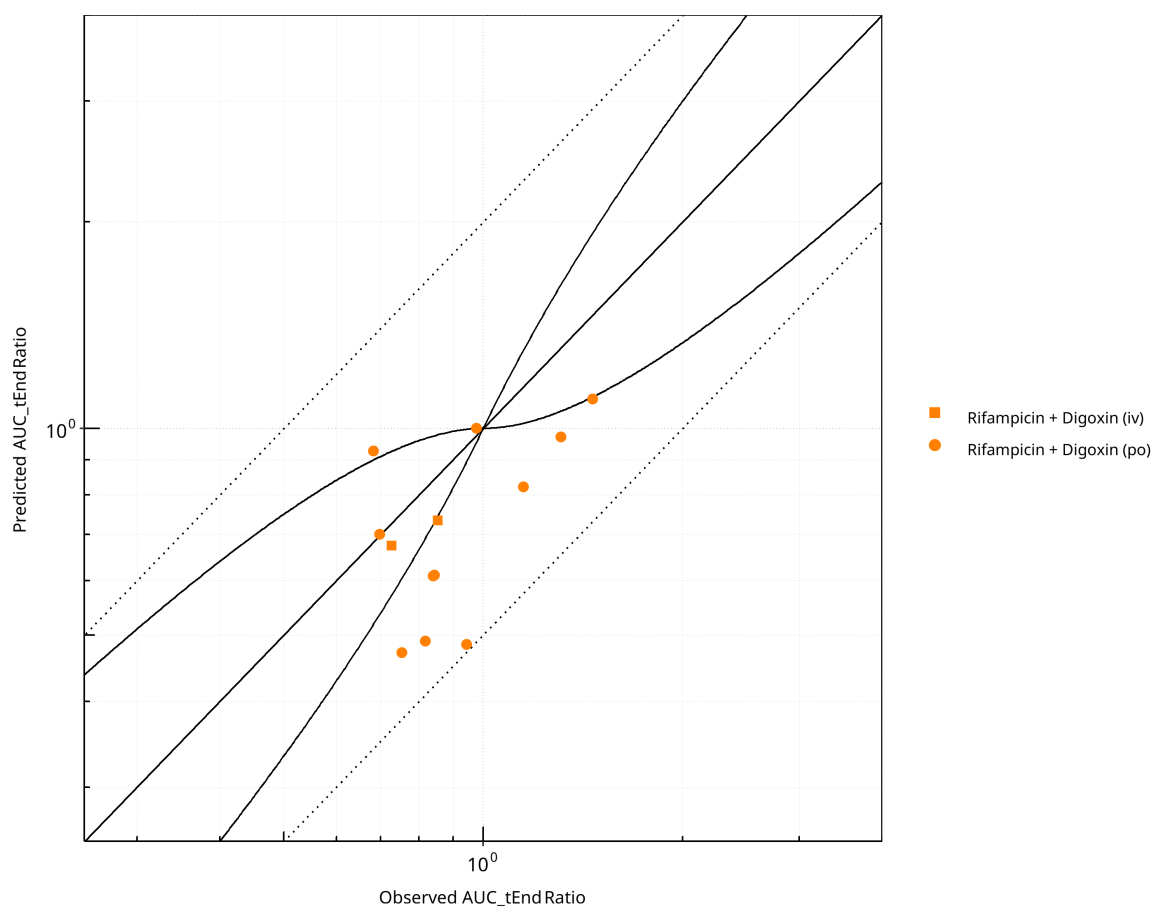


Figure 2-29: Pgp DDI. Perpetrator: Rifampicin. Predicted vs. Observed AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

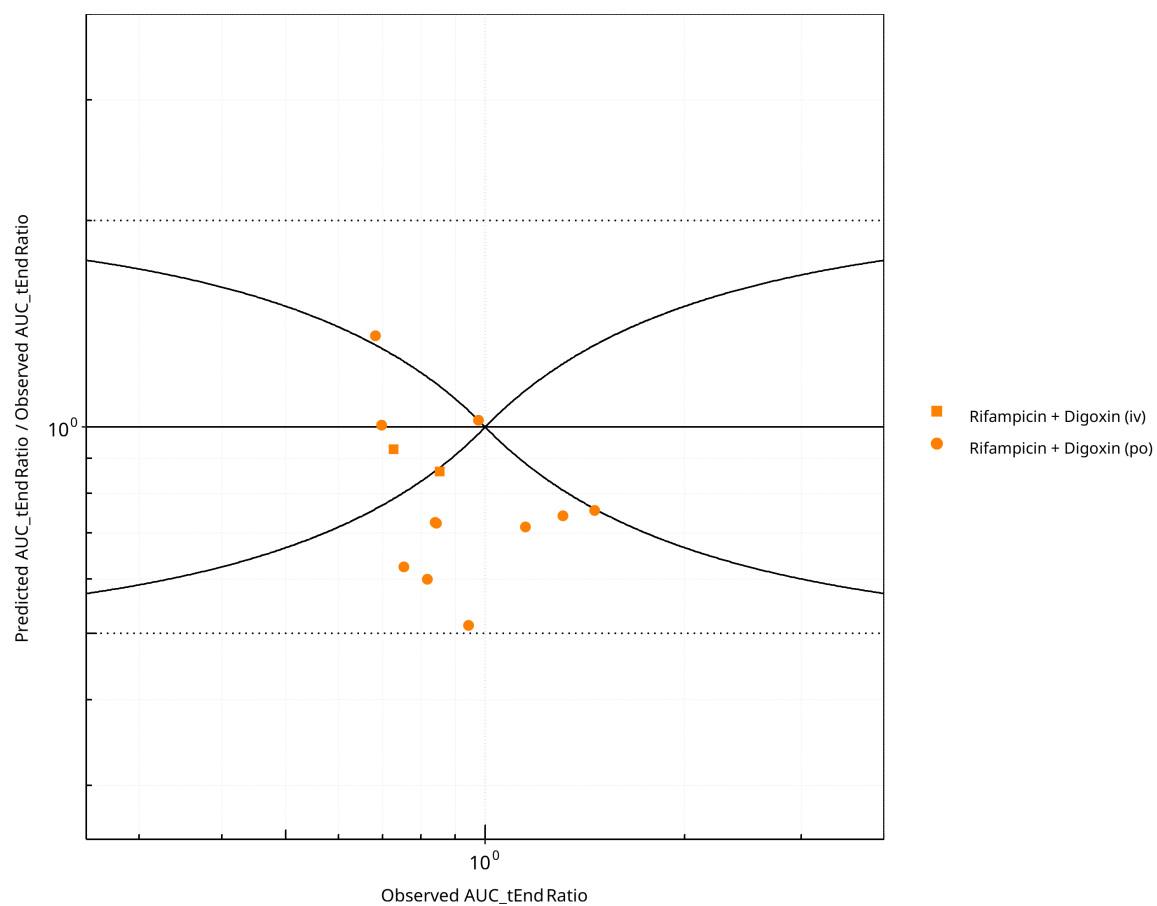


Figure 2-30: Pgp DDI. Perpetrator: Rifampicin. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)



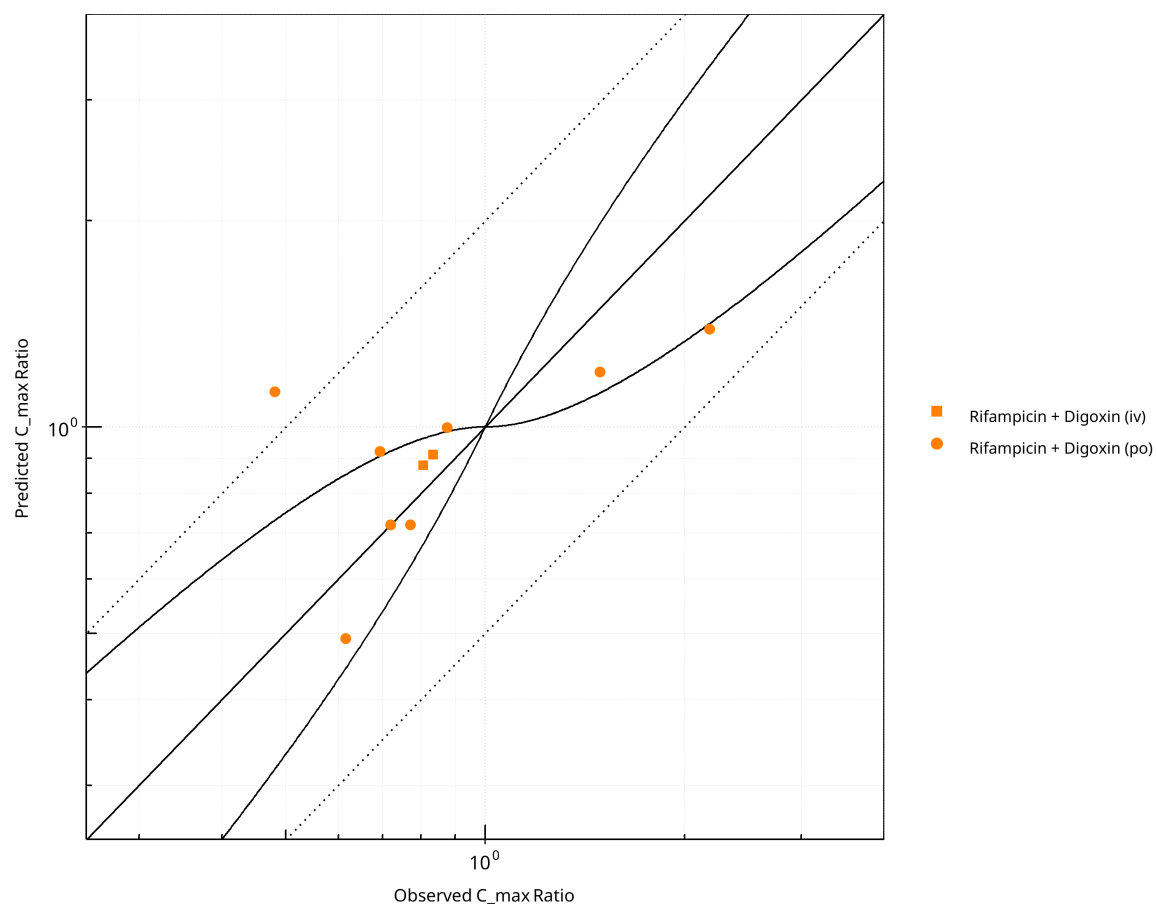
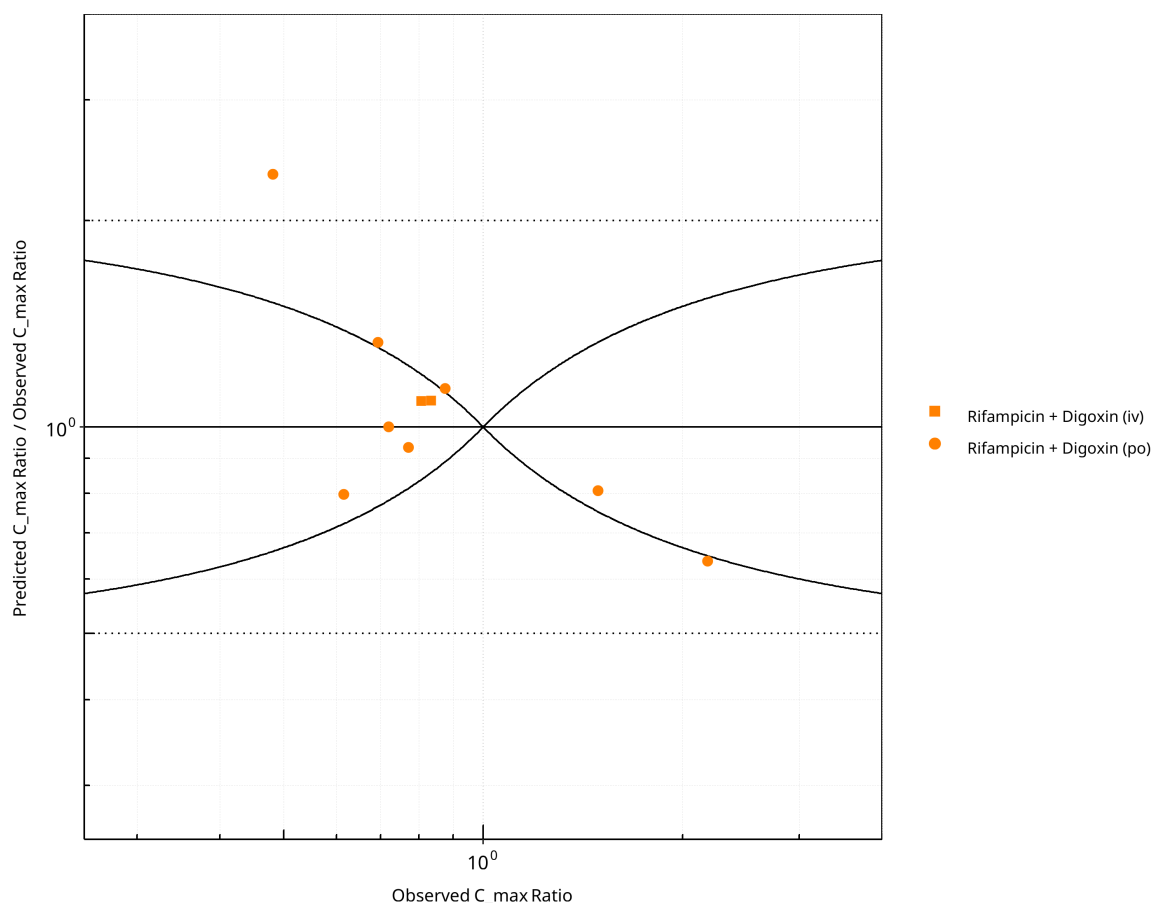


Figure 2-31: Pgp DDI. Perpetrator: Rifampicin. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)



**Figure 2-32: Pgp DDI. Perpetrator: Rifampicin. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

**Table 2-23: GMFE for Pgp DDI Ratio**

| PK parameter        | GMFE |
|---------------------|------|
| AUC <sub>tEnd</sub> | 1.33 |
| C <sub>max</sub>    | 1.27 |

**Table 2-24: Summary table for Pgp DDI - AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| AUC <sub>tEnd</sub>               | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 13     | -         |
| Points within Guest <i>et al.</i> | 2      | 15.38     |
| Points within 2 fold              | 13     | 100.00    |

Table 2-25: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 10     | -         |
| Points within Guest <i>et al.</i> | 6      | 60        |
| Points within 2 fold              | 9      | 90        |

2.2.5 Verapamil

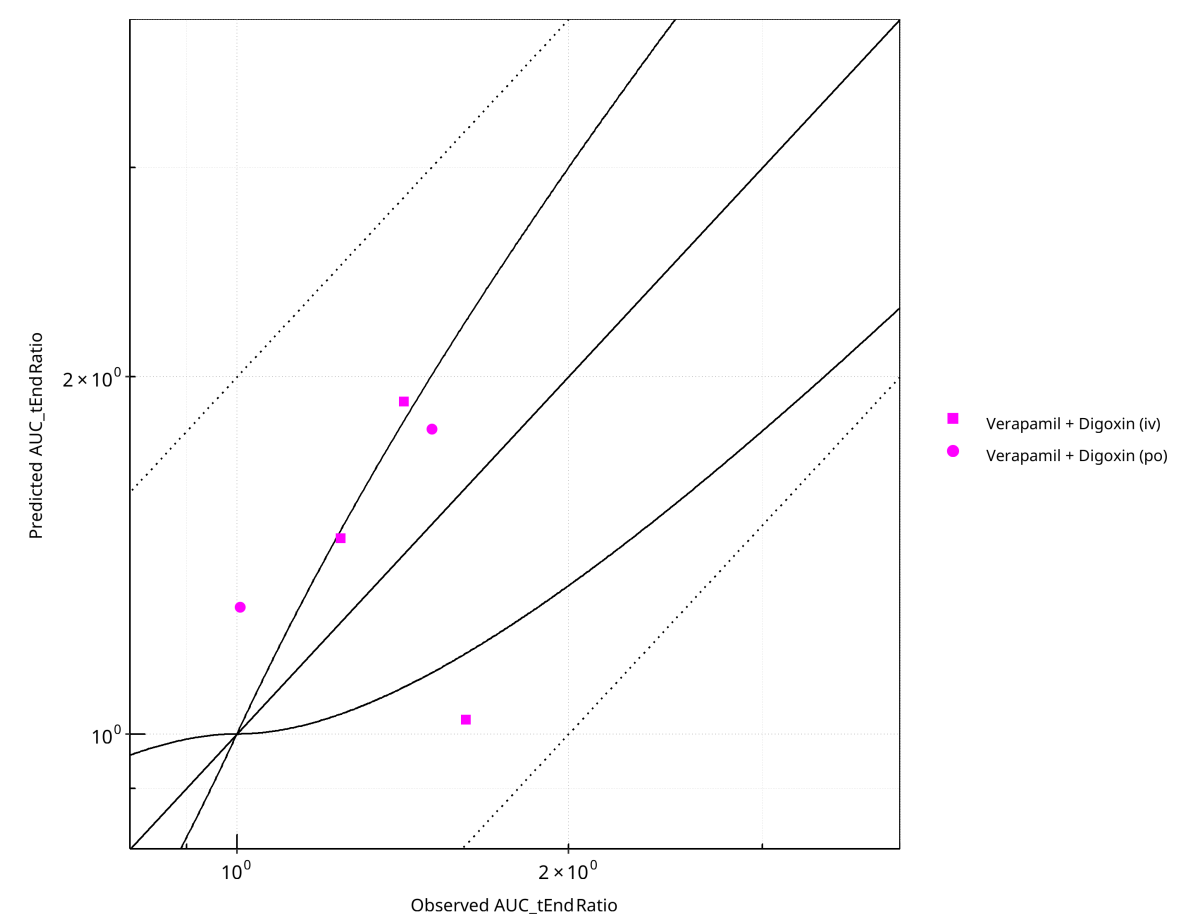
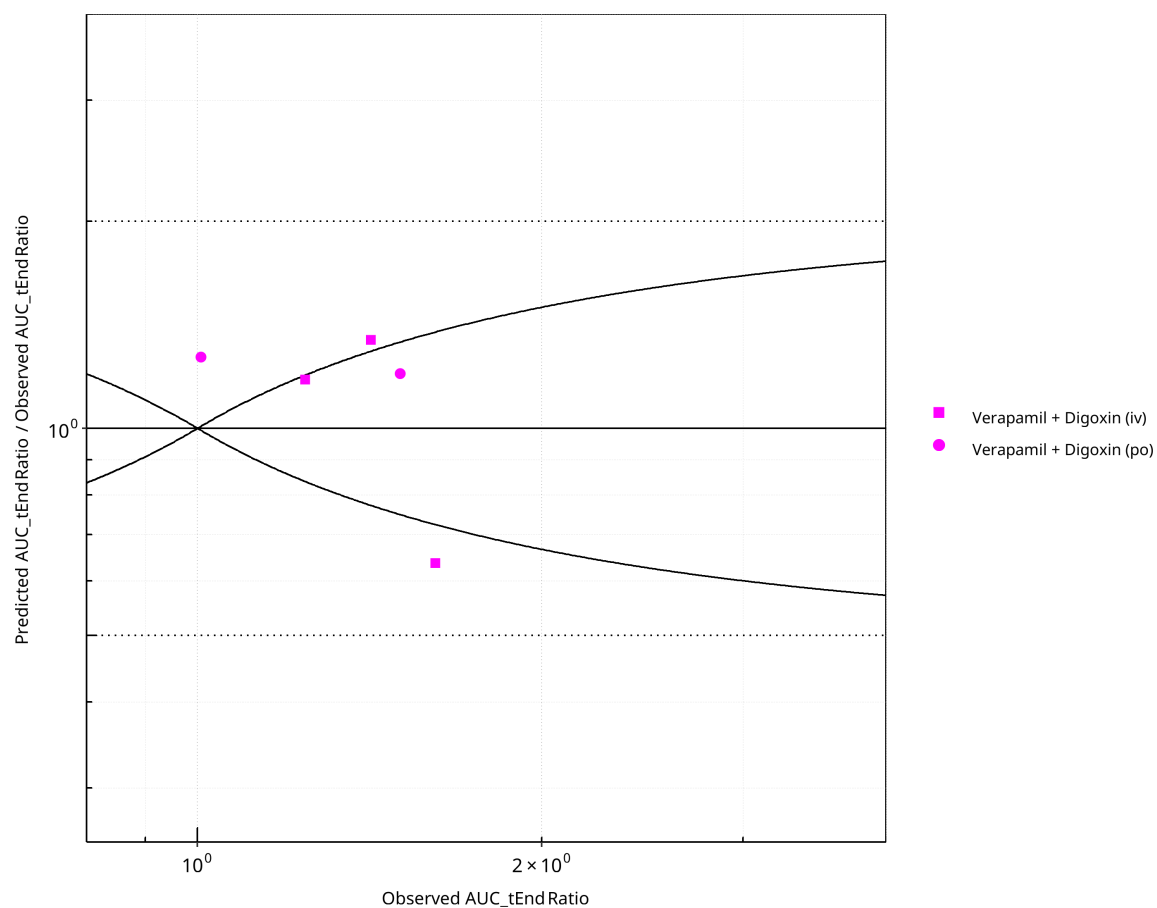
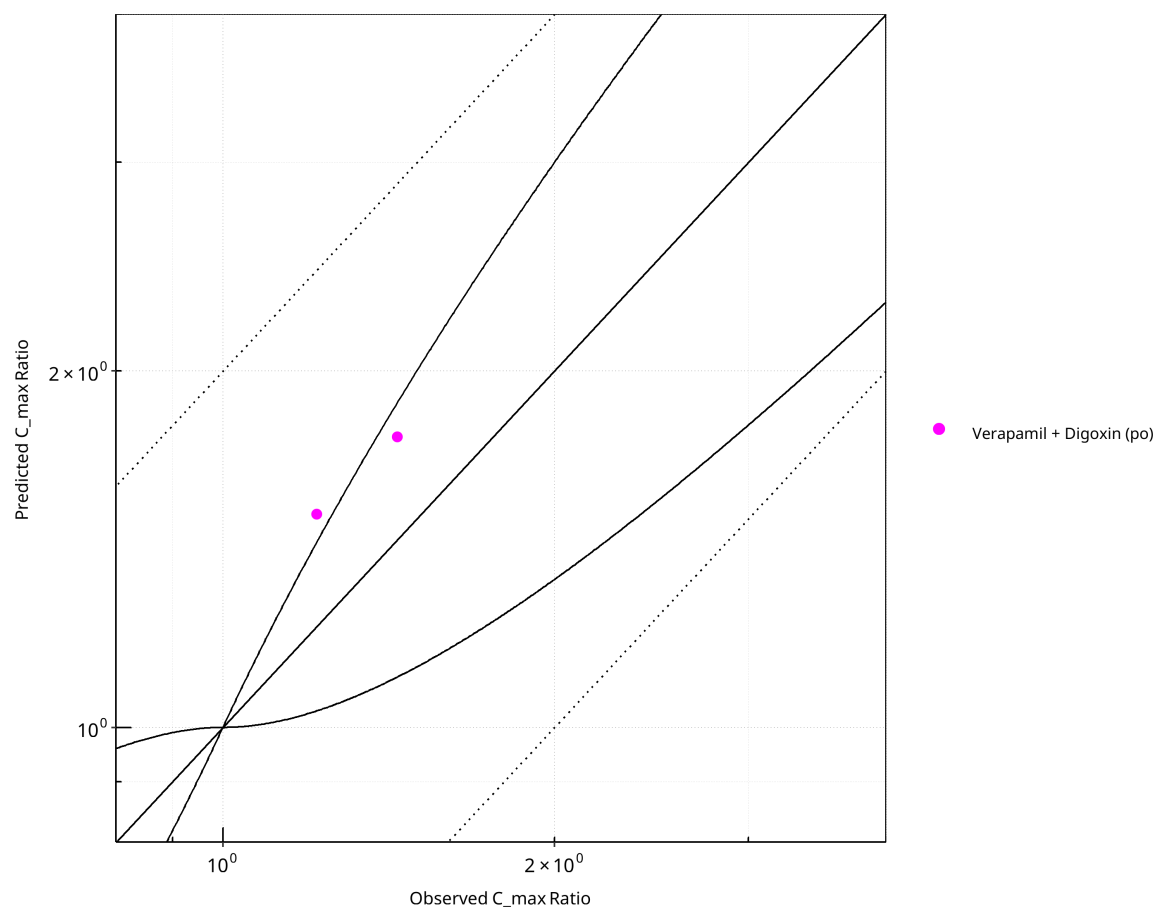


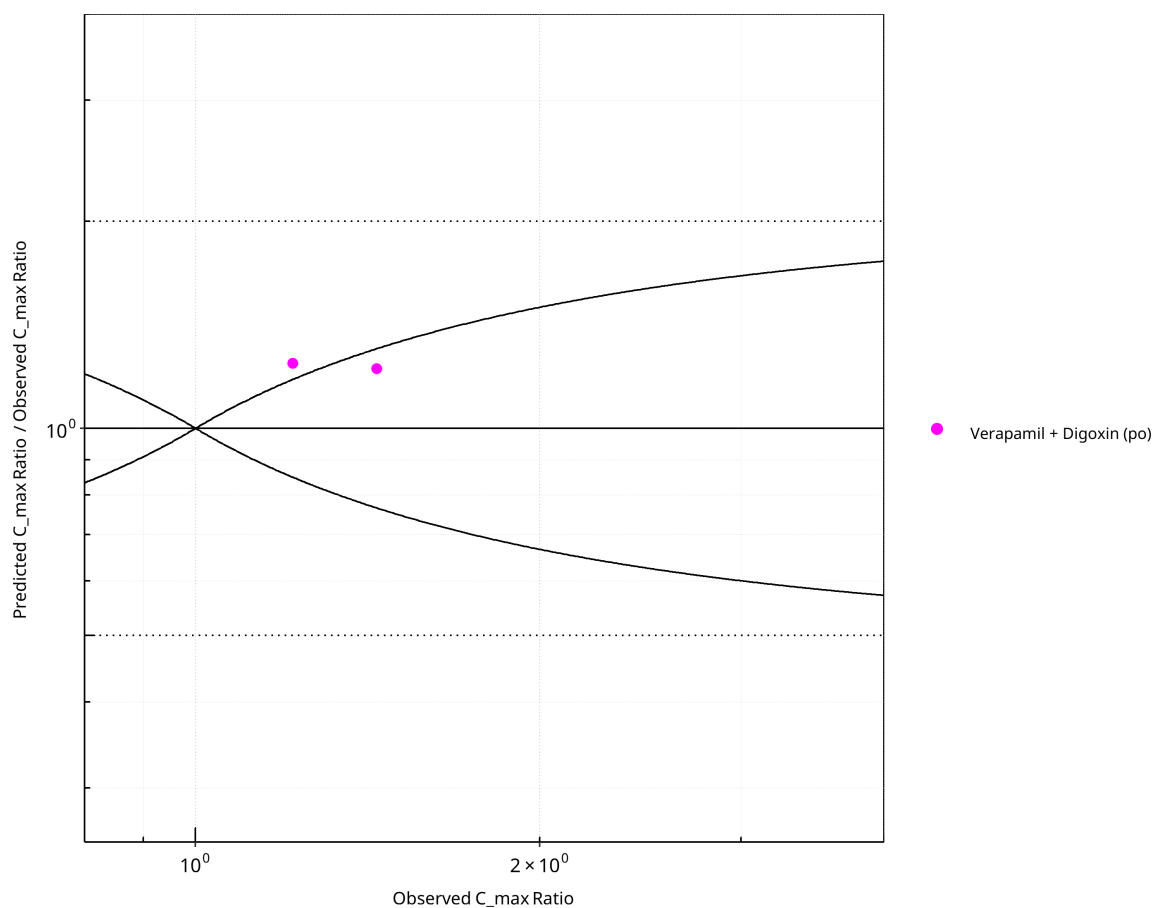
Figure 2-33: Pgp DDI. Perpetrator: Verapamil. Predicted vs. Observed AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)



**Figure 2-34: Pgp DDI. Perpetrator: Verapamil. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-35: Pgp DDI. Perpetrator: Verapamil. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**



**Figure 2-36: Pgp DDI. Perpetrator: Verapamil. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

**Table 2-26: GMFE for Pgp DDI Ratio**

| PK parameter        | GMFE |
|---------------------|------|
| AUC <sub>tEnd</sub> | 1.31 |
| C <sub>max</sub>    | 1.23 |

**Table 2-27: Summary table for Pgp DDI - AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| AUC <sub>tEnd</sub>               | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 5      | -         |
| Points within Guest <i>et al.</i> | 2      | 40        |
| Points within 2 fold              | 5      | 100       |

Table 2-28: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 2      | -         |
| Points within Guest <i>et al.</i> | 1      | 50        |
| Points within 2 fold              | 2      | 100       |

## 2.3 Victim

### 2.3.1 Digoxin

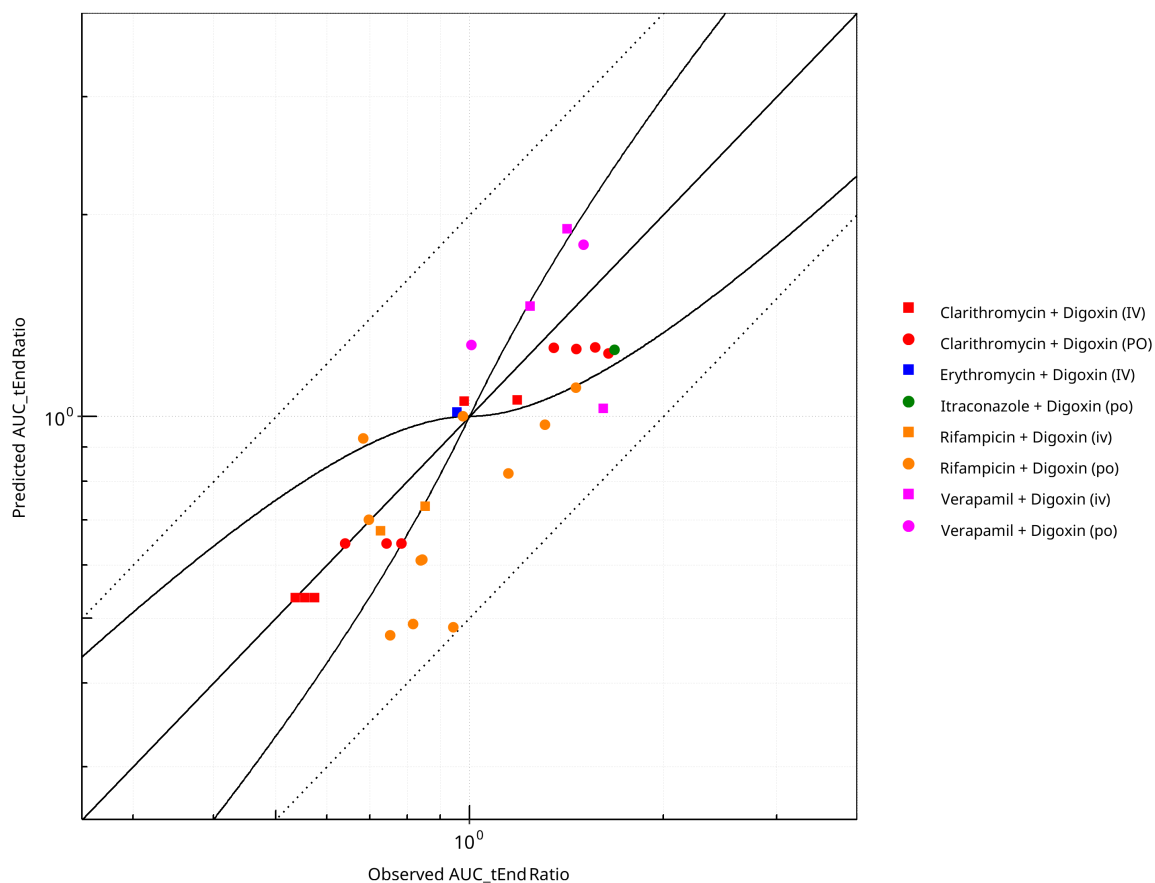


Figure 2-37: Pgp DDI. Victim: Digoxin. Predicted vs. Observed AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)

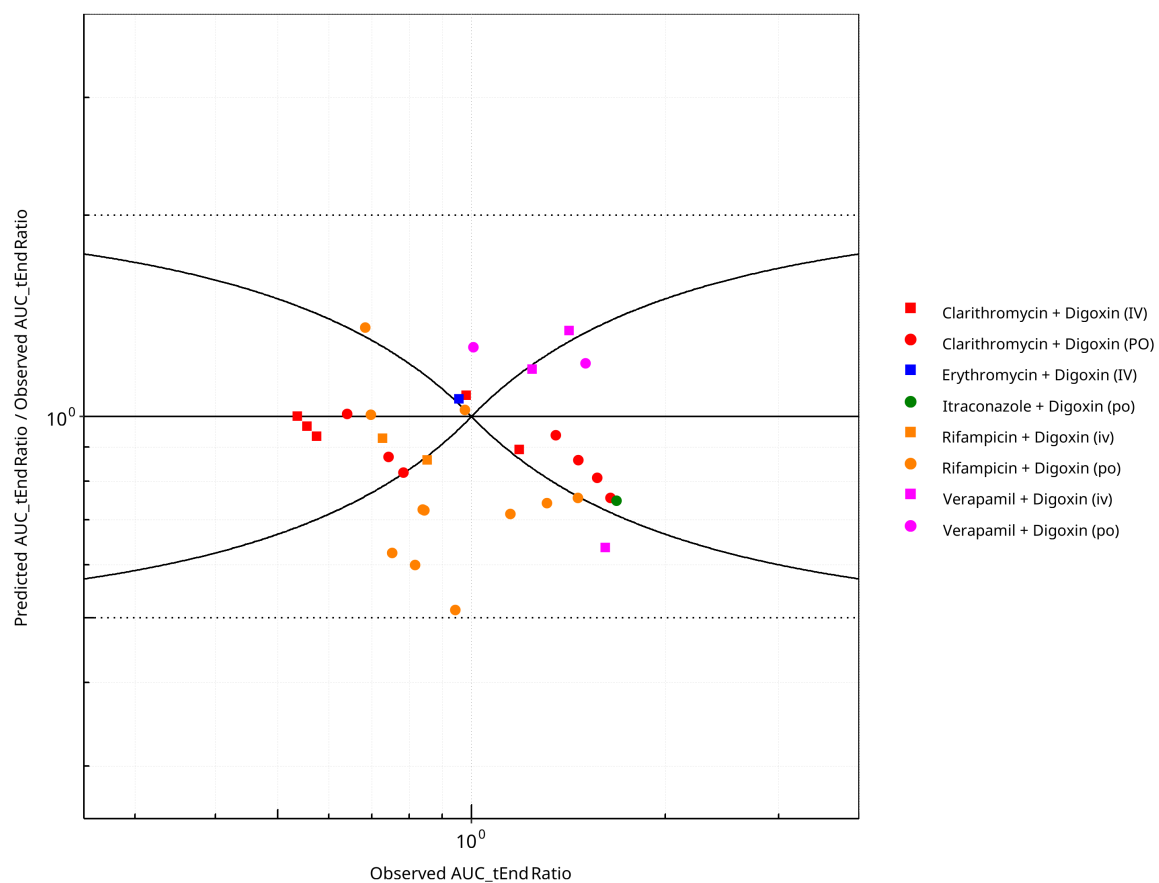


Figure 2-38: Pgp DDI. Victim: Digoxin. Predicted/Observed vs. Observed AUC\_tEnd Ratio. ( $\delta = 1$  in Guest *et al.* formula)



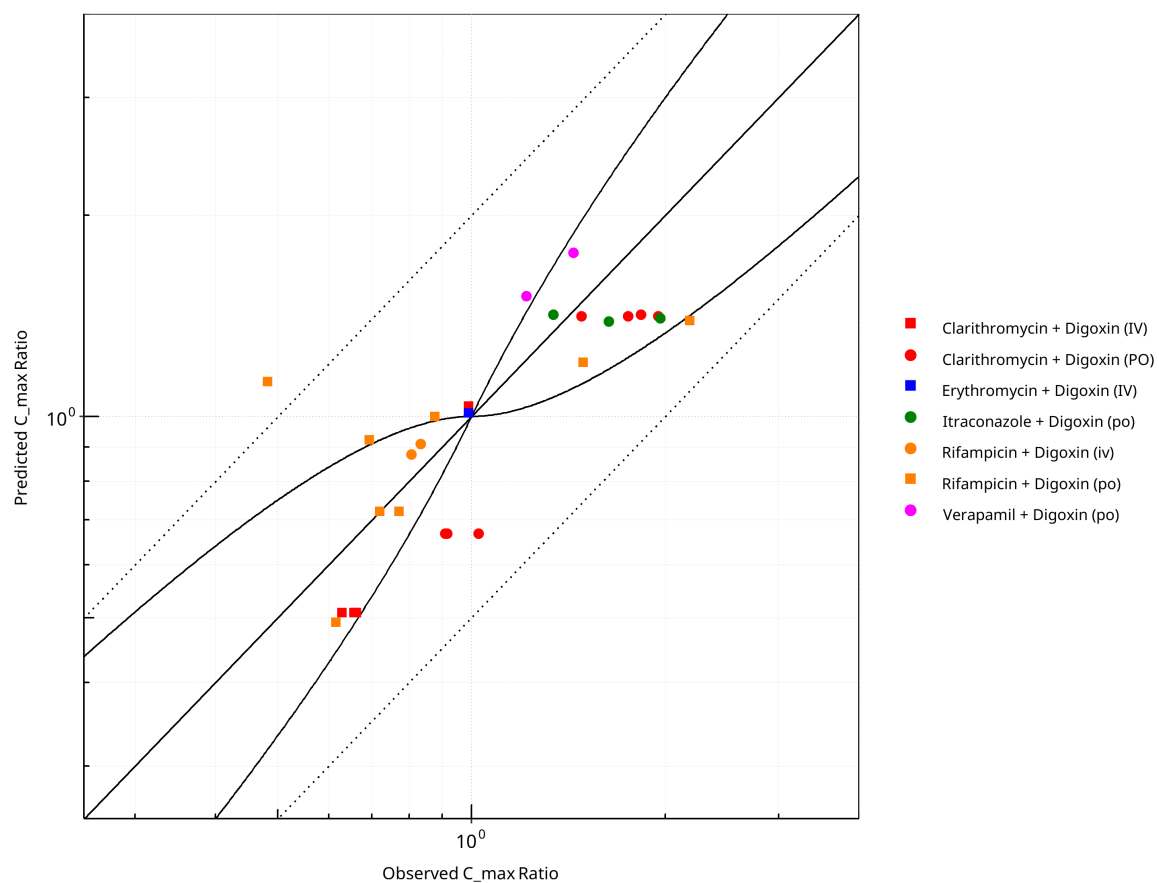
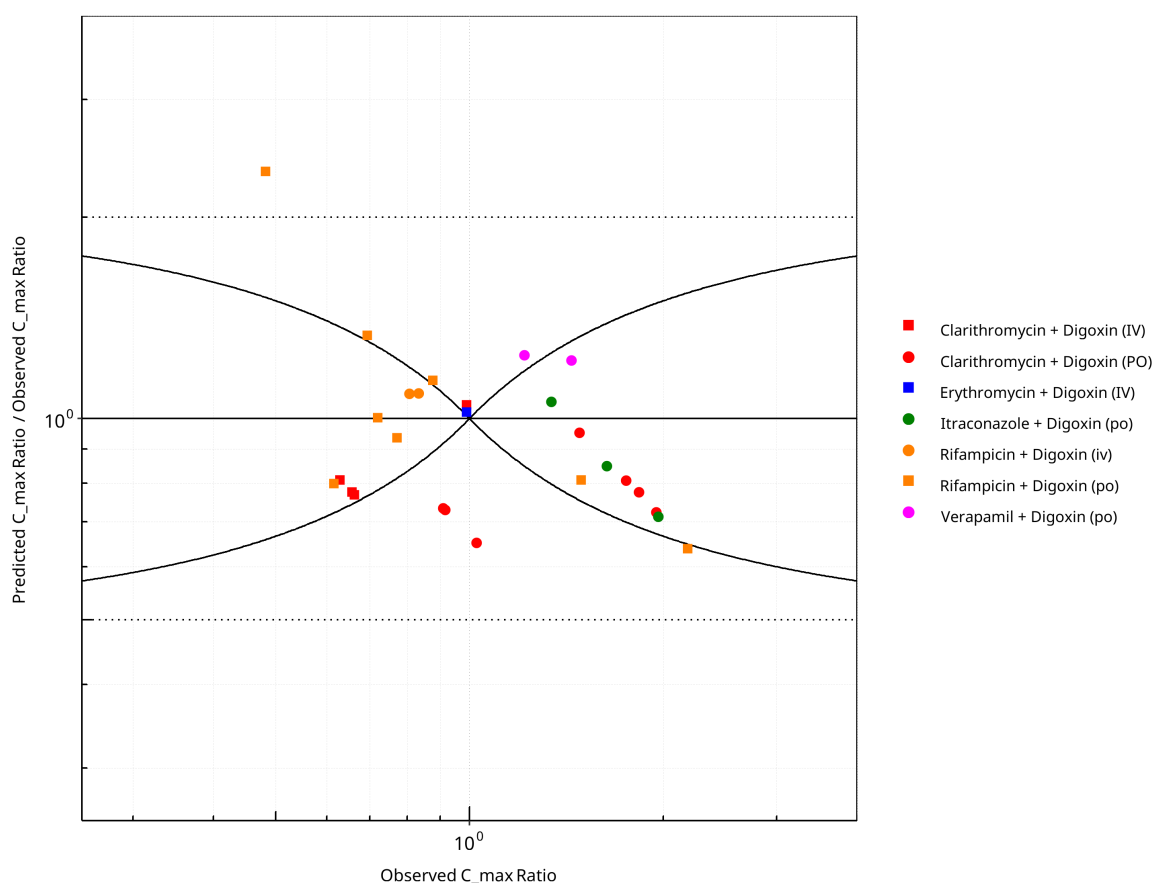


Figure 2-39: Pgp DDI. Victim: Digoxin. Predicted vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)



**Figure 2-40: Pgp DDI. Victim: Digoxin. Predicted/Observed vs. Observed C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

**Table 2-29: GMFE for Pgp DDI Ratio**

| PK parameter        | GMFE |
|---------------------|------|
| AUC <sub>tEnd</sub> | 1.24 |
| C <sub>max</sub>    | 1.25 |

**Table 2-30: Summary table for Pgp DDI - AUC<sub>tEnd</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| AUC <sub>tEnd</sub>               | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 32     | -         |
| Points within Guest <i>et al.</i> | 16     | 50        |
| Points within 2 fold              | 32     | 100       |

**Table 2-31: Summary table for Pgp DDI - C<sub>max</sub> Ratio. ( $\delta = 1$  in Guest *et al.* formula)**

| C <sub>max</sub>                  | Number | Ratio [%] |
|-----------------------------------|--------|-----------|
| Points total                      | 27     | -         |
| Points within Guest <i>et al.</i> | 17     | 62.96     |
| Points within 2 fold              | 26     | 96.30     |

## 3 Concentration-Time Profiles

The following section shows concentration time profiles of the victim drugs of the simulated DDI studies in comparison to observed data (if available).

### 3.1 Clarithromycin - Digoxin DDI

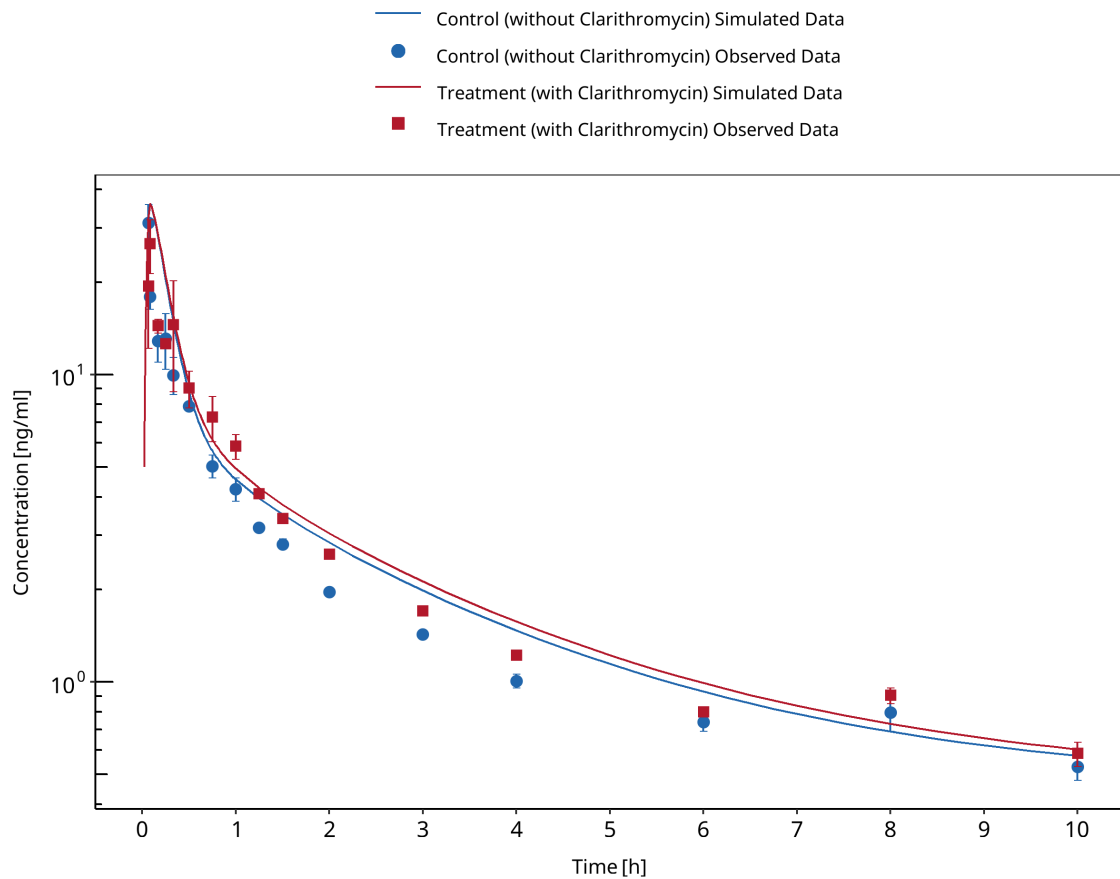


Figure 3-1: Rengelshausen 2003 (0.01 mg/kg IV) - Plasma

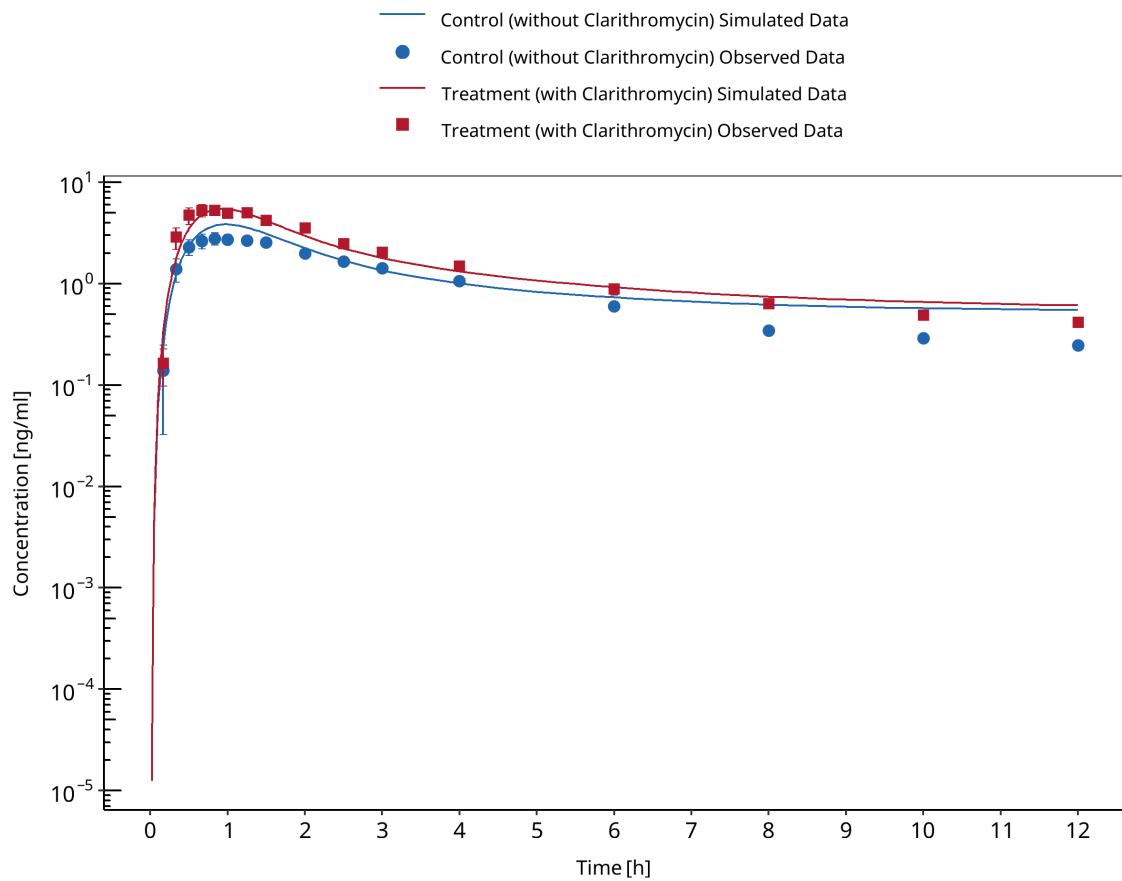


Figure 3-2: Rengelshausen 2003 (0.75 mg PO) - Plasma

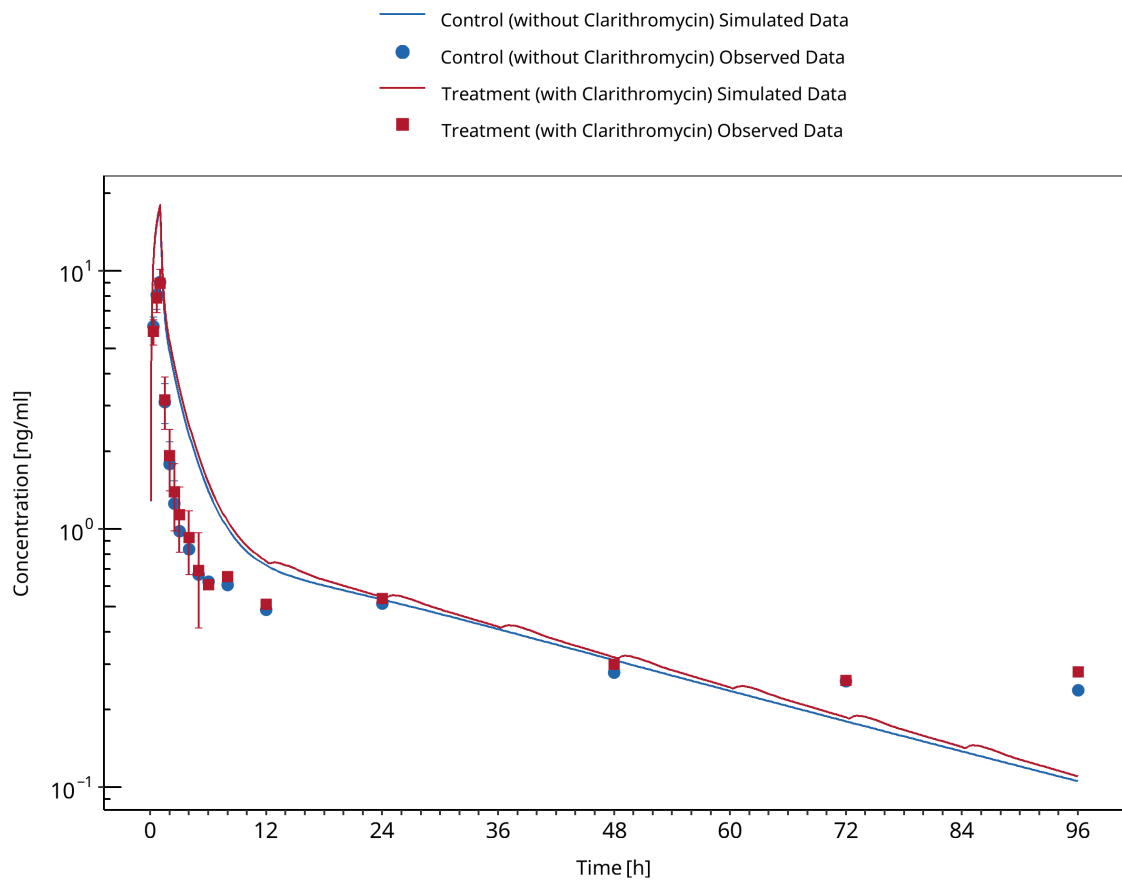


Figure 3-3: Tsutsumi 2002 (0.5mg IV) - Plasma

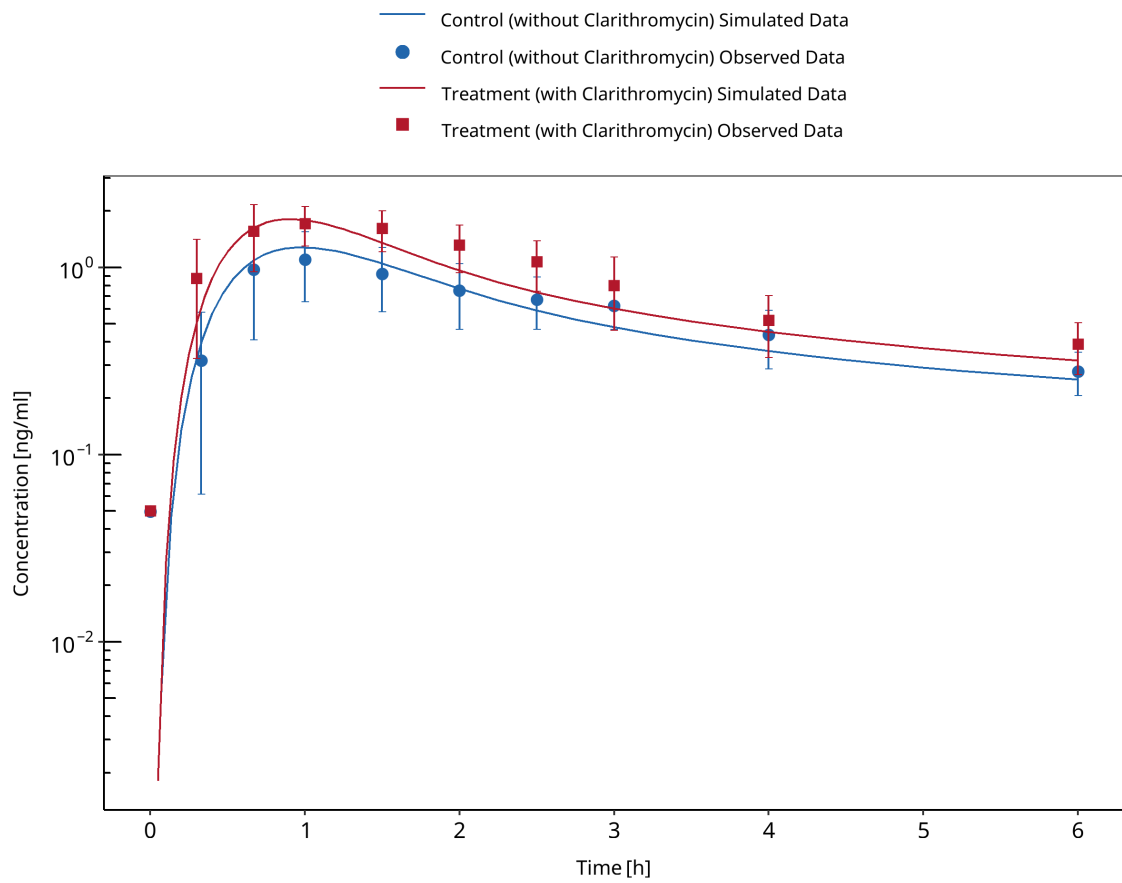


Figure 3-4: Gurley 2008 (0.25 mg PO) - Plasma

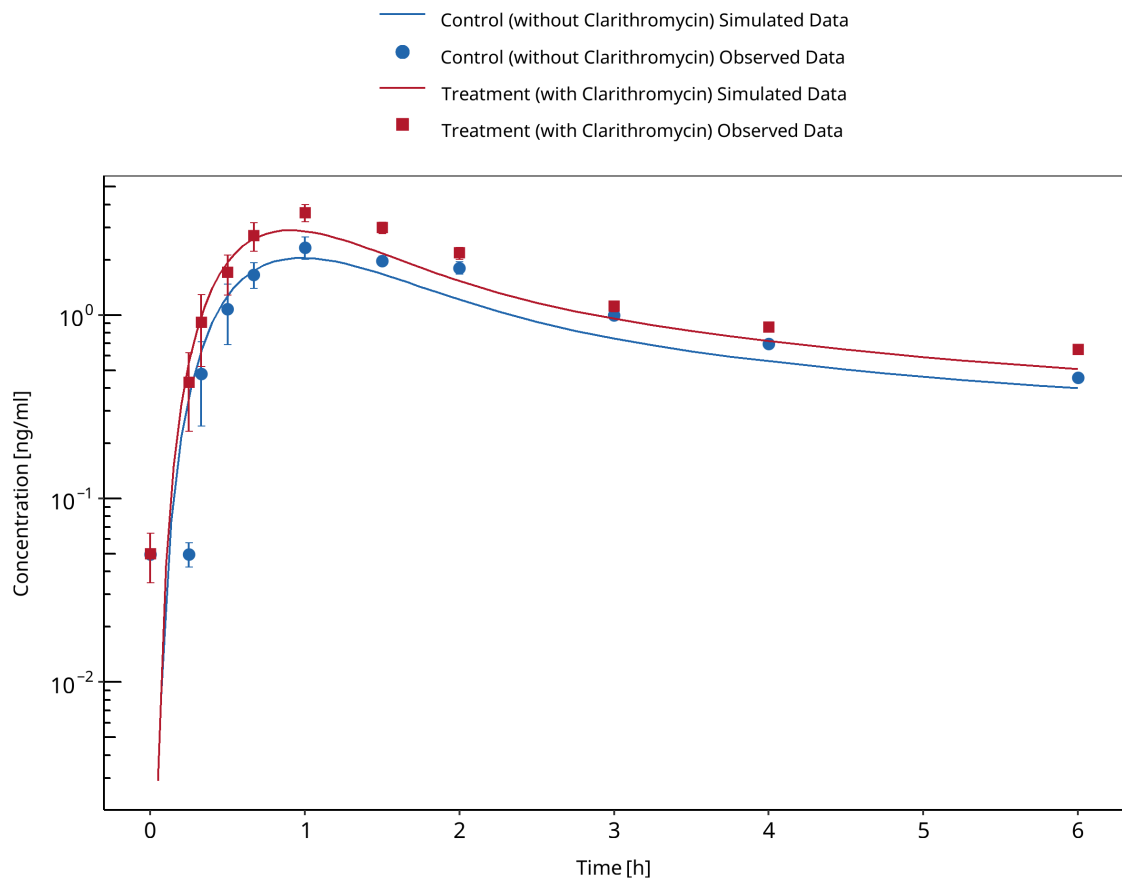


Figure 3-5: Gurley 2006 (0.4 mg PO) - Plasma



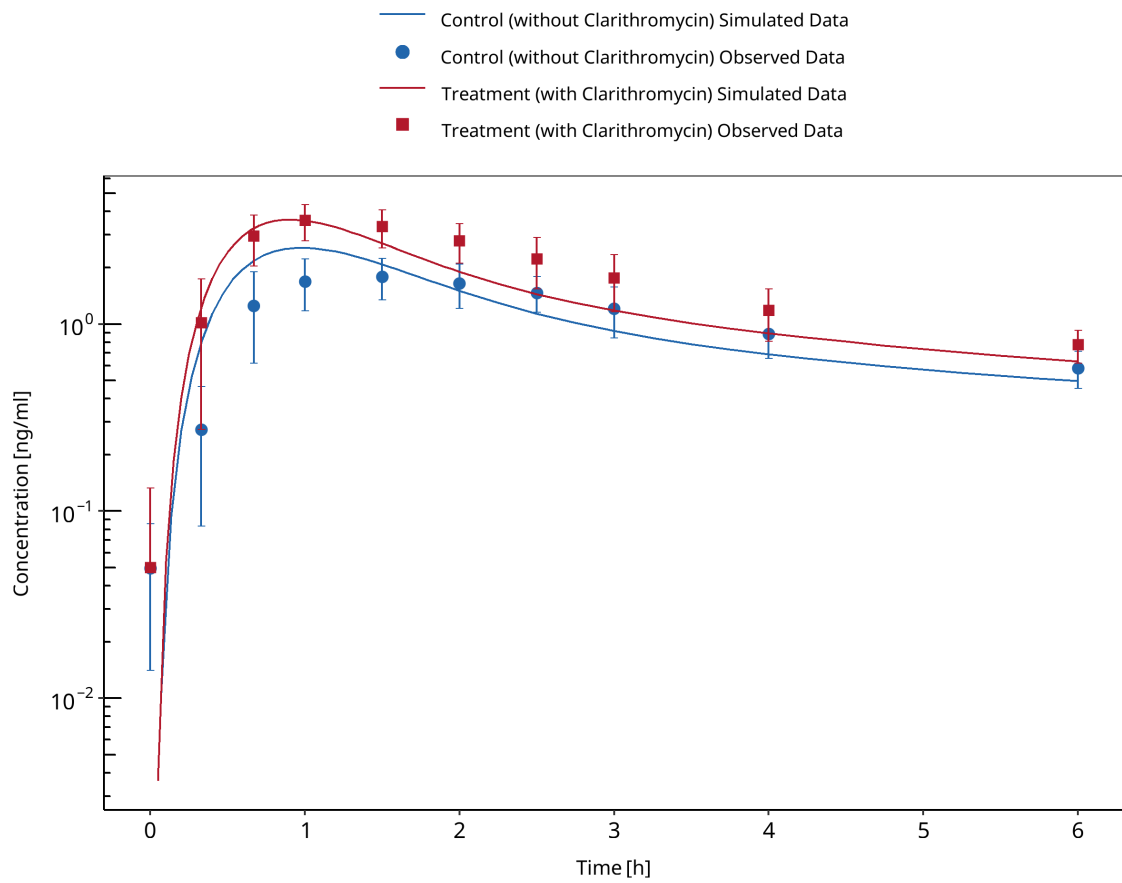


Figure 3-6: Gurley 2007 (0.5 mg PO) - Plasma

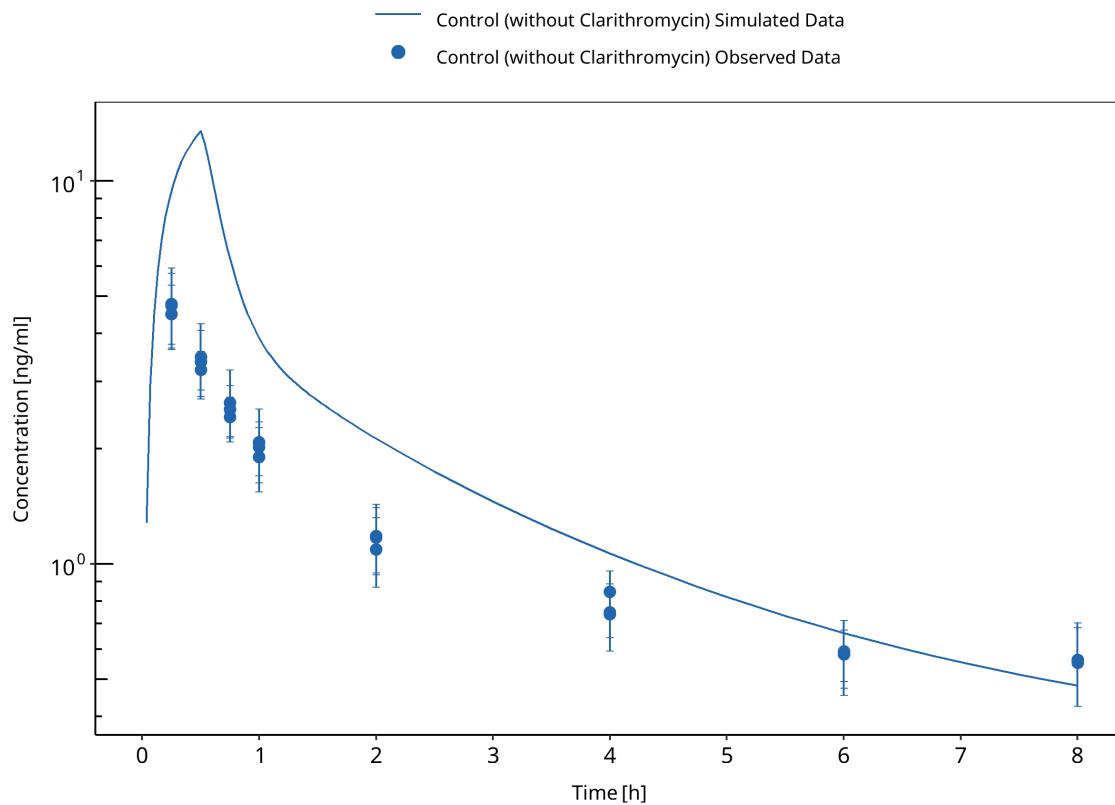


Figure 3-7: Kurata 2002 (0.5 mg IV) - Plasma

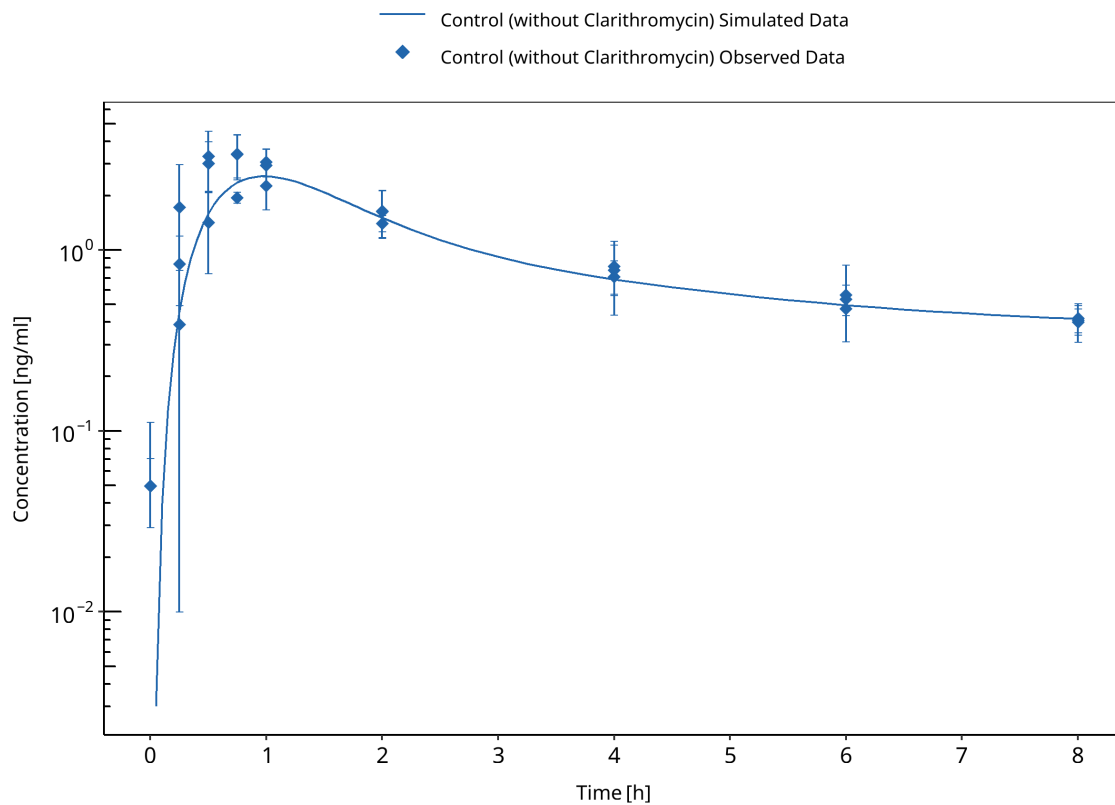


Figure 3-8: Kurata 2002 (0.5 mg PO) - Plasma

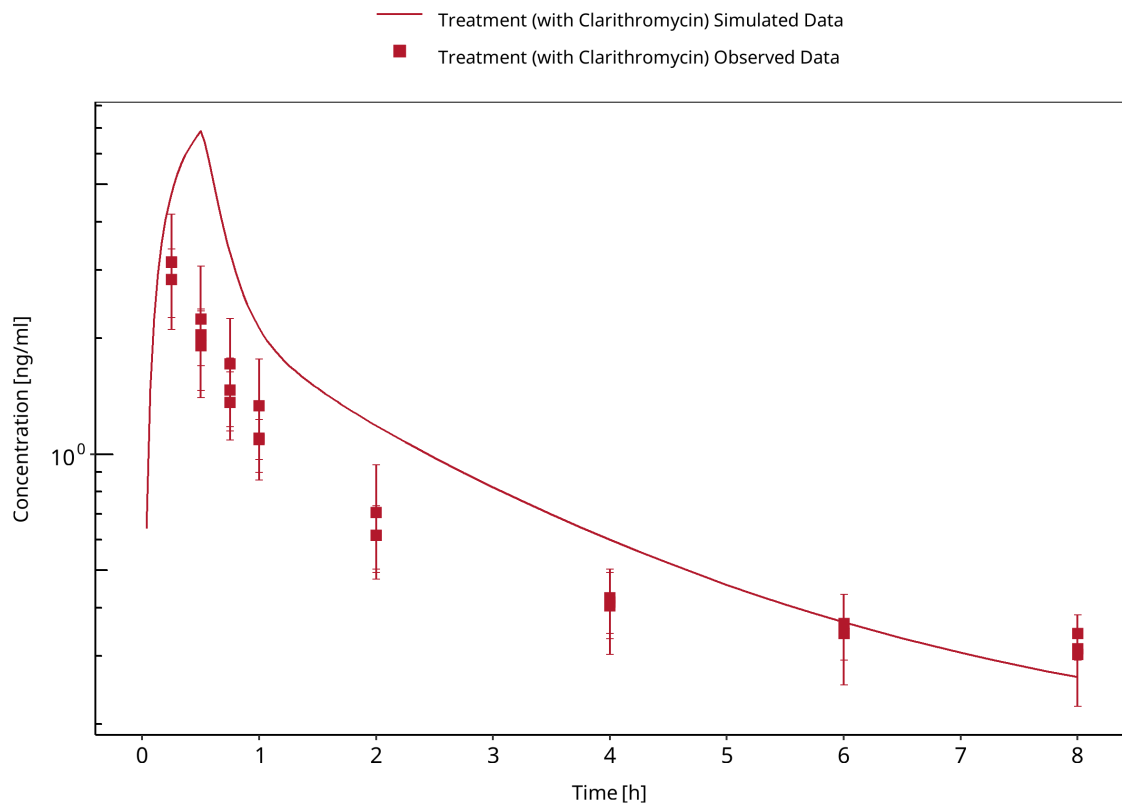


Figure 3-9: Kurata 2002 (0.25 mg IV) - Plasma

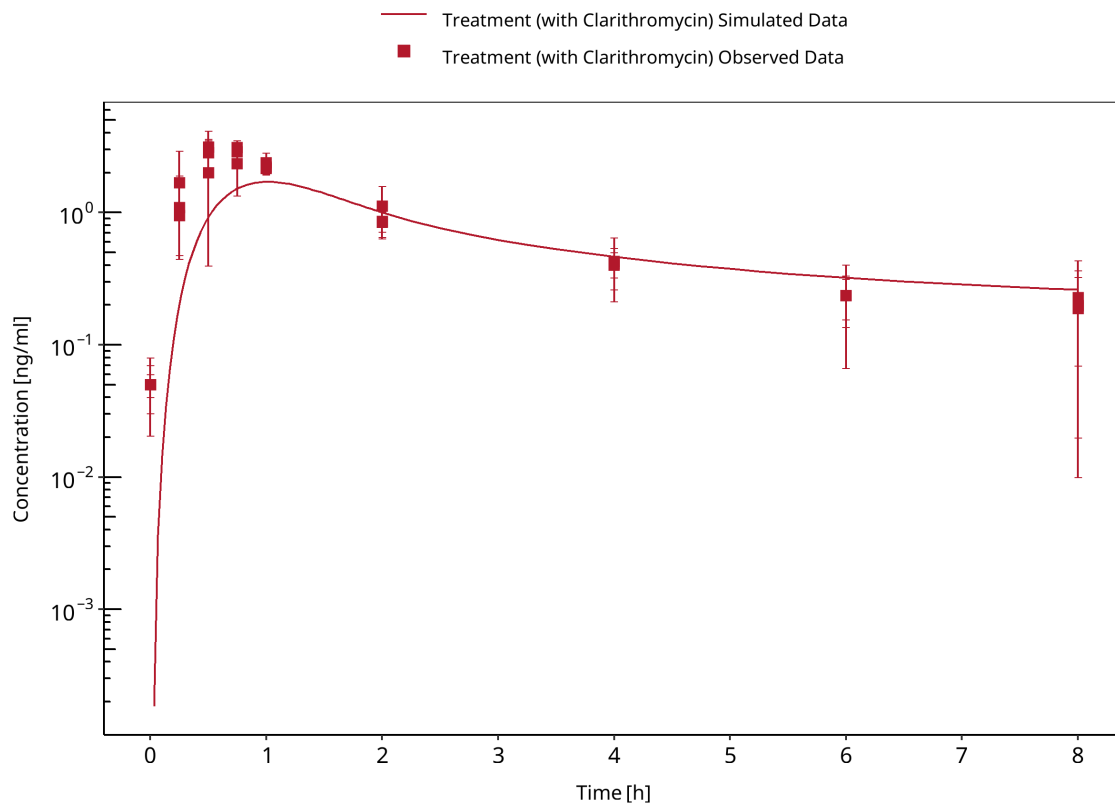


Figure 3-10: Kurata 2002 (0.25 mg PO) - Plasma

## 3.2 Erythromycin - Digoxin DDI

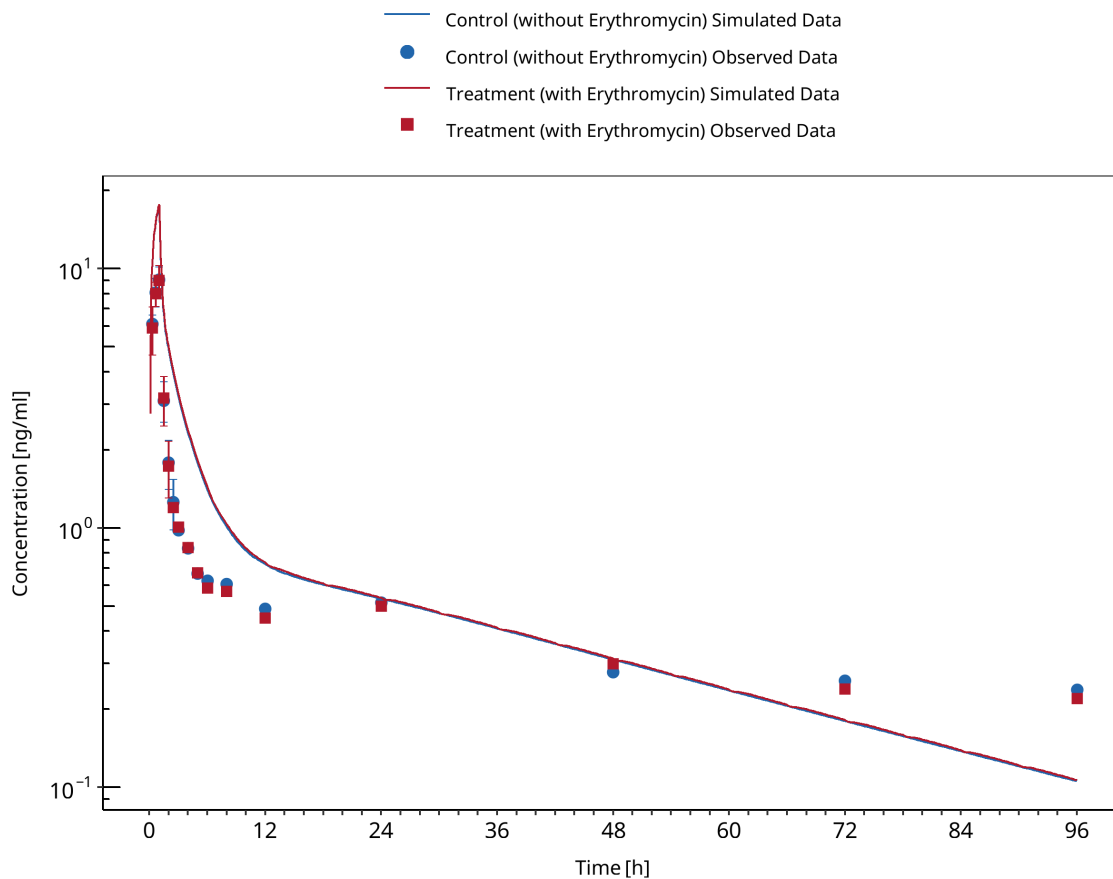


Figure 3-11: Tsutsumi 2002 - Plasma

### 3.3 Itraconazole - Digoxin DDI

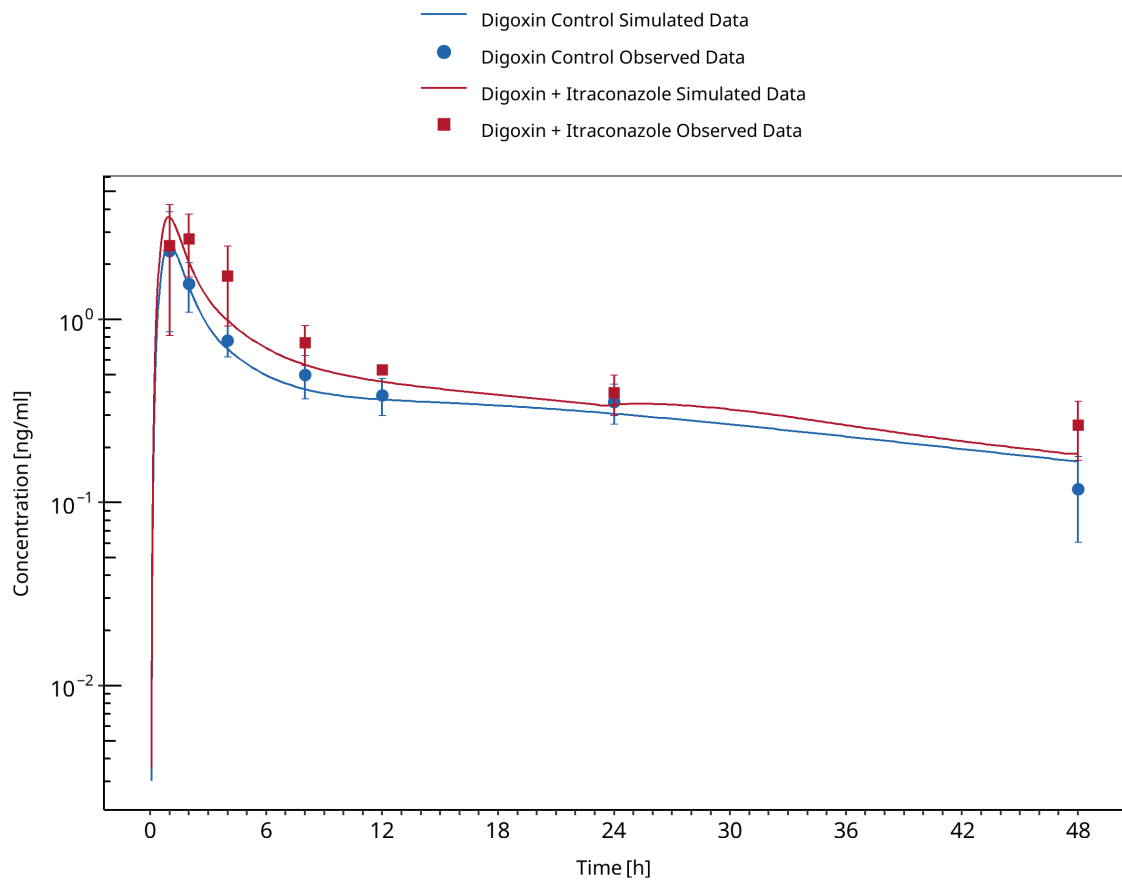


Figure 3-12: Jalava 1997

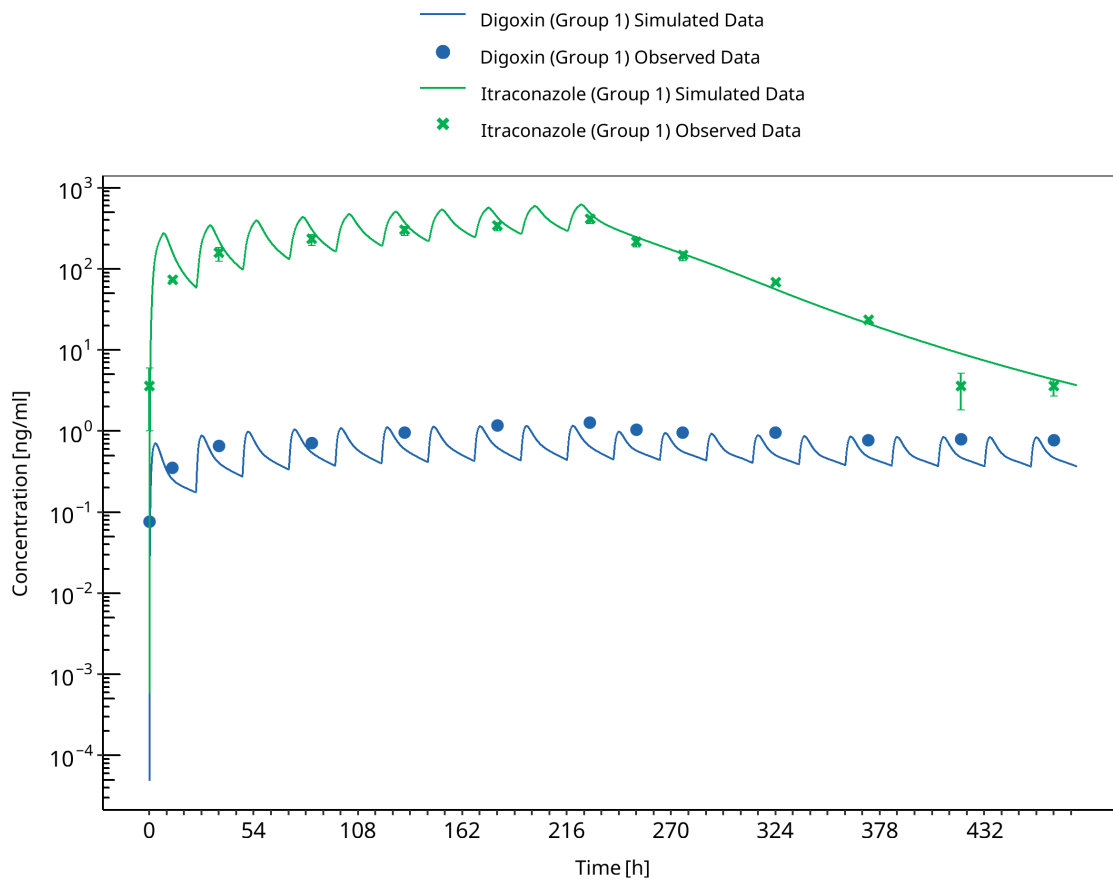


Figure 3-13: Partanen 1996 - Group 1



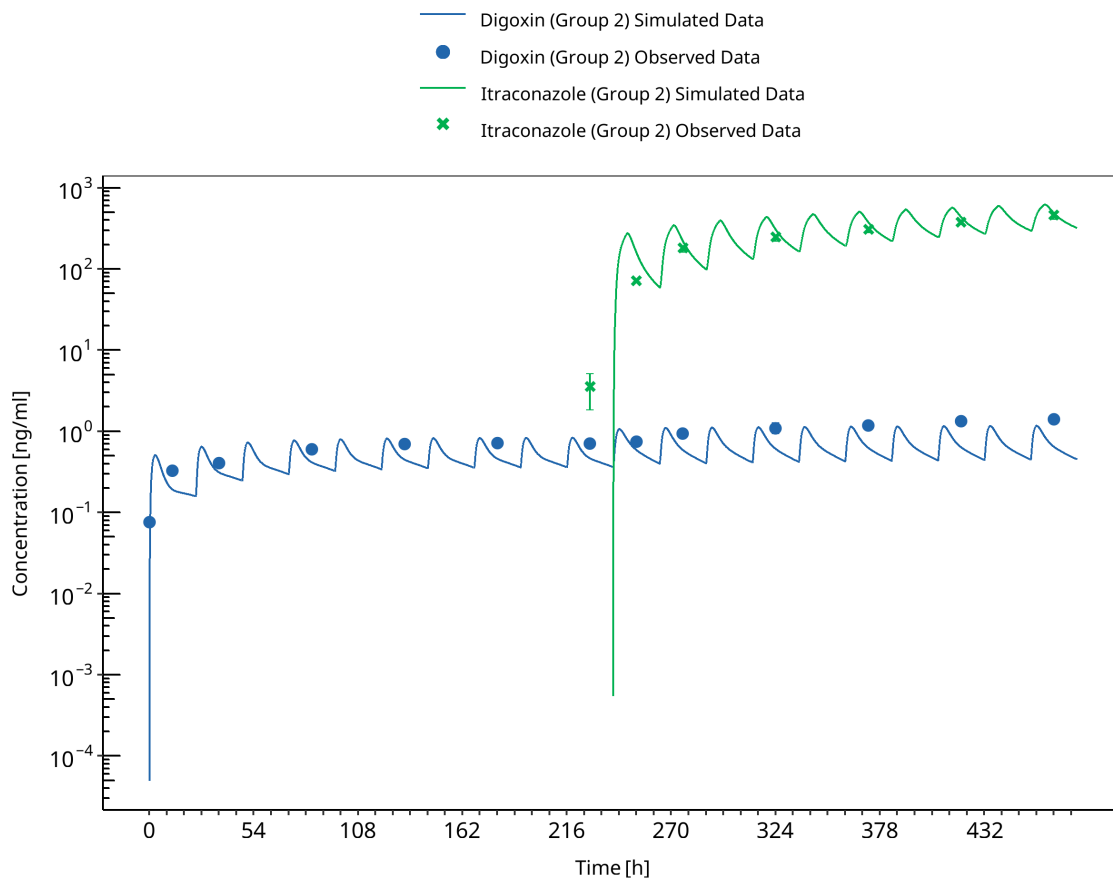


Figure 3-14: Partanen 1996 - Group 2

### 3.4 Verapamil - Digoxin DDI

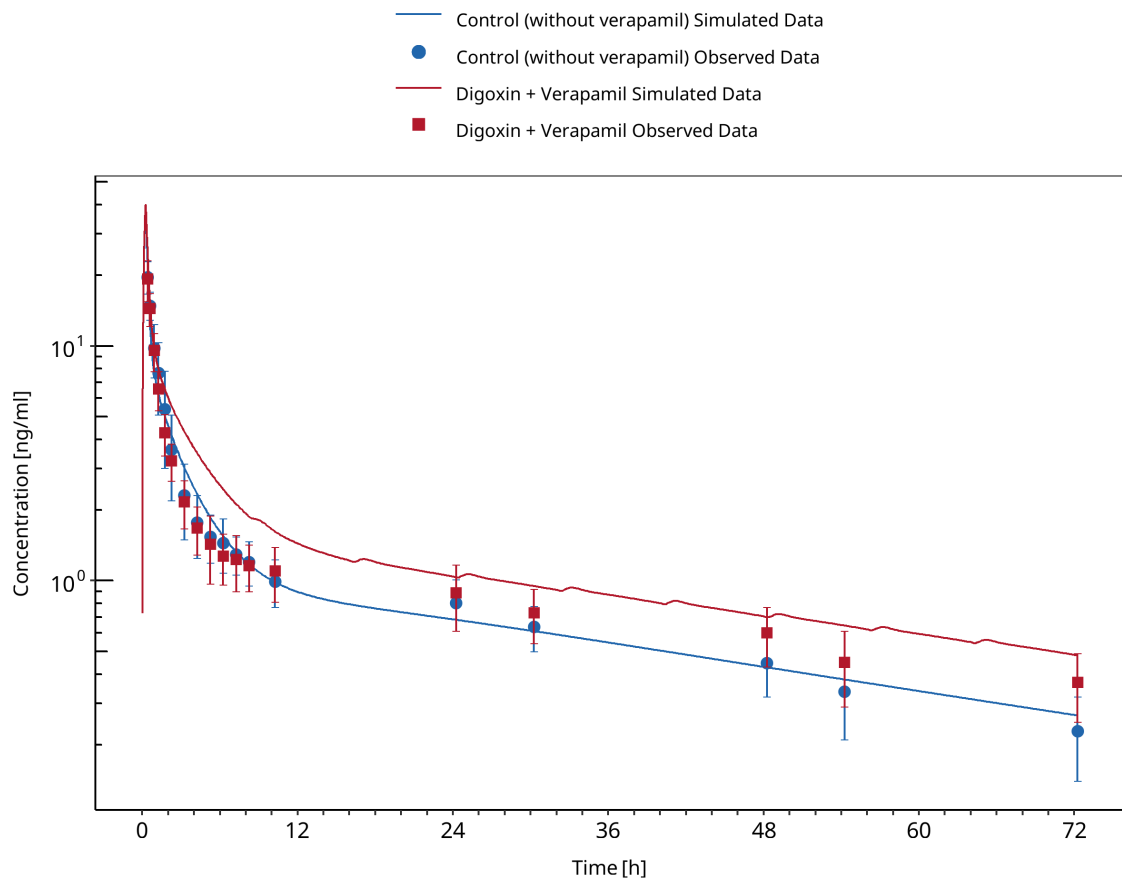


Figure 3-15: Johnson 1987

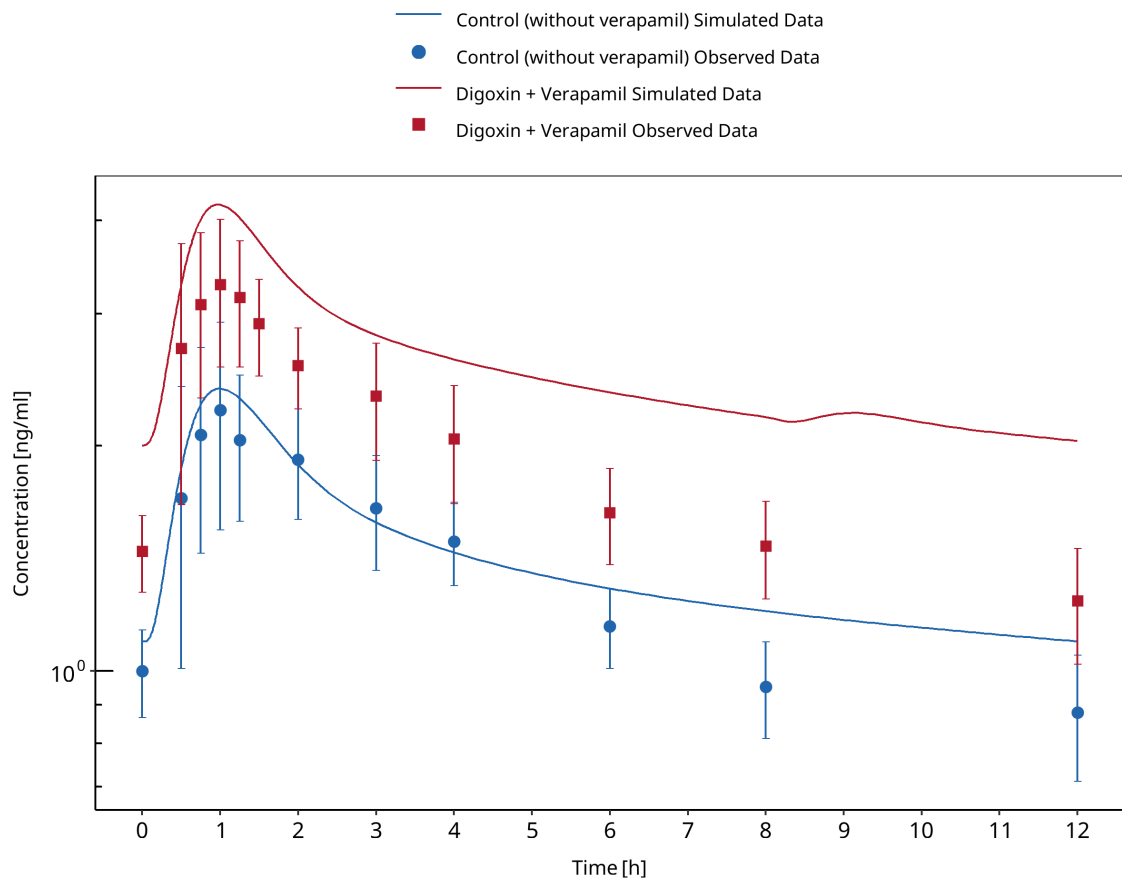


Figure 3-16: Rodin 1988

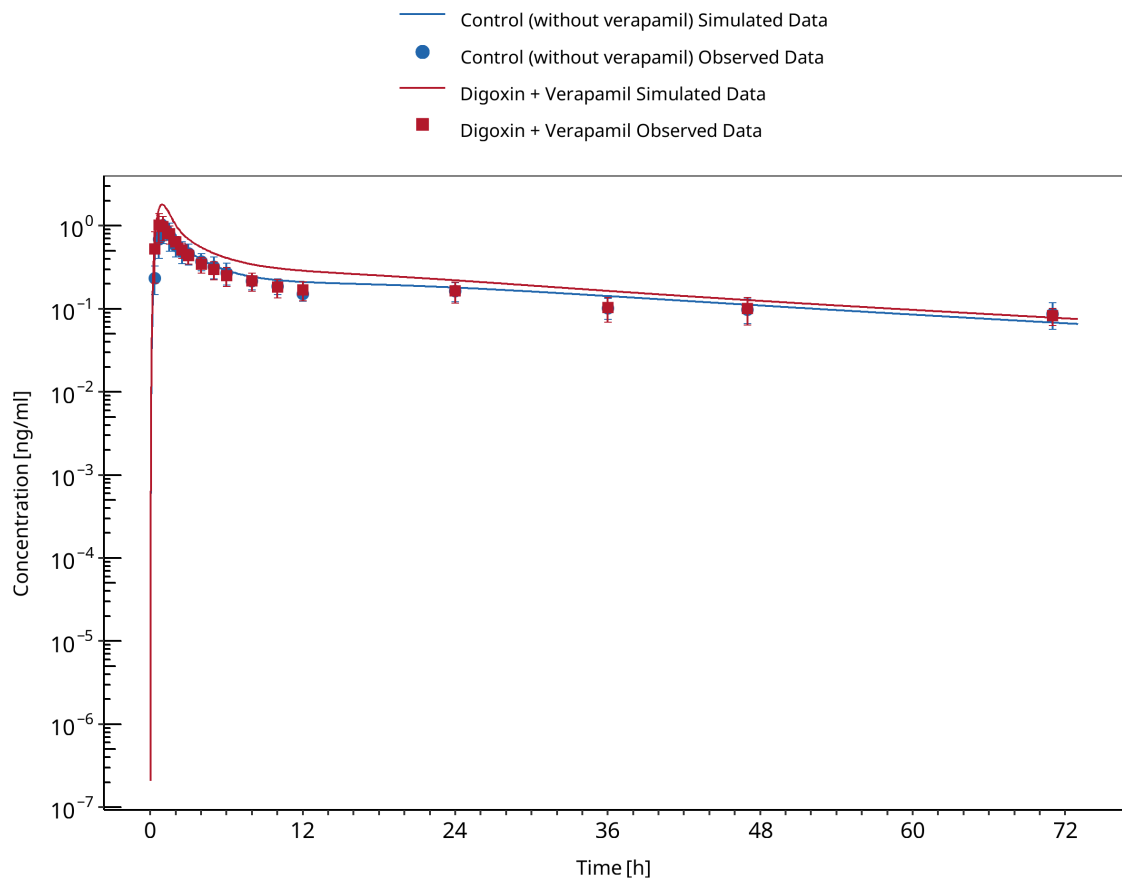


Figure 3-17: Wiebe 2020

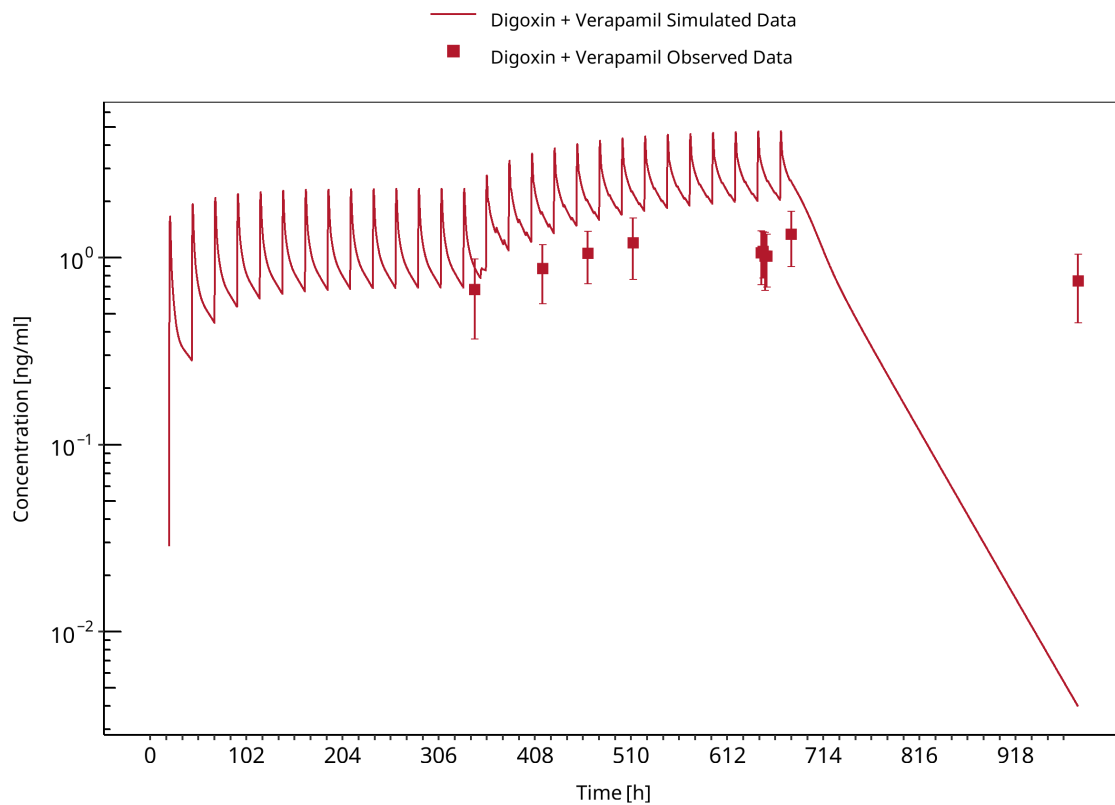


Figure 3-18: Klein 1982

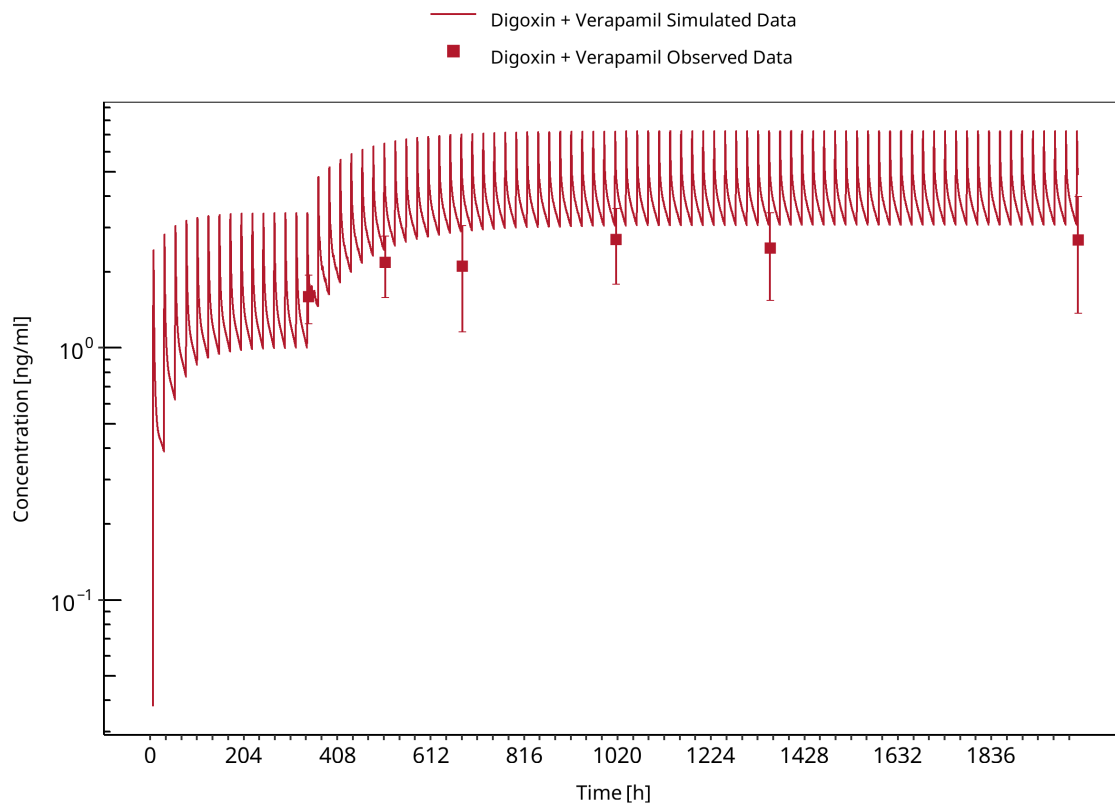


Figure 3-19: Schwartz 1982

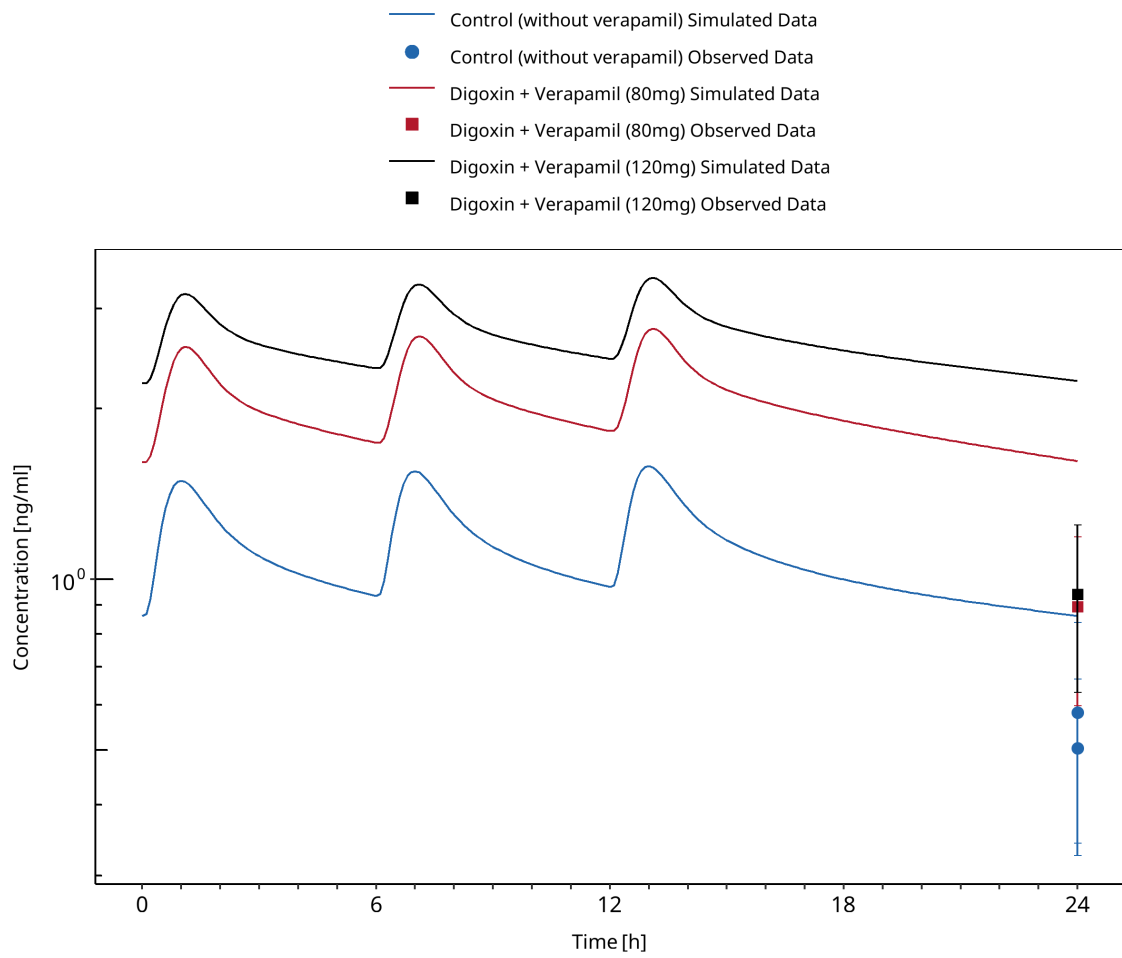


Figure 3-20: Belz 1983

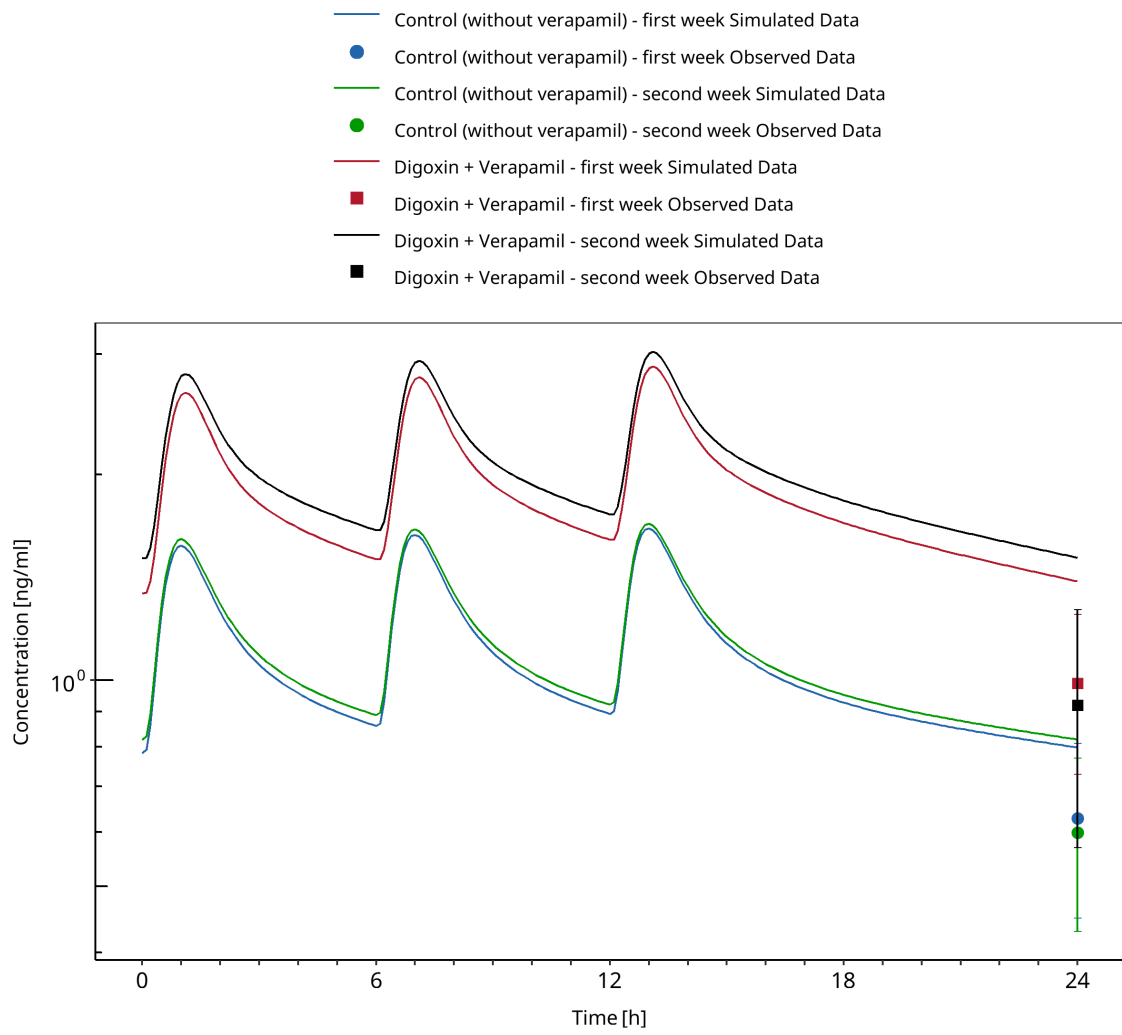


Figure 3-21: Doering 1983



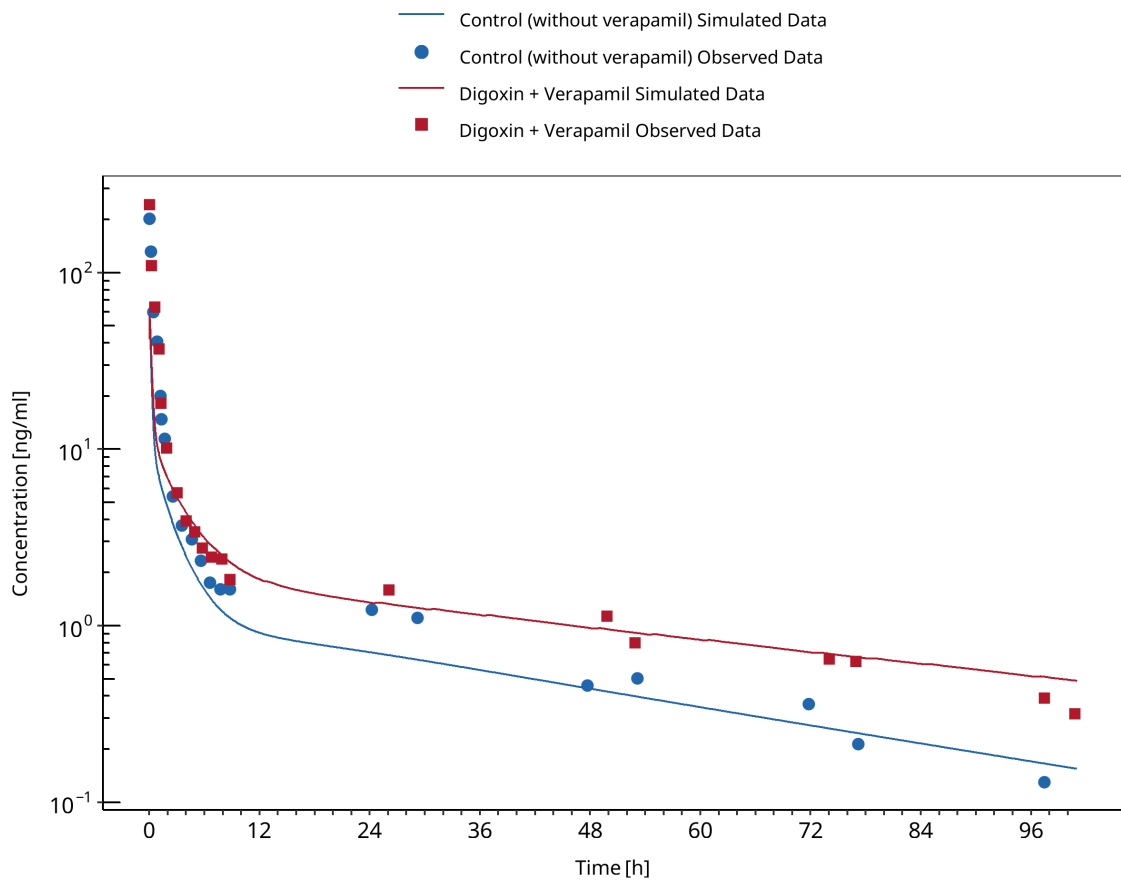


Figure 3-22: Pedersen 1983

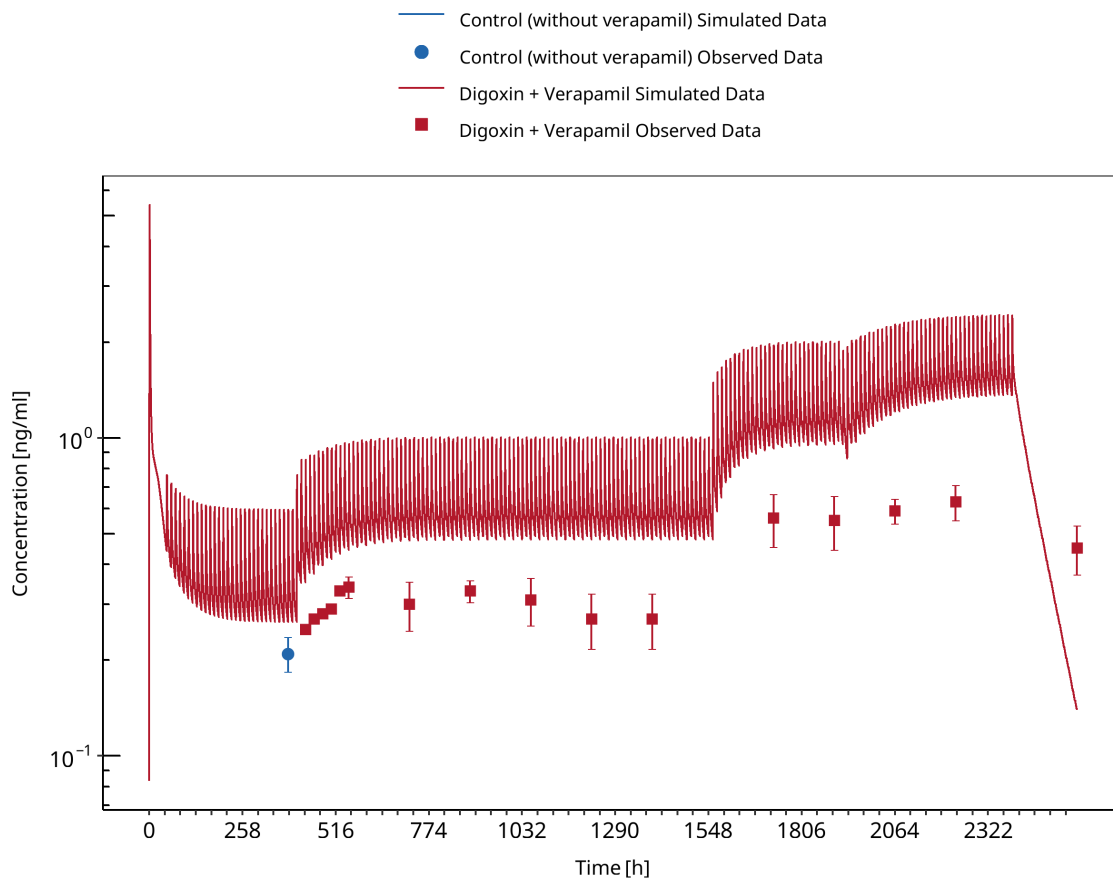


Figure 3-23: Pedersen 1982

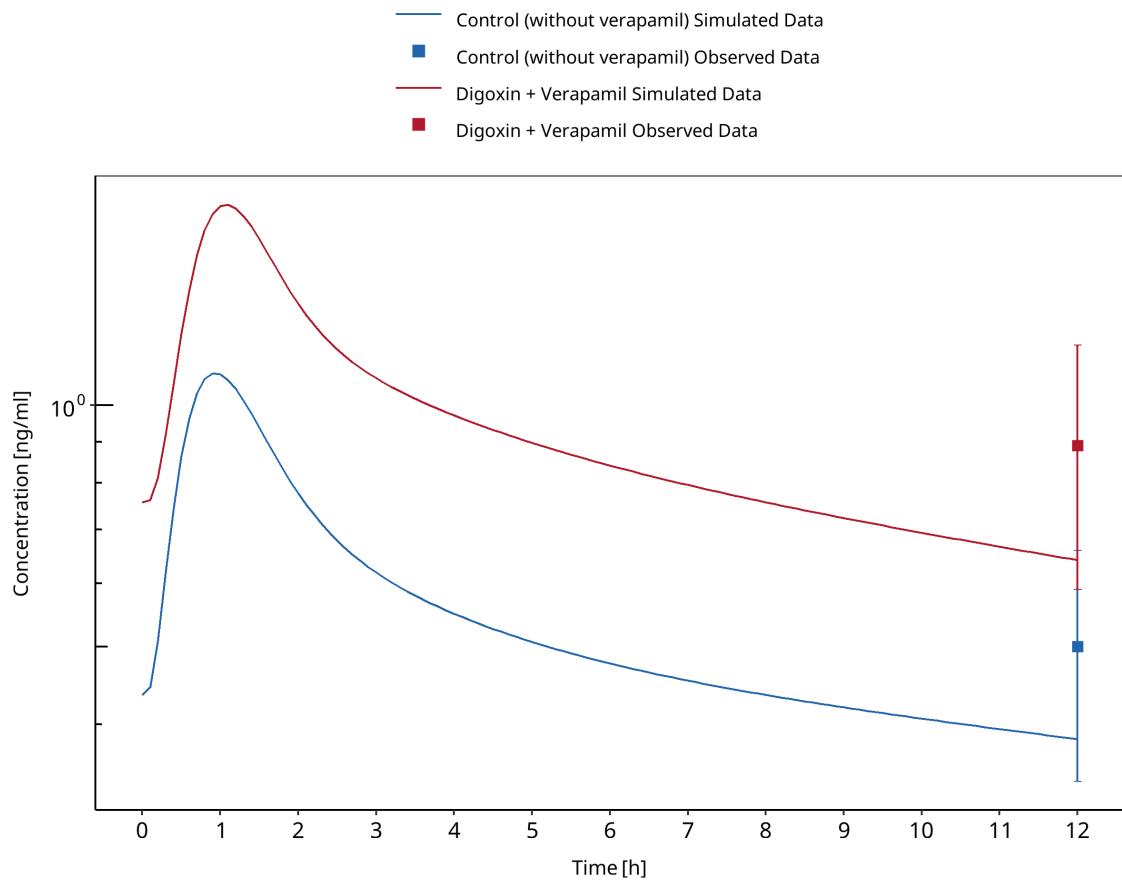


Figure 3-24: Belz 1981

### 3.5 Rifampicin - Digoxin DDI

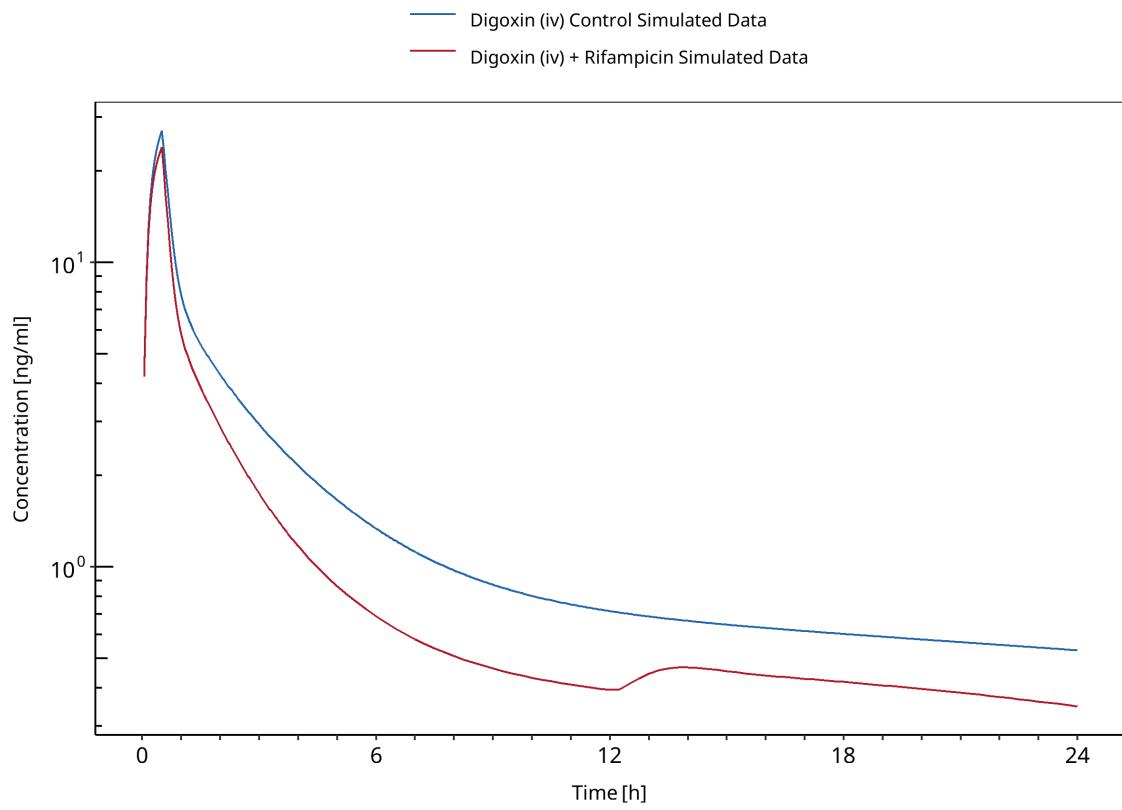


Figure 3-25: Drescher 2003

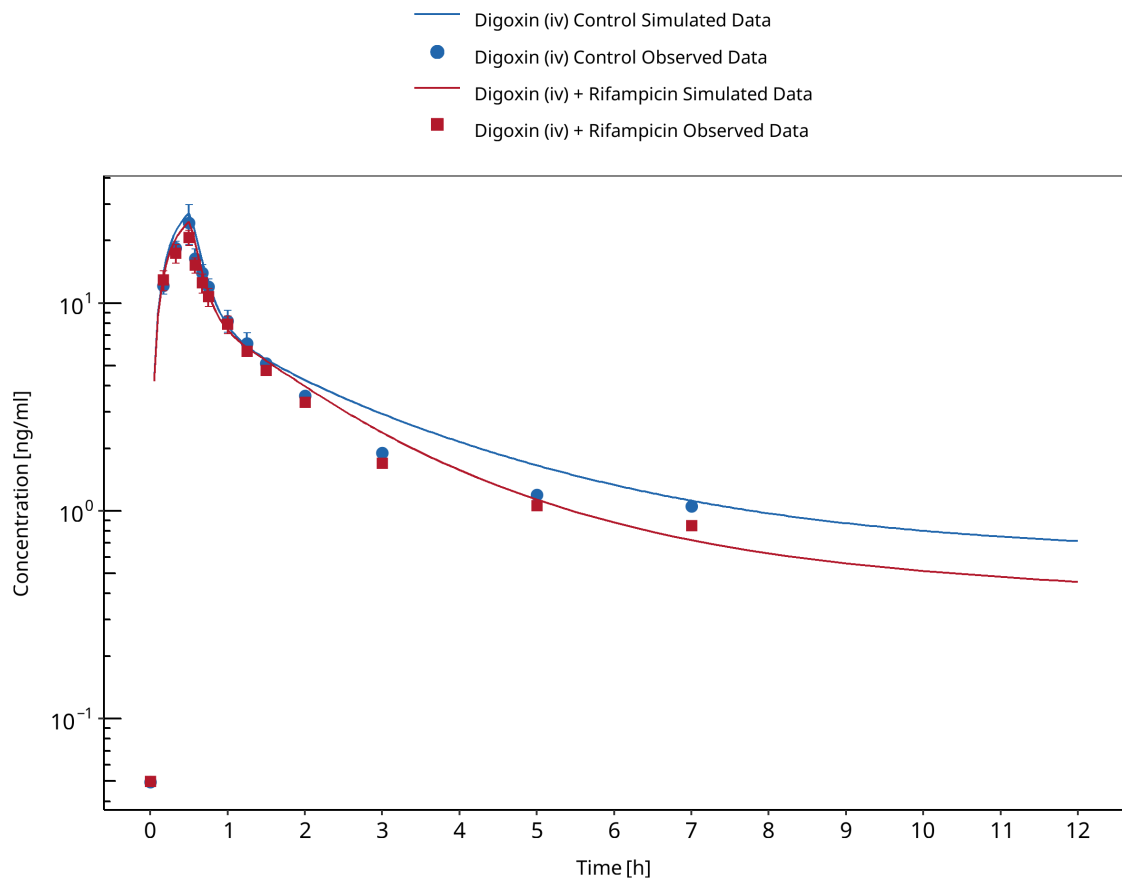


Figure 3-26: Greiner 1999 (iv)

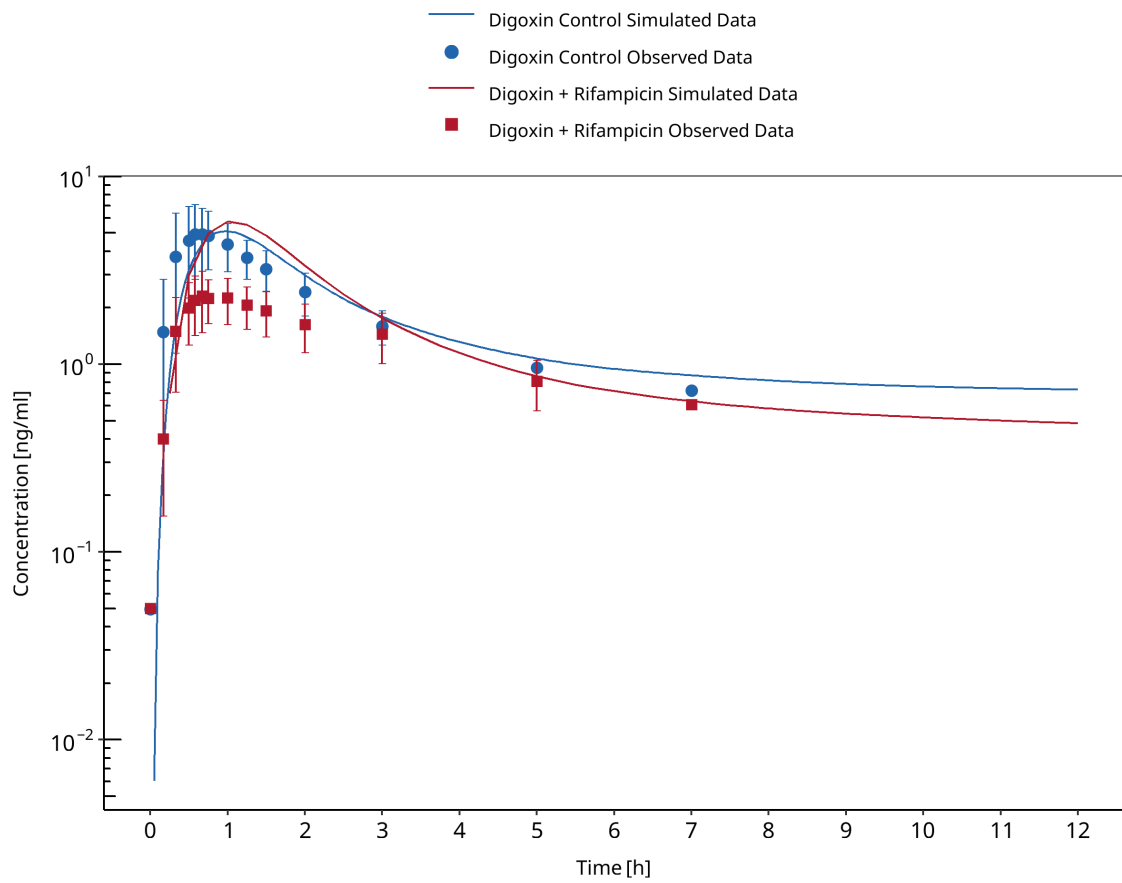


Figure 3-27: Greiner 1999 (po)

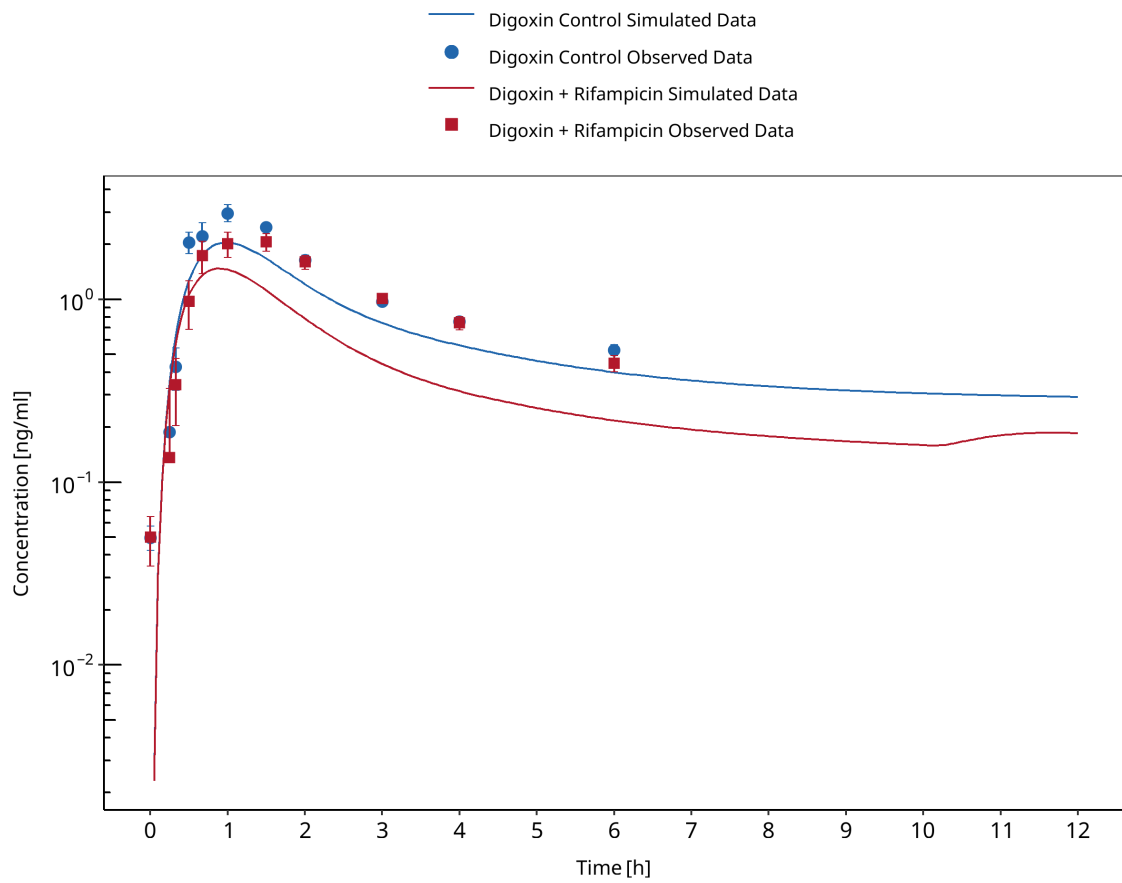


Figure 3-28: Gurley 2006

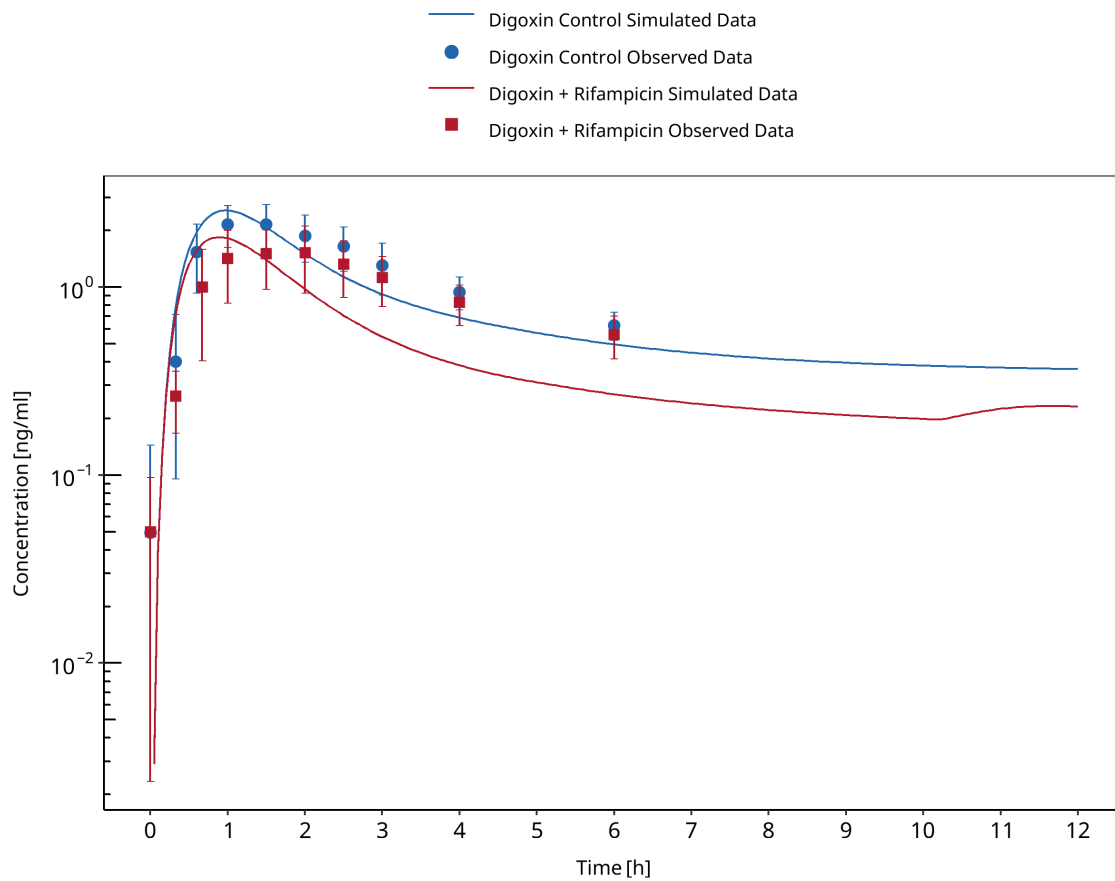


Figure 3-29: Gurley 2007



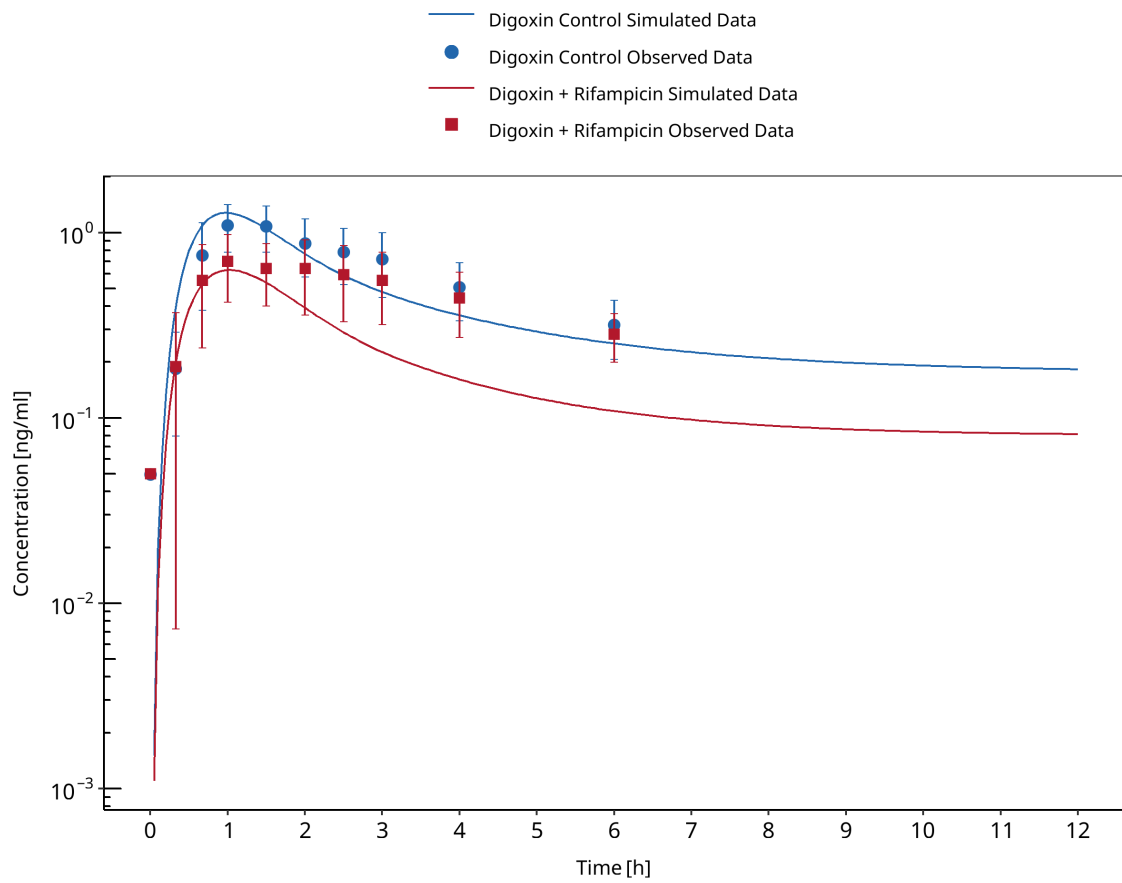


Figure 3-30: Gurley 2008

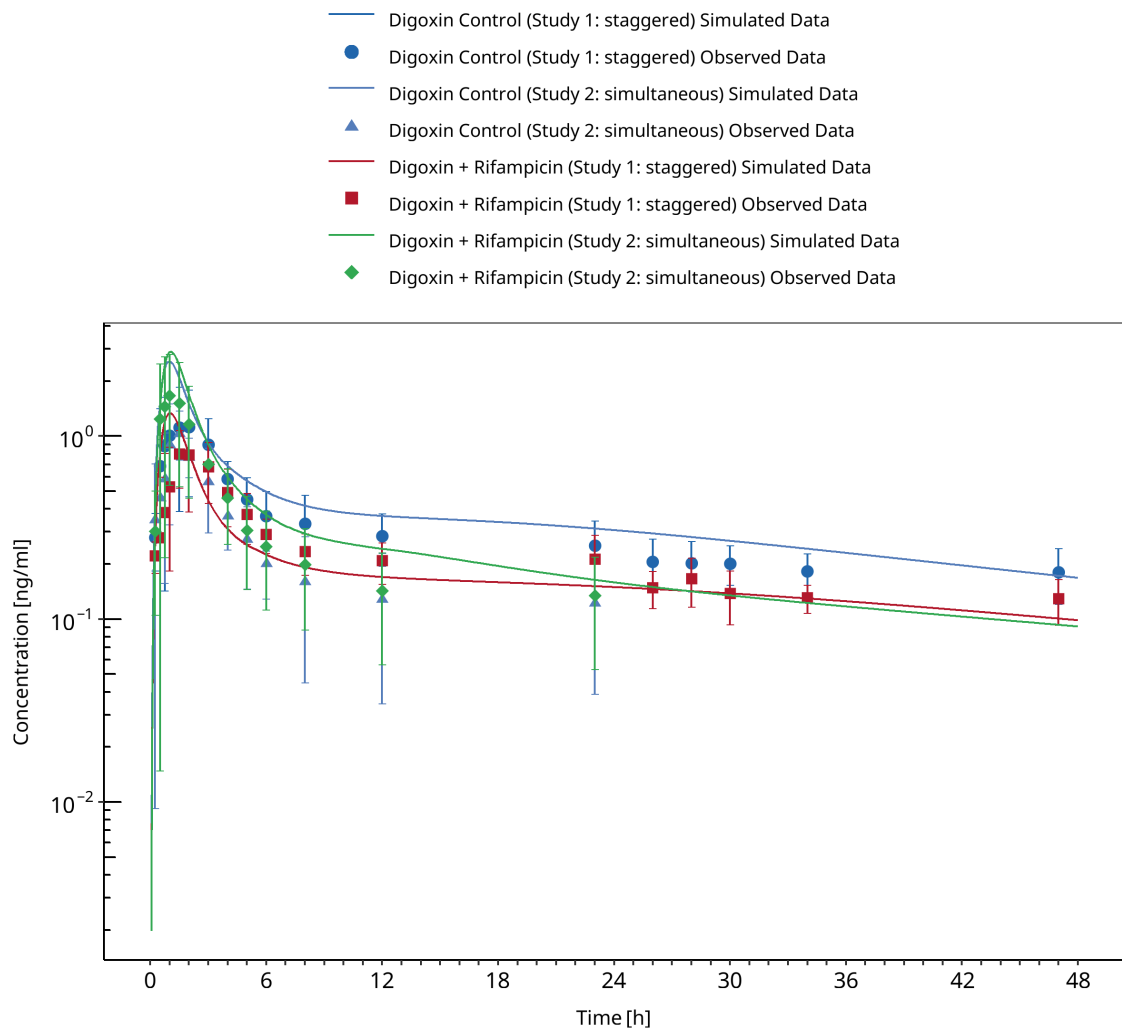


Figure 3-31: Kirby 2012

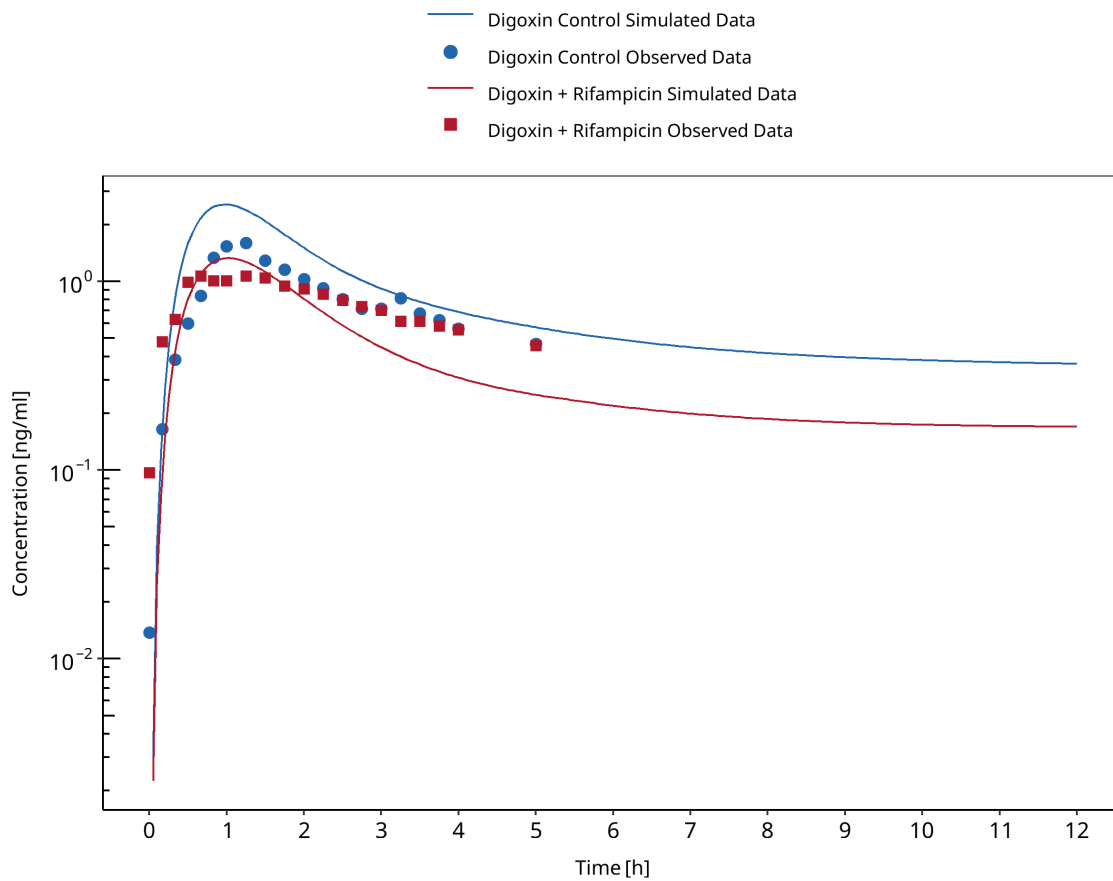


Figure 3-32: Larsen 2007

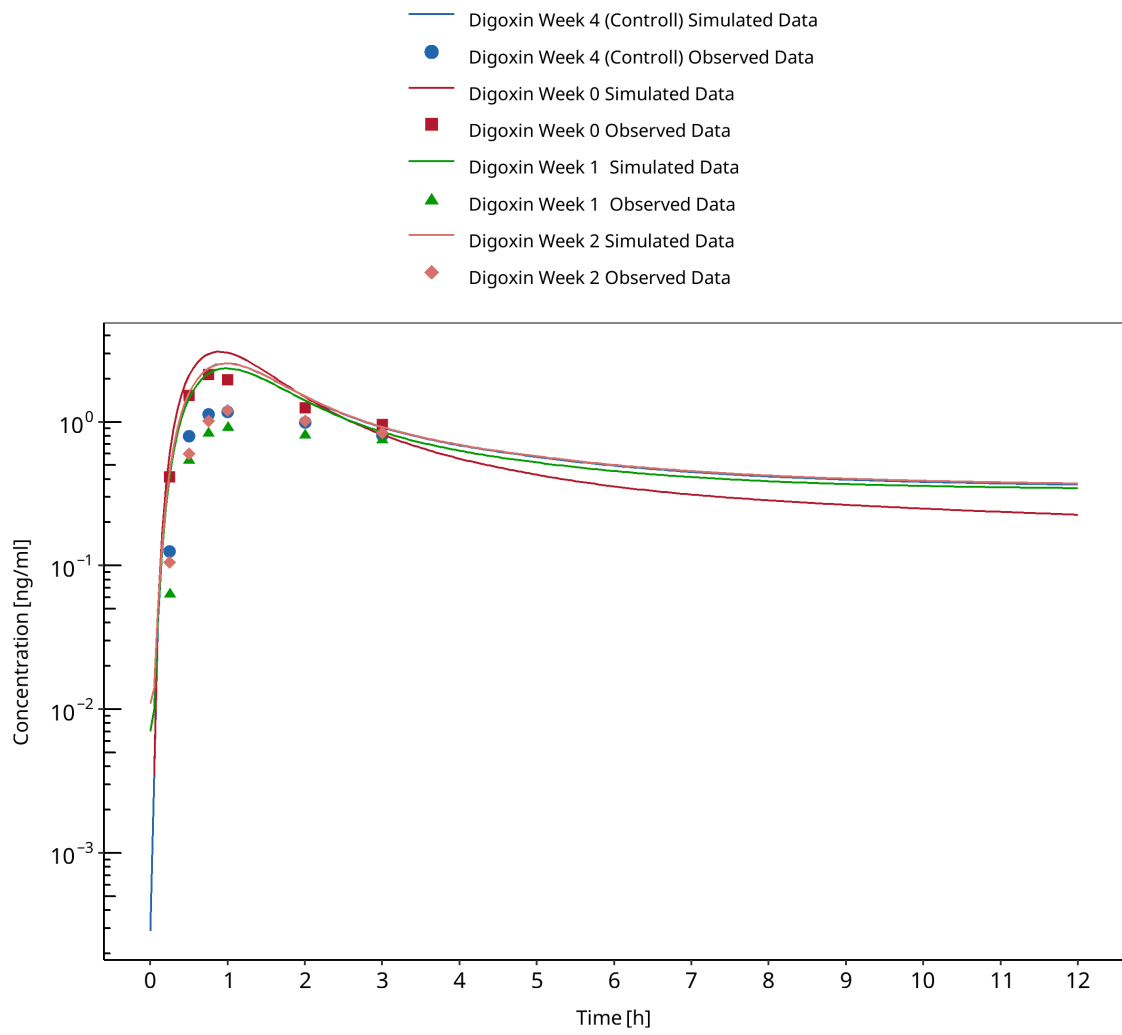


Figure 3-33: Reitman 2011

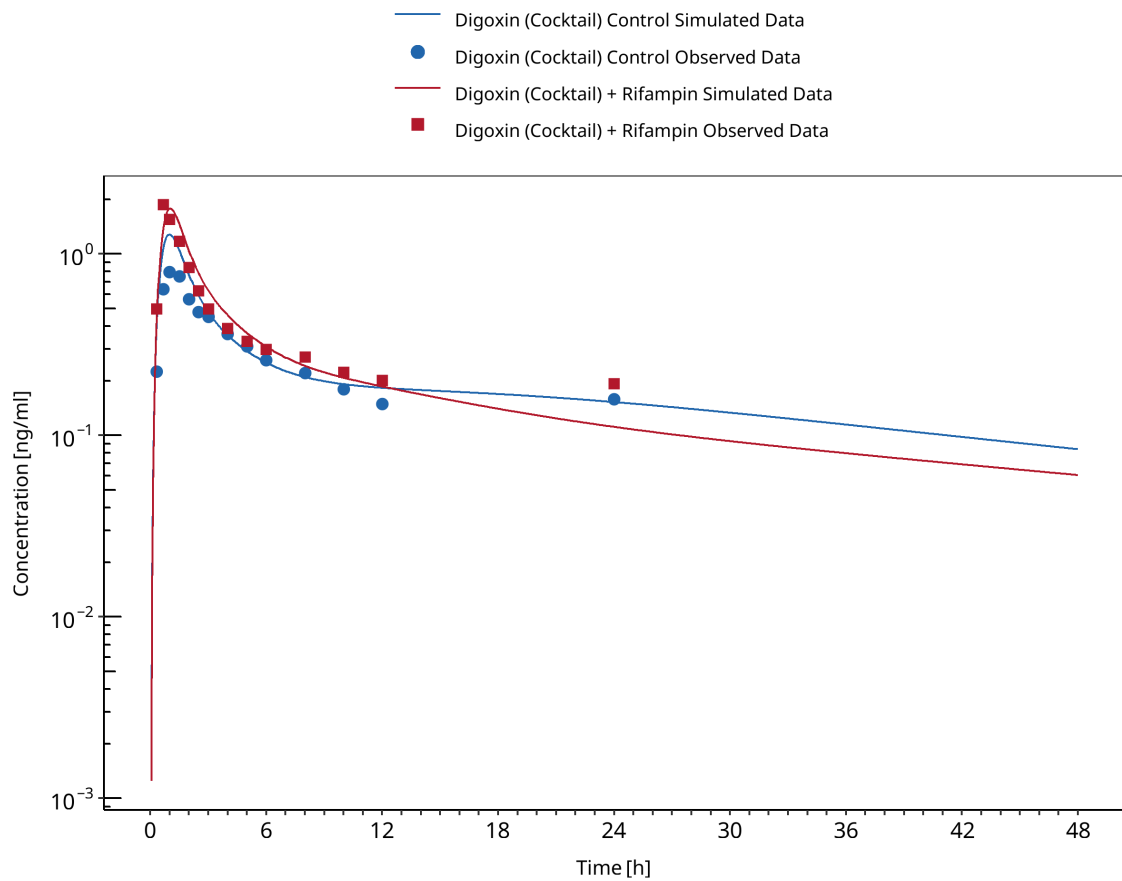


Figure 3-34: Wiebe 2020

## 4 References

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### Clarithromycin - Digoxin DDI

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### Erythromycin-Digoxin-DDI

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## Verapamil - Digoxin DDI

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## 5 Appendix

### 5.1 Open Systems Pharmacology Suite (OSPS) Introduction

Open Systems Pharmacology Suite (OSP suite) is a tool for PBPK modeling and simulation of drugs in laboratory animals and humans. PK-Sim® and MoBi® are part of the OSP suite [1]. PK-Sim® is based on a generic PBPK-model with 18 organs and tissues. One of the main assumptions is that all compartments are well-stirred. Represented organs/tissues include arterial and venous blood, adipose tissue (separable adipose, excluding yellow marrow), brain, lung, bone (including yellow marrow), gonads, heart, kidneys, large intestine, liver, muscle, portal vein, pancreas, skin, small intestine, spleen and stomach, as shown in [Figure Appendix-1](#).

Each organ consists of four sub-compartments namely the plasma, blood cells (which together build the vascular space), interstitial space, and cellular space. Distribution between the plasma and blood cells as well as between the interstitial and cellular compartments can be permeability-limited. In the brain, the permeation barrier is located between the vascular and the interstitial space. PK-Sim® estimates model parameters (intestinal permeability [2] organ partition coefficients (tissue-to-plasma partition coefficients) [3,4], and permeabilities) from physico-chemical properties of compounds (molecular weight, pKa, acid/base properties) and the composition of each tissue compartment (lipids, water and proteins). Partition coefficients can be calculated using a variety of methods available in PK-Sim®, for example the internal PK-Sim® method [3,4] or that of Rodgers and Rowland [5-7].

Physiological databases included in the software incorporate the dependencies of organ composition, organ weights, organ blood flows and gastrointestinal parameters (gastrointestinal length, radius of each section, intestinal surface area, gastrointestinal transit times, and pH in different intestinal segments [2]), with the user-defined body weight and height and ethnicity of the individual [8]. Thereby, PK Sim® allows generating realistic virtual populations. For a detailed description of the PBPK model structure implemented in PK Sim®, see Willmann et al. [2,4,8,9] or the OSP Suite homepage (<https://docs.open-systems-pharmacology.org/mechanistic-modeling-of-pharmacokinetics-and-dynamics/modeling-concepts>).

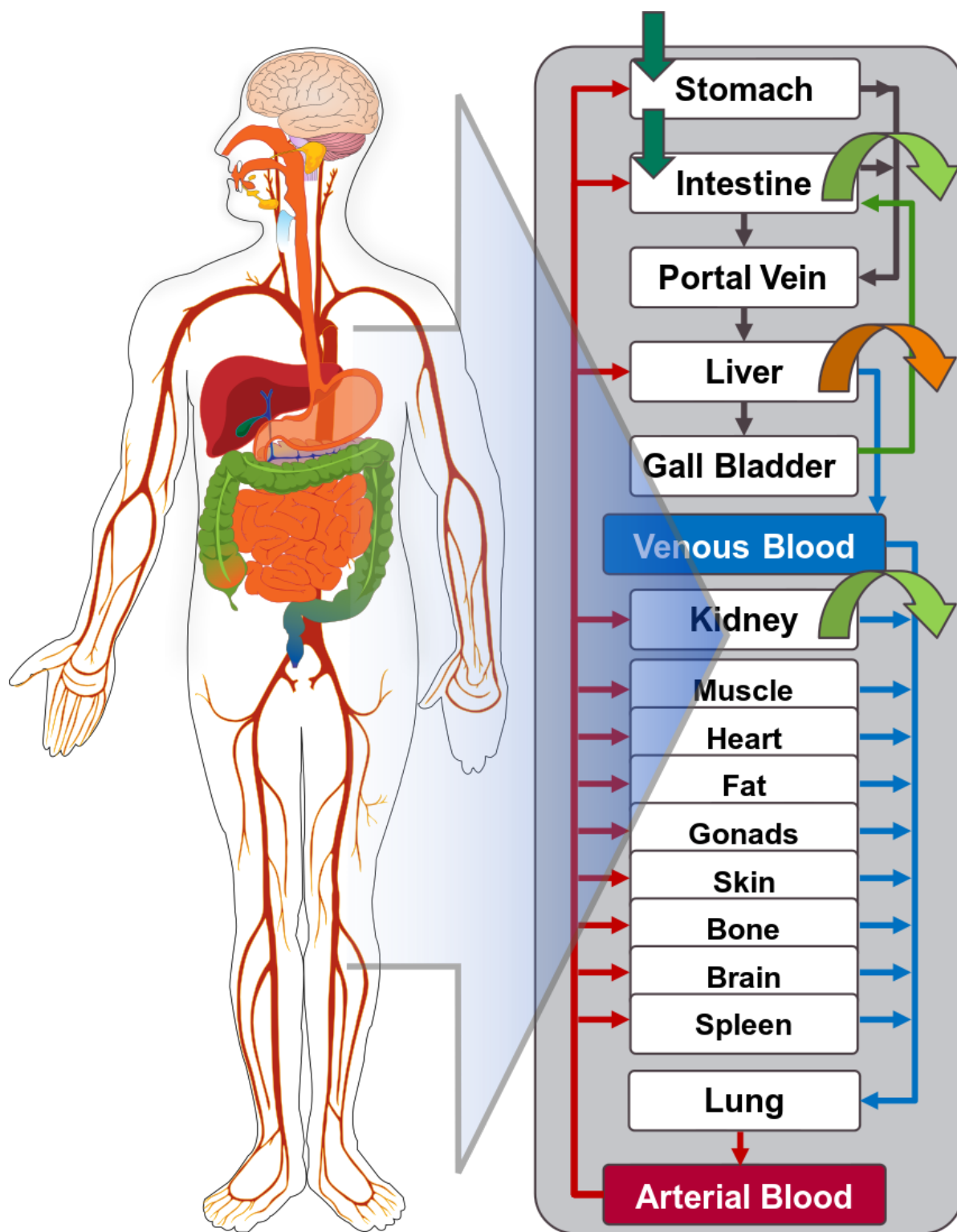


Figure Appendix-1: Structure of the Whole Body PBPK Model integrated in PK-Sim®

## References for OSPS introduction

[1] [www.open-systems-pharmacology.org](http://www.open-systems-pharmacology.org)

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## 5.2 Mathematical Implementation of Drug-Drug Interactions

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### DDI modeling: Competitive inhibition

A detailed representation of the mathematical implementation of competitive enzyme inhibition can be found in the OSP manual (<https://docs.open-systems-pharmacology.org/working-with-pk-sim/pk-sim-documentation/pk-sim-compounds-defining-inhibition-induction-processes#competitive-inhibition-simple-setting-with-one-inhibitor>).

### DDI modeling: Mechanism-based inhibition

A detailed representation of the mathematical implementation of mechanism-based enzyme inhibition can be found in the OSP manual (<https://docs.open-systems-pharmacology.org/working-with-pk-sim/pk-sim-documentation/pk-sim-compounds-defining-inhibition-induction-processes#irreversible-inhibition>).

### DDI modeling: Induction

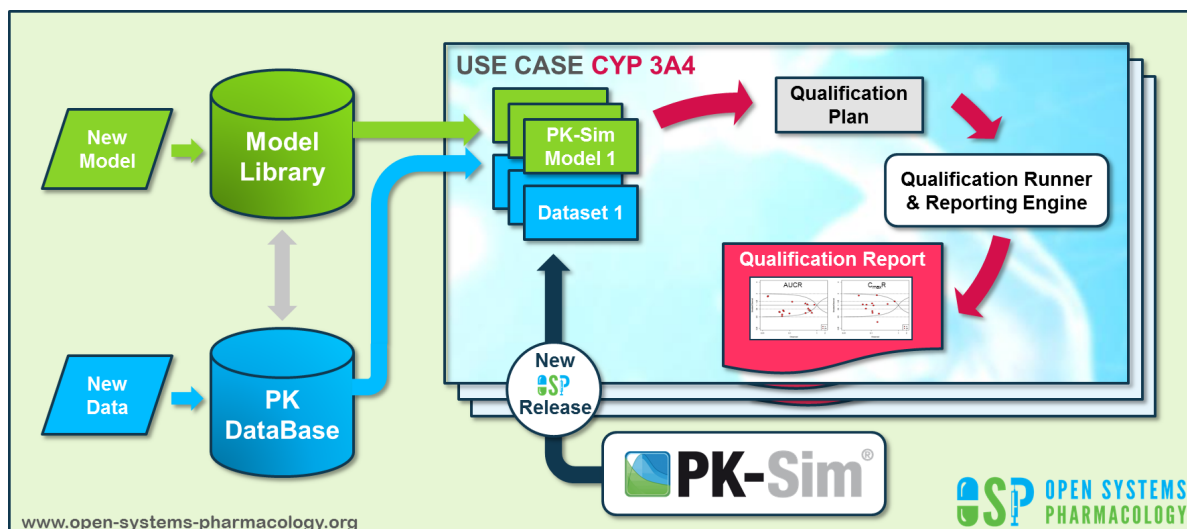
A detailed representation of the mathematical implementation of enzyme induction can be found in the OSP manual (<https://docs.open-systems-pharmacology.org/working-with-pk-sim/pk-sim-documentation/pk-sim-compounds-defining-inhibition-induction-processes#enzyme-induction>).

## 5.3 Automatic (re)-qualification workflow

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Open Systems Pharmacology (<https://www.open-systems-pharmacology.org/>) provides a dynamic landscape of model repositories and a database of observed clinical data. Additionally, a technical framework to assess confidence of a specific intended use has been developed (qualification runner and reporting engine). This framework allows for an automatic (re)-qualification workflow of the OSP suite, comprising the following steps **Figure Appendix-2**:

- PBPK model development and verification with observed data,
- Qualification plan generation,
- Qualification plan execution,
- Qualification report generation.

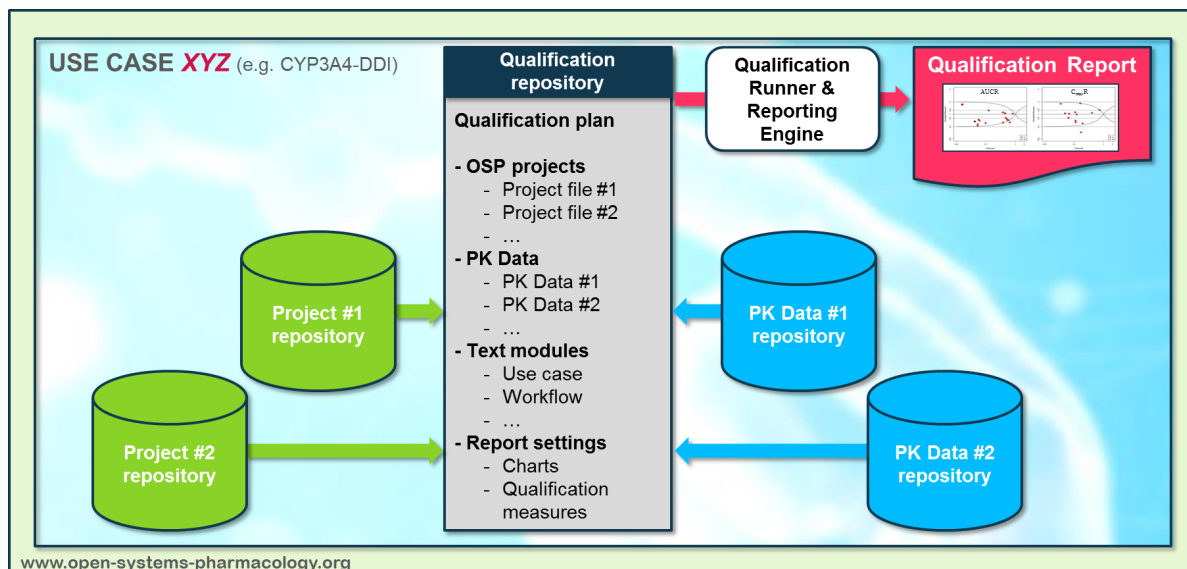


**Figure Appendix-2: OSP suite automatic (re)-qualification workflow**

In a first step, the respective qualification scenario is saved in a special qualification repository on OSP GitHub (<https://github.com/Open-Systems-Pharmacology/>). This qualification scenario repository contains a detailed qualification plan that links and combines respective models and data to address the use case that shall be qualified. Therefore, the qualification plan consists of:

- PK-Sim project files,
- Additional model building steps (if applicable),
- Description of potential cross-dependencies between PK-Sim project files (if applicable),
- Observed data (needed for model development and verification),
- Qualification scenario description text modules
- Detailed report settings to describe the generation of charts and qualification measures.

PK-Sim projects, observed data sets, and qualification scenario text modules are deposited in distinct repositories and are referenced by the qualification plan (**Figure Appendix-3**).



**Figure Appendix-3: Qualification scenario repository landscape on GitHub**

In a second step the qualification runner (<https://github.com/Open-Systems-Pharmacology/QualificationRunner>) processes the qualification plan, i.e. all project parts are exported and prepared for the reporting engine (<https://github.com/Open-Systems-Pharmacology/Reporting-Engine>). The reporting engine provides a validated environment (implemented in R) for model execution and finally generates the qualification report. This report contains

the evaluation of the individual PBPK models with observed data (i.e. standard goodness of fit plots, visual predictive checks) and a comprehensive qualification of the specific use case assessing the predictive performance of the OSP suite by means of a predefined set of qualification measures and charts.

The automated execution of the described workflow can be triggered to assess re-qualification in case new data, changes in model structure or parameterization, or new OSP suite releases arise.